Shrinkflation: What Happens When Package Sizes Change?

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Abstract

Package size is a central characteristic of consumer goods, and changes in size have implications for demand, welfare, policy, and firm behavior. This paper first develops a method to identify size changes at scale. We apply it to retail scanner data and show that size reductions are considerably more common than increases. Next, we estimate dynamic demand responses to size changes with an event study. Following size increases, demand rises immediately and remains stable. In contrast, after size decreases, demand initially falls modestly but continues to decline over time. An incentivized shopping experiment shows that individuals deviate from optimal behavior by responding less to size than to price, and provides evidence of attention as a mechanism. We explain the empirical results with a dynamic demand model that allows for inattention, paired with a supply model in which firms strategically advertise size increases but not reductions. We then derive a sufficient statistic for consumer surplus losses from inattention to sizes and quantify it using estimates from the retail scanner data and the experiment. Following a median size reduction, short-run losses from inattention are equivalent to the deadweight loss from a 6% tax. We also evaluate the policy interventions included in the experiment. Displaying unit prices and making sizes more prominent bring demand responses closer to the optimal benchmark, while size change labels induce overreaction to sizes. Our welfare analyses show that the first two interventions substantially reduce consumer surplus losses from inattention, while the third yields only modest improvements.

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1. Introduction

Package size is a central product attribute: it affects the value consumers derive from packaged goods and the choices they make across products. Understanding how consumers respond to changes in this attribute is therefore important for studying demand, welfare, and firm strategy. Size changes also raise distinct questions about attention, since they may be less noticeable than changes in package prices. If consumers overlook size reductions, they may fail to fully internalize the resulting increase in unit prices (e.g., $/oz).

This has made size changes increasingly prominent in policy debates. Shrinkflation—the increase in unit prices resulting from size reductions—drew criticism from both President Biden, who repeatedly condemned the practice, and President Trump, who criticized it during his most recent presidential campaign (Biden, 2024a, 2024b; Grace, 2024). As a result, policymakers in the U.S. and abroad have sought to address the issue with new laws, bills, and court orders (Casey, 2024; Martin & Yoon, 2023; Niederkofler & Sayahpour, 2026; Omar, 2024; Reuters, 2023; Völk, 2026). These policies include displaying unit prices, mandating labels that highlight size reductions, and even prohibiting size reductions unless they are accompanied by equivalent package-price decreases. However, there is limited evidence with which to predict their potential effects, and the literature in behavioral public economics suggests that even simple disclosure policies can have ambiguous welfare consequences (Allcott et al., 2025).

This paper provides a comprehensive analysis of size changes. We document descriptive patterns and estimate dynamic consumer responses using retail scanner data, explore mechanisms and policy interventions through an online shopping experiment, and develop a theoretical model to explain our empirical findings and quantify welfare implications. The size changes we study involve the replacement of an existing package with one containing a different quantity of the same underlying product.[1] We refer to size reductions as downsizing and size increases as upsizing.

We start by addressing a key data challenge: identifying package size changes. Studying this topic requires a product-level dataset that tracks package sizes over time. However, the commonly used datasets, such as retail scanner data, are structured at the barcode level. This presents a fundamental problem because when manufacturers change package sizes, they typically assign a new barcode, causing different versions of the same product to appear as distinct items. As a result, there is no direct way to link these versions at scale. To tackle this issue, we develop a new algorithm that leverages information on product attributes and sales patterns to identify package size changes. We apply it to the retail scanner module of the NielsenIQ data, which covers a significant share of sales of consumer goods in the U.S. between 2006 and 2024. Validation exercises using package images show that our algorithm is highly effective. This method enables the large-scale analysis of package size changes using retail scanner data.

[1] Size refers to the amount of content in the package, and not to the physical dimensions of the container. We focus on replacements involving relatively similar package sizes, without simultaneous changes to other characteristics of the underlying product (e.g., flavor). Changes in package prices and packaging design are allowed. Size changes are defined at the product level. Appendix Figure A.1 provides examples.

After linking the differently sized versions of the same product, we characterize patterns of size adjustment. We identify more than 12,000 size changes: 72% are size decreases, while the remaining 28% are size increases. Thus, downsizing occurs about 2.5 times as often as upsizing. We provide the first comprehensive distribution of size changes, finding that the median size decrease is 12.5% and the median increase is 16.7%. Package price adjustments only partially offset changes in package size, causing unit prices (e.g., $/oz) to move in the opposite direction. Following size increases, package prices rise by approximately 8% on average. However, because increases in size are usually larger, unit prices fall by an average of 8%. After size decreases, package prices fall by approximately 2%, leading to an 11% increase in unit prices, on average. This pattern supports the use of the term "shrinkflation" for package size reductions, since unit prices tend to rise significantly after downsizing. The time series reveals a strong correlation between the frequency of size changes and inflation.

Next, we study consumer behavior in the observational data by estimating size elasticities over time, defined as the percentage change in demand t months after a 1% change in package size, holding package price and the characteristics of the underlying good fixed. We estimate size elasticities using an event study centered on the introduction of the new package sizes. Identification exploits staggered variation in the timing of size changes across products and stores. For comparison, we also present the price elasticity of demand, which is identified with a Hausman-style instrument.

We find different dynamic patterns in the size elasticities for upsized and downsized products. Following size increases, consumers purchase more packages, and the increased level remains stable over the next twelve months. The size elasticity of packages demanded is 0.54 one month after the change and 0.41 twelve months later; the corresponding elasticities for units (e.g., ounces) demanded are 1.39 and 1.29.[2] The price elasticity is -1.5, regardless of whether demand is measured in packages or units. Downsized products have a different dynamic: the initial decrease in packages sold is small, but sales continue to drop over the following twelve months. The size elasticity of packages demanded rises from 0.41 to 0.83, while that of units demanded rises from 1.46 to 1.94. The price elasticity for these products is -1.85.

The gradual buildup of the demand response to downsizing, as well as the contrast with the immediate and stable response to upsizing, raises questions about the mechanisms underlying these demand responses. Inattention is of particular interest: consumers who overlook size changes may make purchases they would reconsider if they noticed the change. To assess whether the observed responses involve suboptimal purchasing decisions, we need a benchmark for how fully rational consumers would respond in the absence of biases or frictions. The retail scanner data alone cannot provide this rational benchmark. The price elasticity is not an appropriate rational benchmark for the size elasticity because consumers may have preferences over the package size. For example, a consumer who prefers smaller milk cartons to avoid spoilage may not respond to downsizing even if fully aware of the change. The medium-

[2]The roughly one-point difference between the size elasticities for units and packages reflects the fact that units sold equal packages sold multiplied by package size.

term size elasticity is also unsuitable, as consumers may not fully internalize the new package size within the window covered by our estimates. Nor can we use the upsizing elasticity as a rational benchmark: consumers may under- or overreact to size increases, and the upsizing and downsizing elasticities are estimated over different products and potentially different consumers. This lack of a rational benchmark also makes us unable to answer welfare and policy questions based exclusively on the retail scanner data.

We therefore turn to an online shopping experiment to study mechanisms, evaluate policy interventions, and estimate key inputs to our welfare analysis. The central part of the experiment consists of individuals making incentivized consumption decisions over six shopping scenarios. In each scenario, participants receive a $20 budget, and can choose any bundle from the eight products shown to them, subject to the budget constraint. Package prices and sizes are independently randomized at the individual-scenario-product level.

The shopping environment is identical across participants in the first three scenarios, and it visually resembles a real online shopping platform. In the final three scenarios, we introduce between-subject variation by assigning participants to one of four conditions: (i) Control; (ii) Unit Price, which shows unit prices with the same prominence as package prices; (iii) Prominent Size Display, which gives package sizes the same prominence as package prices; and (iv) Size Change Labels, which flags products whose sizes deviate from their baseline values by displaying a label above them. We also elicit beliefs about prices and sizes and directly ask participants whether they noticed changes. In total, 1,951 participants completed the experiment.

Importantly, in the experiment we can derive a clear rational benchmark from the incentive structure, which is what is missing from the retail scanner data. A random subset of the shopping scenarios was implemented. If a scenario was selected for implementation, the participant received the chosen products (via delivery) and the unused budget, subject to an adjustment. The adjustment made it such that incentives depended on the displayed price and size only through their ratio—the unit price. As a consequence, in the experiment, the rational benchmark is the equality of price and size elasticities in absolute value. Participants were informed of the incentive structure at the beginning, and 95% of them correctly answered the two questions testing understanding.

Our demand elasticity estimates in the control group replicate the short-term pattern for downsized products in the observational data: demand is substantially less responsive to product size than to price. Using a Poisson model, we estimate a size elasticity of 0.55 and a price elasticity of -1.62. This is a departure from the rational benchmark of equality of elasticities (in absolute value). Moving to our shopping environment manipulations, we find that displaying the unit prices or the sizes more prominently narrows, but does not eliminate, the gap between price and size elasticities of demand. The ratio of the size to price elasticities (in absolute value) rises from 0.34 in the control group to 0.66 and 0.59 in the Unit Price and Prominent Size Display groups, respectively. The Size Change Label group inverts the relationship between the two elasticities. Under this manipulation, individuals are more responsive to sizes

than to prices, yielding an estimated elasticity ratio of 1.65. This also departs from the rational benchmark of equal elasticities, but in the opposite direction.

The effects of the shopping environment manipulations suggest that attention plays an important role as an underlying mechanism. Two additional results strengthen this interpretation. First, we use a recall measure that asks participants whether they noticed price and size changes across shopping scenarios. Their answers mirror the asymmetries in demand elasticities. Individuals in the control group fail to notice size changes much more frequently than price changes: they underestimate the prevalence of size changes by 36.7 p.p., compared with only 6.7 p.p. for price changes. This gap narrows in the Prominent Size Display group and reverses in the Size Change Label group, consistent with the latter's greater demand responsiveness to sizes than prices. Second, we use the accuracy of participants' beliefs about prices and sizes as a way of distinguishing who paid more attention. Individuals with more accurate price beliefs are more price elastic, and those with more accurate size beliefs are more size elastic, in line with what attention as a mechanism predicts. Beyond providing evidence for attention as a mechanism, the experiment allows us to estimate distributions of attention in each treatment group. We do so with a random-effects Poisson model, and use these distributions in our welfare analyses below.

In the final section, we introduce models that rationalize our empirical findings and allow us to quantify welfare and policy implications through model-derived sufficient statistics. On the demand side, each period consumers decide whether to buy a package of size s at price p. Following a size change, individuals can misperceive the new size. Specifically, the perceived size is a (not necessarily convex) weighted average of the old and the new size. The model features heterogeneity in the attention parameter, which determines the perceived size. Utility depends on actual size, but decisions are made based on the (potentially wrong) perceptions. From this framework, we derive demand elasticities and a sufficient statistic formula for consumer surplus.

In the demand model, the distribution of attention shapes how consumers respond to size changes over time. To explain why size elasticities evolve differently over time for upsized and downsized products in the retail-scanner data, we introduce a supply-side model. Firms receive cost shocks and choose whether to adjust package prices and sizes, with size changes requiring a fixed adjustment cost. We show that firms have an incentive to advertise size increases but not to advertise decreases. If advertising raises attention to size changes, the demand and supply models together can explain the observed asymmetry in demand responses to upsizing and downsizing in the scanner data.

Finally, we use the model-derived sufficient statistic to quantify the consumer surplus losses from inattention to product size. We express them in terms of a common object in public finance: taxes. Specifically, we calculate the tax that, under full pass-through to consumers, would generate a deadweight loss of equal magnitude. We quantify this loss for the median size reduction by combining the price and dynamic size elasticities estimated from the retail scanner data with the distributions of attention across treatment groups estimated in the

experiment, bringing together the paper's empirical and theoretical sections.

Initially, the consumer surplus loss from size inattention is equivalent to the deadweight loss from a 6% tax. This declines monotonically over time as consumers internalize the new size. Moving to the shopping environment manipulations included in the experiment, prominently displaying package sizes or unit prices cuts these losses by about half, while size change labels yield only modest welfare improvements. These findings have direct policy relevance, as disclosing unit prices is already required in many jurisdictions, and size change labels are increasingly being pursued (Niederkofler & Sayahpour, 2026; Omar, 2024; Völk, 2026). Our results support unit price disclosure and more prominent size displays, while cautioning that size change labels can induce excessive responsiveness to size.

This paper is related to several strands of the literature. First, a literature primarily in management and marketing has studied consumer responses to package size changes. Most of this work has focused on specific categories, typically using structural demand models (Çakır, 2022; Çakır & Balagtas, 2014; Kim, 2024; Meeker & von Ditfurth, 2024; Wan et al., 2026; Webb et al., 2022; Yonezawa & Richards, 2016).[3] We differ from this work in some important respects. First, our scope is broader. We consider all products in the NielsenIQ retail scanner data, identify more than 12,000 instances of size changes, and estimate consumer responses across these events. Second, our empirical approach differs. We estimate demand responses using a reduced form approach, whereas these papers mostly estimate demand structurally.

More closely related are Janssen and Kasinger (2026) and Lee (2024), which are concurrent with our paper and use reduced form methods to study demand responses to size changes across a broad set of products. Our algorithm to identify size changes and the validation procedure are broadly similar to those independently developed by Janssen and Kasinger (2026). We complement and extend this work in several ways. These differences also further distinguish our paper from the structural demand literature discussed before. First, we estimate dynamic responses to size changes, finding differences between the short- and medium-term size elasticities for downsized products. Second, we pair the scanner-data evidence with an incentivized online shopping experiment, which provides a rational benchmark not available in the observational data. This allows us to show that consumers respond suboptimally to sizes. Third, using the experiment, we provide direct evidence that attention is a mechanism mediating demand responses to sizes. Fourth, we evaluate the effects of policy interventions, and are the first to use a sufficient-statistics approach to quantify how inattention to sizes affects consumer welfare under different shopping environments.

Our paper also contributes to the empirical literature on limited attention in consumer markets. Prior work shows that consumers often underweight non-salient attributes, including sales taxes, shipping charges, energy costs, and the right digit of prices (Allcott & Taubinsky,

[3]Less directly related literature on size changes includes theoretical models of shrinkflation (Bonell, 2026; Chalioti & Serfes, 2024), work on descriptive patterns and implications for price indices (Budianto, 2024; Rojas et al., 2026), and marketing studies of package-size perceptions, attitudes toward quantity reductions, and judgments about unit-price-increase tactics (Chandon & Ordabayeva, 2009; Kachersky, 2011; Ordabayeva & Chandon, 2013; Yao et al., 2020).

2015; Brown et al., 2010; Chetty et al., 2009; Kraft & Rao, 2026; Lacetera et al., 2012; List et al., 2023; Strulov-Shlain, 2023; Taubinsky & Rees-Jones, 2018). Closest to our setting, Bauner and Hossain (2023) show that unit-price information affects consumers' choices between smaller and larger package sizes. We extend this literature by showing that attention plays an important role in consumer responses to package size changes. We also recover the distribution of attention to sizes and show how it varies across different policy environments.

Furthermore, while our experiment is not designed to fully distinguish among theories of attention, its results align more closely with some models than with others. Salience-based models can predict overreaction to attributes that stand out (Bordalo et al., 2013, 2020, 2022). The results from our size change label environment are consistent with this logic: participants deviated from the rational benchmark of equality between price and size elasticities by responding too much to sizes. A second class of models views attention as a scarce resource that agents allocate optimally across attributes, taking into account the expected benefits and costs of information processing (Maćkowiak et al., 2023; Sims, 2003). This perspective suggests that attention to size should increase when size variation is more consequential for choice. In the experiment, however, more frequent size changes do not systematically increase the size elasticity of demand. We also document substantial overconfidence in the accuracy of price and size recall, which is difficult to reconcile with classes of rational inattention models in which consumers correctly perceive their own attentional constraints.

We also contribute to industrial organization and quantitative marketing work on endogenous product characteristics, shrouded attributes, and disclosure. A broad literature treats non-price product characteristics as endogenous firm choices with implications for demand, competition, and welfare (Draganska et al., 2009; Fan, 2013; Ito & Sallee, 2018; Wollmann, 2018). Models of shrouding and obfuscation show that firms can benefit when consumers do not attend to, or cannot easily compare, economically relevant product attributes (Ellison & Ellison, 2009; Gabaix & Laibson, 2006; Heidhues et al., 2017). A related disclosure and labeling literature studies policies that make product attributes more visible to consumers and shows that such policies can affect both consumer choice and firm responses in quality, reformulation, or communication (Ale-Chilet & Moshary, 2022; Ater & Avishay-Rizi, 2024; Avishay-Rizi & Reshef, 2025; Barahona et al., 2023; Bollinger et al., 2011; Dranove & Jin, 2010; Jin & Leslie, 2003; Kiesel & Villas-Boas, 2013; Lim et al., 2020). Our contribution is to study these forces for package size, a product attribute that is ubiquitous in consumer markets. We show how limited attention to size changes affects demand, derive the resulting incentives for firms' product-design and communication strategies, and examine how disclosure policies can redirect attention in ways that may either improve choices or create new distortions.

Finally, we add to the literature on behavioral public economics (Allcott et al., 2019, 2022, 2025; Allcott & Taubinsky, 2015; Ambuehl et al., 2022; Bernheim & Taubinsky, 2018; Chetty et al., 2009; Taubinsky & Rees-Jones, 2018). We combine a theoretical framework with estimates from both scanner data and our experiment. This allows us to estimate the consumer welfare effects of inattention to size changes and of disclosure policies. More broadly, we provide a

portable framework for studying the consumer welfare effects of changes to non-price product attributes that consumers may imperfectly internalize, including changes in quality. The sufficient-statistics formula we derive applies to this broader class of settings and requires only context-specific estimates of consumer inattention and demand responses.

The rest of the paper is organized as follows. In section 2 we explain our methodology for identifying size changes, provide descriptive statistics, and estimate consumer responses using observational data. Section 3 presents the experimental design and results. In section 4 we introduce our demand and supply models, and do welfare analyses. Finally, section 5 concludes.

2. Size Changes and Demand Elasticities in Retail Scanner Data

In this section, we use data from a large fraction of the U.S. consumer packaged goods retail market to identify package size changes, document key descriptive patterns, and study consumer behavior. Given the broad coverage of the data, our analyses in this section offer strong external validity.

We identify more than 12,000 instances of size changes, with downsizing being 2.5 times more common than upsizing. We present the distribution of size changes, and show that there is a very high correlation between the frequency of size changes and inflation. Studying prices, we find that following size increases, package prices increase but not as much as the size increases, leading to reductions in the price per unit. For downsized cases we see small reductions in package prices after size changes, leading to significant increases in per unit prices. Our elasticity estimates reveal interesting asymmetries in the responses to upsized and downsized products. For upsized products, we find that consumers adjust their purchases immediately after the size change, and the size elasticity is stable over time. In contrast, for downsized products, consumers respond less in the short term, yielding a smaller initial size elasticity that gradually grows over time.

2.1 Data Description

We use the NielsenIQ Retail Scanner data, which cover over 70,000 retail stores and approximately 4.5 million products from 2006 to 2024. The data includes products sold through food, drug, liquor, pet, and convenience stores, as well as mass merchandisers. For each covered location, the module reports the number of packages sold and the average price at the week-store-UPC-version level.[4] It also includes selected product and store characteristics.

On the product side, the 2006–2020 data include a master file with basic information for

[4]UPC stands for Universal Product Code. Although the terms are technically distinct, a UPC can be thought of as a product's barcode.

each UPC-version. These variables include a short product description, brand, product category, size, unit of measurement, and multipack information.[5] Product categories are organized in three nested levels, ranging from 10 broad categories to roughly 1,400 sub-subcategories. A new version is assigned to a UPC when it is observed with a different sub-subcategory, size, brand, or multipack value. This may reflect a change in product characteristics by the company or the reuse of the same UPC for a different product.

In 2021 there were changes to the structure of the data. Primarily, a master file of all UPC-versions is no longer provided, and versioning was discontinued. Information about each UPC is still available at the yearly level. However, the numerical codes assigned to product attributes (e.g., brands) were replaced in 2021, and the corresponding text descriptions for these attributes often differ from the earlier data. Consequently, we cannot reliably track products across these structural breaks, forcing us to treat the 2006–2020 and 2021–2024 data separately when identifying package size changes.

Throughout the sample period, UPCs for private-label products are masked in the data, and multiple distinct products may be mapped to the same masked UPC. We exclude private-label products from the analysis for this reason. Store information is reported at the yearly level and includes store, retailer, and parent-company identifiers. Importantly, it also assigns each store to a designated market area region (DMA region(s)).

2.2 Methodology for Identifying Size Changes

We first address the challenge of connecting the different versions of a product at scale. To study size changes, researchers need a product-level dataset that links the different versions of the same product—namely, the ones before and after the new size is introduced. However, because companies issue new barcodes, or UPCs, for new package sizes in more than 80% of cases, the different versions mostly appear as distinct products in the data. This limitation is a major contributor to the relative lack of research on this issue. To overcome this challenge, we developed an algorithm that uses information on product attributes and sales to link different versions of the same underlying product. Our validations using package images confirm this method is highly effective. This process is similar to the one developed independently by Janssen and Kasinger (2026).

Matching of Product Characteristics. In the 2006–2020 data, we start by using the master file that contains information on each UPC-version. The process begins by identifying, for each UPC-version (row) in this master file, all the other UPC-versions that could be a new upsized or downsized version of it. To do so, for each row (which we call the candidate for the old version), we identify all other rows with the same sub-subcategory, brand, multipack, and a similar but different size. We consider products to be of similar size if the new version differs by no more than 34% from the old version. For each of these candidates, we measure the string

[5] The multipack variable reports the number of items in a package—for example, 6 for a six-pack of soda—although it often takes the value 1 even when the package contains multiple items.

similarity between their descriptions and the description of the row that is the candidate for the old version. We only keep the top 3 most similar ones (plus ties), conditional on them having a minimum level of similarity.[6] The procedure up to this point delivers, for each UPC-version, a set of candidate UPC-versions that may represent upsized or downsized versions of the same product.

Identifying Product Transitions. Next, we use sales data to refine the set of candidates by checking for product transitions. The underlying idea is that, when one package is replaced by another, stores should initially sell only the old version, then at most have a short transition period during which both versions are sold, and should eventually sell only the new version. This step formalizes that idea.

The graph in Figure 1 illustrates the intuition behind this step. In each plot, columns correspond to months and rows correspond to stores, using a random subset of stores for visualization. Each cell therefore represents a store-month. In each cell, we show if only the candidate for the old version was sold (in red), if both versions were sold (in blue), if only the candidate for the new version was sold (in green) or if neither was sold (in white).[7] In Panel (a), initially, only the old version is sold. This is followed by a short period in which most stores selling the product carry both versions, after which only the new version is sold. Panel (b) shows a similar pattern, but without overlap. In this case, the absence of overlap arises mechanically because the comparison is between two versions of the same UPC. Different versions of the same UPC may correspond to distinct versions of the same product, as defined in this paper, or to cases in which a firm reuses a UPC for a different product. Because product characteristics in the data are updated annually, the change is assigned to the end or beginning of the year, and the exact timing of the transition is not observed.

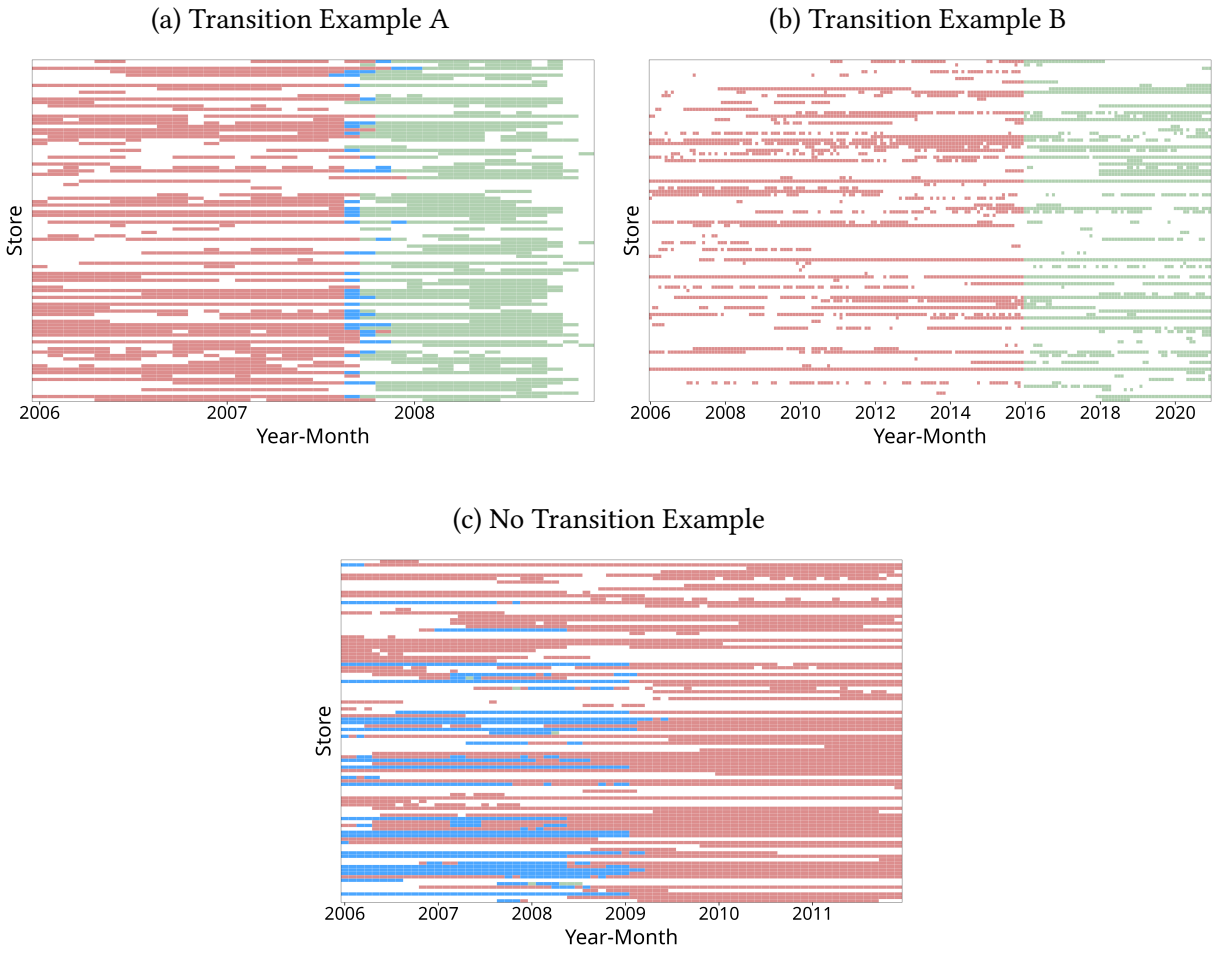
Finally, Panel (c) illustrates a type of pattern observed in cases without a product transition. In this example, both products are sold simultaneously for many years, after which one of them is discontinued. This step of the algorithm is designed to distinguish valid transition patterns such as those in Panels (a) and (b) from non-transition patterns such as the one in Panel (c). The details on how it is formalized are provided in Appendix subsection B.1.

2021-2024 Time Period. The algorithm for 2021–2024 is almost identical. Because the newer data does not include a master file of all UPC-versions, we construct one from the annual files using the same definition of a version as in the earlier data. The main differences concern the string-similarity steps. Product descriptions in the newer data are more detailed,

[6] We use the Indel normalized similarity on the slightly cleaned product descriptions. It starts by calculating the Indel distance, which is the minimum number of insertions or deletions needed to turn one string into the other. It is normalized to take values between 0 and 1 by dividing by the sum of the lengths of the two strings. Finally, the normalized similarity is obtained as 1 - the normalized distance, and it is also in the $[0, 1]$ range. Higher numbers indicate more similar strings. The minimum similarity threshold we impose is 0.6.

[7] The apparent overlap in these plots may overstate the true overlap because the data are aggregated to the monthly level. For example, if a store sells only the old version during the first two weeks of a month and then switches immediately to the new version, the month will appear as an overlap even though the two versions were never on the shelves simultaneously.

Figure 1: Store-Level Transition Patterns Between Old and New Product Versions



Notes: Each row is a store and each column is a month. Each cell shows whether only the candidate for the old version, only the candidate for the new version, both or neither were sold in a given store-month pair. A random subset of stores is plotted. Each panel displays a different pair of UPC-versions.

creating more scope for minor differences across otherwise similar products. We therefore apply additional string cleaning before computing similarities, use a lower minimum similarity threshold for filtering out candidates, and carry forward the five most similar candidates, plus ties, that survive the previous filters into the sales-pattern analysis. All remaining steps are unchanged.

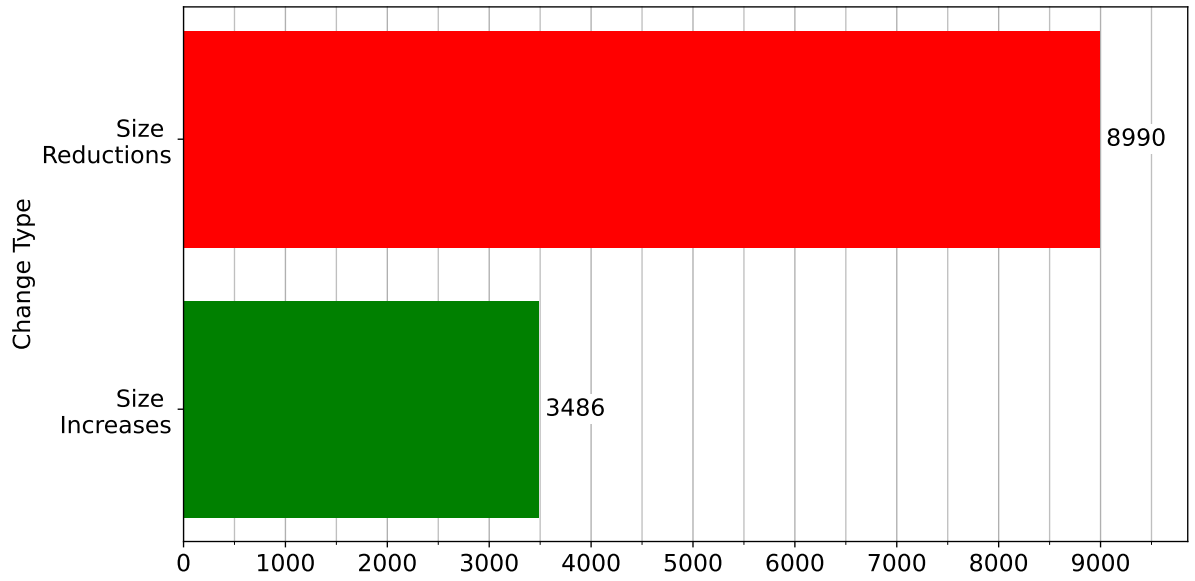
Validation. A natural next question is whether our algorithm accurately identifies product size changes. We validate the algorithm using images of products that it identifies as having changed size and show that it has high precision. For this exercise, we obtained product images from two public barcode lookup databases, Go-UPC (go-upc.com) and UPCitemdb (upcitemdb.com). When using images, the criterion for classifying them as the same product was very strict. If any differences in the underlying product were noticeable (e.g., different flavors,

different calories for the same portion, etc.) that pair is classified as not being the same product. The authors and six RAs independently did more than 2,000 classifications in total. Among those, 83% were determined to be correct identifications of size changes. The remaining 17% were false positives.[8] An informal review of the false positives suggests that the most common error involves matching different flavors of the same product. We also find high agreement across classifiers, further supporting the reliability of the algorithm. This validation exercise shows that our methodology identifies product size changes with a high success rate.

2.3 Descriptive Statistics of Size Changes

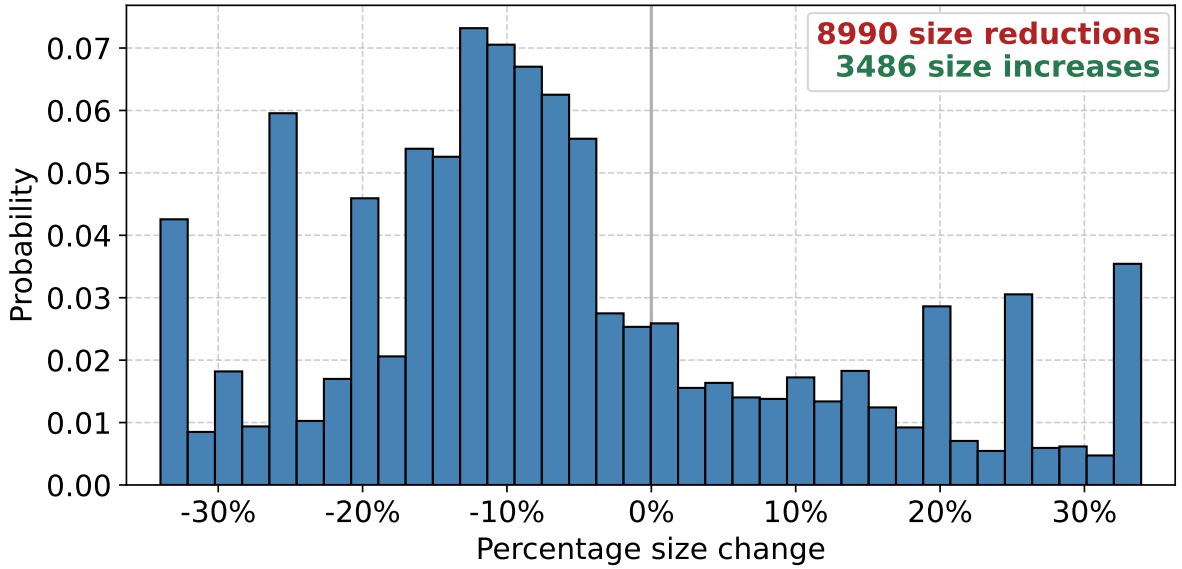
Using the procedure described above, we identify 12,476 size changes: 8,990 reductions and 3,486 increases, making downsizing approximately 2.5 times more frequent than upsizing. Figure 2 presents that information. Appendix Figure B.1 reports instances of up and downsizing by subcategory, showing that downsizing is more common than upsizing in almost all cases. The subcategories with the most size changes are, in order: snacks, candy, pet food, dressings/salads/prep foods-deli, and detergents. In Figure 3, we present what is, to the best of our knowledge, the first comprehensive distribution of the magnitude of within product size changes. The median size change is a 9.1% reduction. Size increases are relatively evenly distributed, with a median increase of 16.7%. In contrast, size decreases have a median of 12.5% and a more asymmetric distribution, with greater mass concentrated in smaller reductions.

Figure 2: Instances of Up and Downsizing Identified



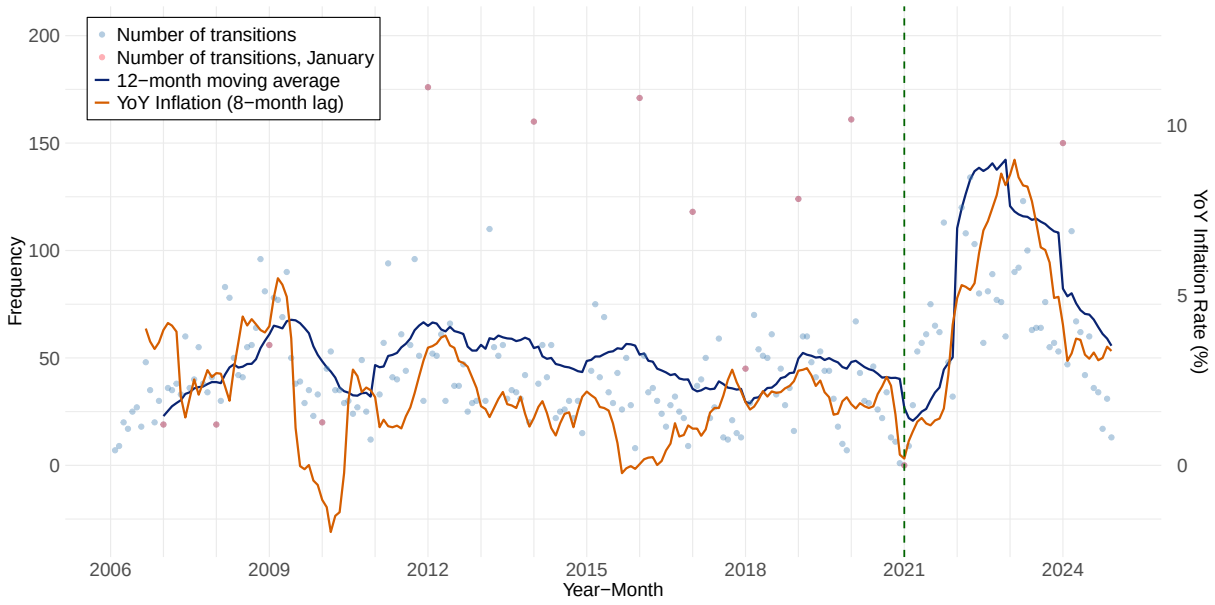
[8]For a substantial number of pairs we were unable to verify whether they were the same product because images were not available for at least one of the versions, or they were not of high enough quality.

Figure 3: Distribution of the Magnitude of Size Changes (in %)



We next examine how the prevalence of package size changes varies over the years. Because stores may transition to newer versions at different times, the timing of a version change is defined at the product-store level. For each product-store pair, we define the transition time as the month when a store first sold the new version. To measure the number of products that change size in each period, we aggregate to the product level. We define the event time as the median event month across stores for a given product. In Figure 4, we present this frequency and its 12-month moving average. We also overlay the year-on-year inflation calculated using the CPI, and we lag it by eight months for the plot. The two series show a very high correlation (0.75), and both spike in recent years, when inflation was higher. In Appendix Figure B.3 we show the upsizing and downsizing time trends separately. The two series exhibit broadly similar patterns, although the prevalence of size decreases varies more over time when compared to size increases.

Figure 4: Number of Size Changes Per Month and Inflation



Notes: This plot shows instances of size changes (circles), the 12-month moving average (blue line), and year-on-year CPI inflation (orange line) lagged by eight months. The instances of size changes in January of each year are plotted in red to highlight that due to the nature of the data many of the size changes that happened in the previous year are being assigned to January of the next year. The top 2% of circles are not plotted because they are out of the y-axis range.

2.4 Methodology for Estimating Demand Elasticities

This subsection explains how we estimate demand elasticities, a measure of how consumers respond to changes in prices and sizes. Following DellaVigna and Gentzkow (2019), to identify the price elasticity we instrument the price with the prices of the same product, in the same time period and chain, but in different markets. To identify the size elasticity and its time dynamics, we use an event study specification, exploiting the fact that different products and different stores transition into the new version at different times.

Main Specification. For our main specification, we estimate separately for upsized and downsized products the following equation:

$$ln(\text{sales}_{gjt}) = \beta^P ln(\text{price per package}_{gjt}) + \sum_{k \in \mathcal{K}} \beta_k^S D_{gjt}^k \cdot \frac{\text{new size}_g - \text{old size}_g}{\text{old size}_g} + FEs + u_{gjt} \tag{1}$$

where g indexes products, j stores and t time, measured at the monthly level. Let k denote time relative to the event, the first month in which the new size is sold, with $\mathcal{K} = \{\underline{k}, \ldots, -4, -2, -1, 0, \ldots, \bar{k}\}$. $(\underline{k}, \bar{k})$ are "binned" endpoints. In the baseline specifications, the omitted period is $k = -3$. We choose this period because sales decline in the months immediately preceding the size change, especially at $k = -1$, consistent with stockouts or

limited availability preceding the transition.[9]

D^k_{gjt} are the event time dummies, indicating that store j transitioned into the new size of product g k periods ago. We interact the event time dummies with the proportional change in package size, equivalent to the intensity of treatment. This turns the β^S_k, for $k \geq 0$, into (approximated) elasticities. Our preferred specifications include the following fixed effects: month, product sub-subcategory by year and sub-subcategory by month of the year (capturing different yearly shocks and seasonality patterns for different classes of products), store by product, weeks of display, and weeks of feature in that month (these variables capture, for example, if a product was displayed prominently in the store).

The outcome variables are the natural logarithm of packages sold and the natural logarithm of units sold. Here, "units" denotes the quantity of the underlying good, measured in ounces, pounds, fluid ounces, or the product's relevant unit of measurement.

To address the potential endogeneity of prices, we use an instrument previously established in the literature. The instrument is the natural logarithm of the average of the prices of the same product in stores of the same chain in the same period but in different markets, $ln(\overline{\text{price per package}_{g,j\notin\mathcal{M},ct}})$, where c indexes chains and $j \notin \mathcal{M}$ refers to the stores in different markets. A chain is defined by the combination of parent and retailer codes, as in DellaVigna and Gentzkow (2019), and for the market definition we take Nielsen's designated market area regions (DMA region(s)). This approach builds on the classic instrument proposed by Hausman (1996). A similar instrument has been used with the NielsenIQ data by DellaVigna and Gentzkow (2019), Strulov-Shlain (2023) and Allcott et al. (2019). The identification assumption behind this instrument is that chain-level price changes are conditionally uncorrelated with local demand shocks.

The demand effects of size changes are identified from the staggered timing of package size changes across products and stores, conditional on permanent product-store differences, common calendar-month shocks, sub-subcategory-specific annual shocks and seasonality, display and feature controls, and (instrumented) contemporaneous package prices. Because the estimation sample includes only products that eventually change size, the comparison is not with never-treated products. The relevant counterfactual is instead constructed from product-store observations at different points in event time.

Intuitively, the design asks whether sales move systematically around the month in which a product changes size, and whether larger size changes are associated with larger sales responses, relative to the sales changes of other product-store observations facing the same aggregate and product-class time shocks. The identifying assumption is that, had package sizes remained unchanged, product-stores with different size-change dates would not have experienced differential shifts in residual demand around those calendar dates. The magnitude of

[9]This decline is unlikely to reflect a shock that induced the size change. Changing package sizes requires firms to select the new size, redesign packaging, adjust production, and distribute the new version to retailers. The timing therefore makes it unlikely that a sales decline in the month before the transition both triggered the change and allowed the new version to reach shelves in the following month.

the size change must also not be systematically related to how residual demand would have evolved in its absence. The event-study coefficients before the transition provide a direct check on this assumption in the pre-change period.

The main parameters of interest from Equation 1 are β^P, the price elasticity of demand, and the β_k^S for $k \geq 0$, which capture the size elasticity of demand k periods after the size change. The estimates of β_k^S for $k < 0$ are used to assess whether products exhibit differential pre-trends.

Alternative Specification. For robustness, we also estimate:

$$ln(\text{sales}_{gjt}) = \beta^P ln(\text{price per package}_{gjt}) + \sum_{k \in \mathcal{K}} \beta_k^S D_{gjt}^k + FEs + u_{gjt} \tag{2}$$

Here, the notation is the same as above, and we use the same price instrument. The difference is that in this equation the event dummies are not scaled by the intensity of treatment. Hence, the β_k^S coefficients for $k > 0$ are not interpretable as elasticities. Instead, $100 \cdot \beta_k^S$ is the approximate percent change in demand k periods after the introduction of the new version.

This specification allows us to estimate dynamic effects and assess whether larger size changes lead to larger demand responses without relying on the constant elasticity assumption implicit in Equation 1. To achieve this, we split the upsized and downsized products into three groups each based on the magnitude of the size change. Then, we estimate Equation 2 separately for each group. Comparing coefficients within a group over time allows us to assess whether the dynamics mirror those in the main specification. By comparing the coefficients within a period across groups, we assess whether the magnitude of the size change influences the demand response.

Price Movements. Finally, to understand the behavior of prices around the time of a size change, we estimate—using in turn package price and unit price—the following event study:

$$ln(\text{price}_{gjt}) = \sum_{k \in \mathcal{K}} \beta_k D_{gjt}^k + FEs + u_{gjt} \tag{3}$$

where the fixed effects and notation are the same as in Equation 1.

Aggregation. Our primary estimation strategy treats upsizing and downsizing events separately. Due to computational restrictions given the scale of our data, we further partition the sample within these groups and estimate regressions by category and time period (2006–2020 and 2021–2024). We then aggregate these results using a weighted average of the coefficients. In the main specification, we weight by the number of products in each regression. As a robustness check, the appendix reports estimates weighted by the number of observations in each regression.

Formally, using c to index categories, tp to index time periods (2006–2020 and 2021–2024), and ω to denote the weighting variable, our final estimate is obtained as:

$$\hat{\beta}_{\omega} = \frac{\sum_{c,tp} \omega_{c,tp} \hat{\beta}_{c,tp}}{\sum_{c,tp} \omega_{c,tp}} \tag{4}$$

In each regression, we cluster standard errors at the product level. Assuming that unobserved shocks are uncorrelated across products (which is an implication of the assumption underlying clustering at the product level), the variance of the aggregated estimator is:

$$Var(\hat{\beta}_{\omega}) = \frac{\sum_{c,tp} \omega_{c,tp}^2 \cdot Var(\hat{\beta}_{c,tp})}{(\sum_{c,tp} \omega_{c,tp})^2} \tag{5}$$

To estimate this variance, we use the squared clustered standard error from each regression as the estimate of the corresponding coefficient variance, $Var(\hat{\beta}_{c,tp})$. The only exception to this partitioning and aggregation method is the analysis that splits products by the magnitude of the size change. In these cases, the subsamples are sufficiently small to allow for estimation across all categories and time periods jointly for each of the groups.

Sample Restrictions. We impose several restrictions on the estimation samples. These restrictions are designed primarily to strengthen the internal validity of the estimates. At the product level, we retain only products that undergo exactly one size change within 2006–2020 or 2021–2024. We treat these time periods separately because changes in the data beginning in 2021 prevent us from tracking products consistently across them. The resulting sample therefore consists of products that are eventually treated. We also exclude products that change size while retaining the same UPC. As discussed above, for these products we observe that size changed between years, but not the month in which the new version was introduced in each store. Because the event month is not observed, these products are not used in the event-study estimation. At the store level, we exclude stores with missing DMA region(s), or that have different DMA region(s) in different years, which may reflect Nielsen market redefinitions. We also drop observations for which we cannot construct the price instrument (e.g., in a given month, a chain sold that product in only one market).

Next, we impose restrictions at the product by store level. First, for each product, we keep only stores in which the last sale of the old version and the first sale of the new version occur no more than one month apart. This restriction minimizes the number of months with partial treatment, in which both versions are sold in the same store.

We then address a limitation of the data: for a given UPC-store, price is observed only if the product is sold in that period. This missingness is not random, since sales are likely to be lower when prices are higher. We therefore impose a balance requirement around the time of the size change. For each product, we keep only stores with positive sales in the event month—the first month with sales of the new version—as well as in each of the 12 months before and after the event. Furthermore, we require stores to have sold at least two packages in each of the 12 months prior to the event. This last restriction allows demand to decline after the size change without mechanically causing the product-store to drop out of the sample. We impose no additional restrictions outside this 25-month window. This requirement is stringent and ex-

cludes a large share of product-store observations. Other papers using the same data address missing prices in zero-sales periods by focusing on frequently purchased products, thereby reducing the incidence of missing data (DellaVigna & Gentzkow, 2019; Strulov-Shlain, 2023). Our approach is conceptually similar. Appendix Table B.1 compares basic sample characteristics before and after imposing these sample restrictions.

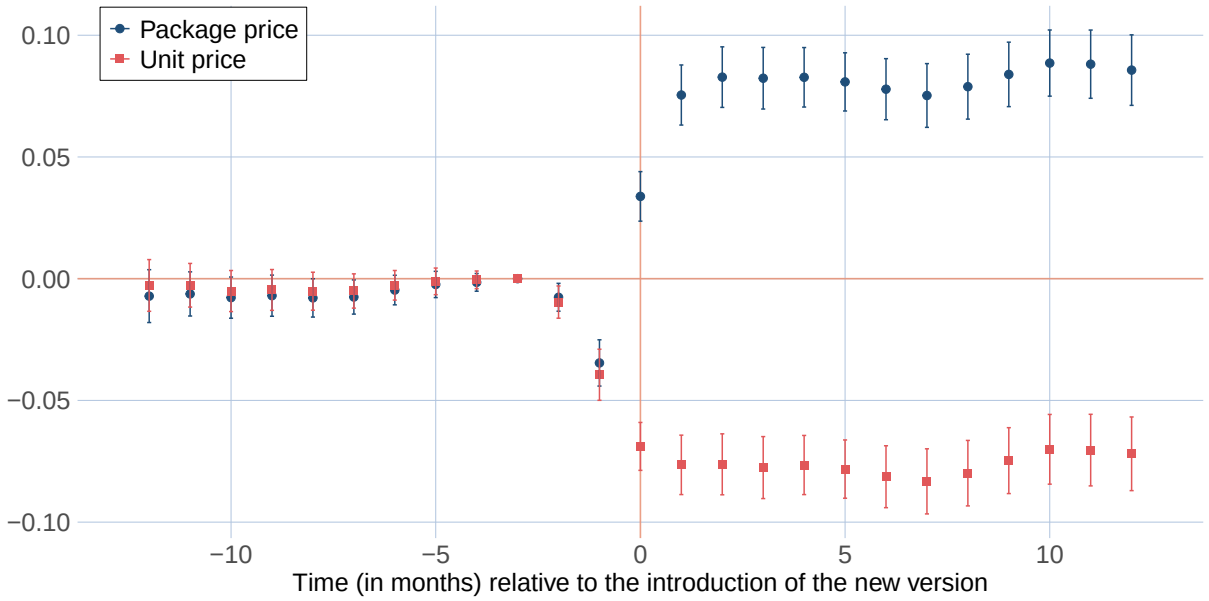
2.5 Results

This section presents the main results from the observational data. We first examine the relationship between size changes and prices. We find that size changes are followed by changes in package prices in the same direction, but of a smaller magnitude. As a result, unit prices increase after downsizing and decrease after upsizing. We then study consumer responses. For upsized products, consumers respond immediately: the size elasticity of units demanded is stable over time and similar in magnitude to the price elasticity. The dynamics are different for downsized products. There, the short-run size responsiveness is smaller, with the elasticity increasing over time. Twelve months after the size change, the size and price elasticities of units demanded are similar.

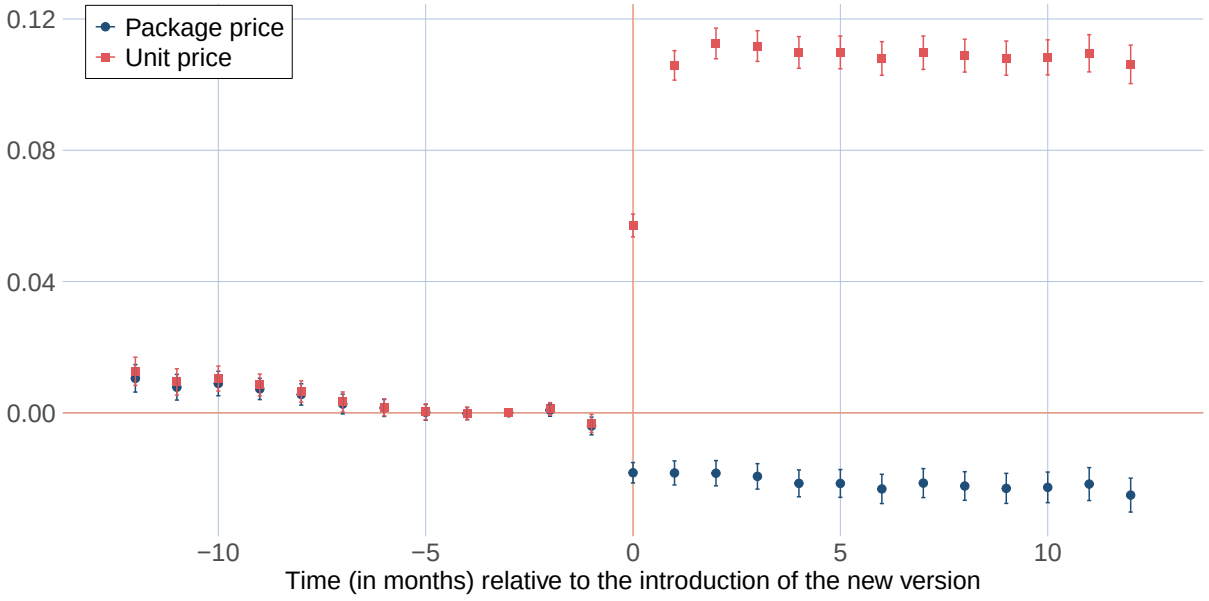
Price Movements. We begin by examining price dynamics around the time of size changes. These event study results are presented in Figures 5a and 5b, as well as in Appendix Table B.3. For upsized products, package prices increase by 8% on average after the size change. Because increases in package size tend to be larger than increases in package price, unit prices fall by 8% on average. Downsized products exhibit a similar pattern, but in the opposite direction. Package prices decrease by 2% on average after the size change, but unit prices increase by 11% because package sizes fall by more than package prices. This finding shows that size decreases, on average, fit the pattern implied by the "shrinkflation" label: they translate into unit price increases.

Figure 5: Price Dynamics Around Package Size Changes

(a) Upsized Products



(b) Downsized Products



Notes: Estimated coefficients and 95% confidence intervals. These aggregate results use the number of products in each regression as weights. Separate regressions were estimated for each product category in each time period following Equation 3, with standard errors clustered at the product level. Panel (a) presents the estimation for upsized products, while panel (b) does so for downsized products. The blue circles plot the estimates using the natural logarithm of package prices as the outcome variable, while the red squares plot the estimates using the natural logarithm of the price per unit as the outcome variable. The coefficients of the binned endpoints are not plotted.

Demand Elasticities. We next turn to consumer responses, beginning with effects on packages sold. These results are presented in Appendix Figure B.4 and Appendix Table B.2. The main dynamic patterns are already visible in these estimates. For upsized products, the response is relatively stable over time: the size elasticity of packages demanded is 0.54 in the month after the size change and remains between 0.4 and 0.6 over the next 11 months. For downsized products, the response is more gradual. The size elasticity is 0.41 in the month after the change and increases steadily over time, more than doubling over a year to reach 0.83 after 12 months. In both cases, the size elasticities are well below the corresponding price elasticities, estimated at 1.52 and 1.85 in absolute value for upsized and downsized products, respectively.[10]

Our preferred specifications use units sold, rather than packages sold, as the outcome, where units are measured in the product's underlying quantity, such as ounces. This distinction does not affect the price elasticities. It is central for the size elasticities, however, because the effect on units sold combines the change in packages demanded with the mechanical change in package size. Focusing on packages sold alone can therefore be misleading because it omits the mechanical change in quantity per package. For example, a decline in packages sold after upsizing need not imply a decline in total demand for the product: consumers may buy fewer packages while purchasing the same or larger total quantity. Conversely, a stable number of packages sold after downsizing can mask a decline in total quantity purchased. Thus, units sold can provide a more relevant measure of demand as it combines information on consumers' package choices with the amount of product contained in each package.

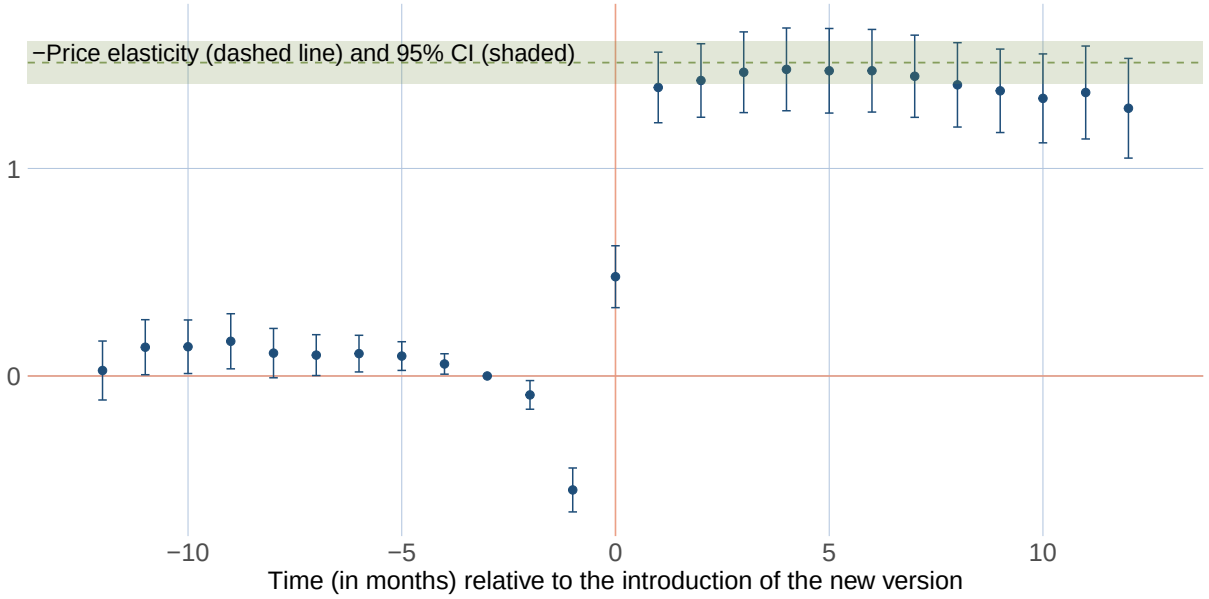
The estimates using the natural logarithm of units sold as the outcome variable are presented in Figures 6a (upsized products), 6b (downsized products), 7 (size elasticities for both kinds of products, rescaled by their long term value), and Appendix Table B.3. For upsized products, the response to size changes is immediate and stable, with a size elasticity close to the corresponding price elasticity. The size elasticity is 1.39 one month after the size change and 1.29 twelve months after the event. The price elasticity is -1.51. For downsized products, the initial response is smaller and increases over time, approaching the magnitude of the price elasticity. The size elasticity is 1.46 one month after the size change and rises to 1.94 twelve months after the event. The estimated price elasticity for these products is -1.85.

These results indicate asymmetric responses to package size changes. One interpretation is that upsizing is immediately salient because firms have incentives to make larger package sizes visible to consumers. Indeed, larger packages are profitable only if consumers notice the additional quantity. These incentives are not in place for downsizing scenarios. The gradual response to downsizing is consistent with consumers that only partially internalize the size reduction and therefore underreact to it. As consumers notice the change over time, the demand response increases. We provide further evidence for this attention interpretation below through a randomized online shopping experiment, and formalize this intuition in section 4.

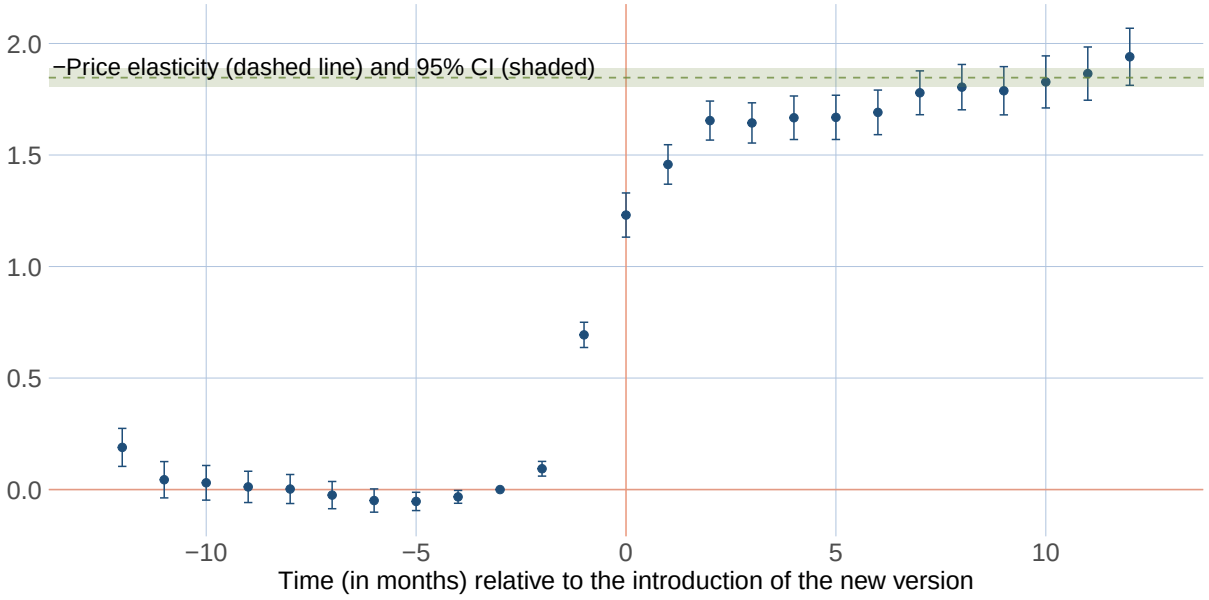
[10] In general, there is no theoretical reason for price and size elasticities to coincide, as consumers may value these two dimensions differently. We formalize this distinction in subsection 4.1.

Figure 6: Size and Price Elasticities of Units Demanded

(a) Upsized Products

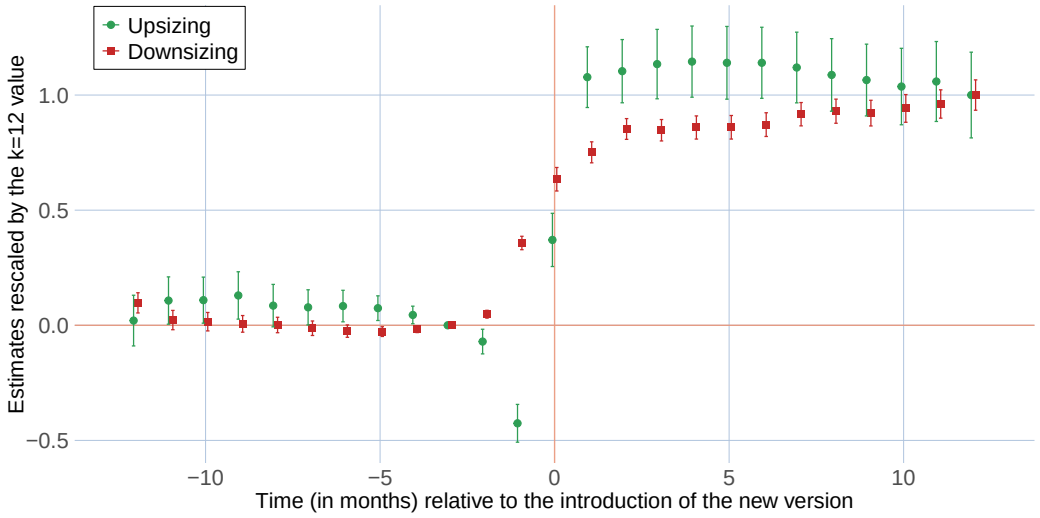


(b) Downsized Products



Notes: Estimated coefficients and 95% confidence intervals. The coefficients of the binned endpoints are not plotted. The dashed line plots the negative of the price elasticity, $-\widehat{\beta}^{P}$, with the shaded area showing its 95% confidence interval. The sample in the top panel (a) includes only upsized products, while the sample in the bottom panel (b) includes only downsized products. The outcome variable is the natural logarithm of units sold. Separate regressions were estimated for each product category in each time period following Equation 1, with standard errors clustered at the product level. These aggregate results use the number of products in each regression as weights.

Figure 7: Size Elasticities of Units Demanded Rescaled by Their Long Term Values



Notes: Estimated coefficients and 95% confidence intervals. This figure plots the pre-event event study coefficients and the post-event size elasticities of units demanded for both upsized and downsized products, each series rescaled by their respective estimated values 12 months after the size change. The coefficients of the binned endpoints are not plotted. The outcome variable is the natural logarithm of units sold. Separate regressions were estimated for upsized and downsized products for each product category in each time period following Equation 1, with standard errors clustered at the product level. These aggregate results use the number of products in each regression as weights.

Robustness. We conduct a series of robustness checks. To begin, we show the results obtained by aggregating the estimates using as weights the number of observations (instead of the number of products) in each regression. The findings are presented in Appendix Figures B.5 and B.6. This alternative weighting leaves the qualitative patterns unchanged.

Next, we estimate the specification without rescaling by the magnitude of the size change, given by Equation 2, using the natural logarithm of packages sold as the outcome variable. For this exercise, we divide upsized and downsized products into three groups, each containing approximately the same number of products, based on their percentage size change. These subsamples are small enough to estimate jointly across categories and time periods without running into computational constraints, so we estimate one regression per group, for a total of six regressions. This exercise serves three purposes. First, it confirms that the asymmetric dynamics between upsized and downsized products are not artifacts of the outcome variable rescaling or the treatment intensity adjustment used in our primary specification. Second, it provides direct evidence that larger size changes generate larger demand responses. Third, it shows that the main results are not dependent on our aggregation strategy.

The results for downsized products are clear and presented in Appendix Figure B.7b. We find no evidence of pre-trends. After the size change, packages demanded decline for all groups, with larger percentage reductions in size generating larger declines in demand. Furthermore, the dynamics are in line with our main results: there is a smaller short term response that grows over time.

The estimates for upsized products are noisier, reflecting the smaller number of products in these regressions. Appendix Figure B.7a presents these results. Overall, groups with larger size increases tend to exhibit larger demand responses, although this pattern is not uniform across all event months. The response is generally stable over time, except for the group with the smallest size increases, where the point estimates decline. If anything, this pattern implies a declining size elasticity, in contrast to the increasing elasticity observed for downsized products.

Implications. The evidence in this section establishes a robust asymmetry in demand responses to package-size changes. For upsized products, the response is immediate and remains stable over time. For downsized products, demand is less elastic with respect to size in the short run, but the elasticity increases over time. However, these results by themselves do not provide direct evidence of mechanisms, nor do they allow us to derive policy and welfare implications. To address these questions, we turn to an online experiment designed to complement the observational evidence, isolate mechanisms, and inform the welfare and public policy discussion.

3. Consumer Behavior in an Online Shopping Experiment

We designed an online shopping experiment to complement the previous evidence. It provides experimentally identified demand elasticity estimates, offers direct evidence on attention as a mechanism, allows us to evaluate policy interventions, and generates the inputs needed for welfare analysis. The experiment was conducted in an environment designed to mimic an online retail marketplace. Relative to an in-person experiment, this setting allows for tighter control over product characteristics and the shopping environment.[11]

Participants chose bundles in incentivized shopping scenarios with randomized prices and package sizes. In the second half of the experiment, we randomized participants into four shopping environments: control, unit-price disclosure, prominent size display, or size-change labels. Throughout, the experimental incentives implied a rational benchmark of equal price and size demand elasticities. The experimental results replicate the short-run scanner data findings for downsized products: in the baseline condition, participants are less responsive to package sizes than to prices, implying a deviation from optimal behavior in our setting. Elicitations of beliefs on product attributes point to attention as a mechanism. Consistent with this interpretation, displaying sizes more prominently and showing unit prices reduce the gap between price and size elasticities. This evidence supports the effectiveness of these interventions. By contrast, size-change labels reverse the gap: individuals become more responsive to sizes than to prices. This suggests that such a policy may induce bias in the opposite direction.

[11] Furthermore, online grocery shopping is relevant in its own right. It is a large and rapidly growing market in the United States, with annual sales exceeding $250 billion (Capital One Shopping, 2024). Thus, this experimental setting provides a natural complement to the observational evidence from brick-and-mortar stores.

3.1 Experimental Design

Screening and Product Selection. The survey began with a consent form that described the study and stated the eligibility criteria: participants had to be at least 18 years old and reside in the United States. Participants then answered three screening questions. They reported their country of residence, whether they were interested in purchasing any of the four product categories featured in the study—pasta, chips, hand soap, and cereal—on an upcoming grocery trip, and whether delivery services were available in their area. Participants who did not reside in the United States, were not interested in purchasing any of these product categories, or did not have access to delivery services were screened out of our survey.

Each participant completed six shopping scenarios. In each scenario, participants had a budget and could choose any combination of the displayed products. In the shopping scenarios, each participant encountered products from two categories. To enhance the external validity, these categories were drawn from the participants' purchase interests in the screening stage. Participants who indicated interest in purchasing products from one category were shown products from that category and from one randomly assigned category. Participants who indicated interest in two categories were shown products from both. For participants who selected three or four categories, we randomly chose two from their selection.

For each of their two assigned product categories, participants first viewed six products and indicated whether they would consider buying each one at the usual price. In the shopping scenarios, participants were shown four products from each category. These products were selected as follows. If a participant expressed interest in four or fewer of the six products in a category, all products of interest were included, and any remaining slots were filled by randomly selecting from the products in which the participant had not indicated interest. If a participant expressed interest in more than four products, four were randomly selected from that set.

Incentive Structure. Next, we described the survey structure and experimental incentives to the participants, and assessed their understanding of the latter. A random 5% of participants had one of their shopping scenarios implemented. For these participants, we randomly selected one of the six shopping scenarios and shipped the products chosen in that scenario to an address they provided. Participants also received the unspent portion of their budget in that scenario, subject to a price adjustment.

For each product in our survey, only one of the package sizes displayed in the experiment was available for purchase in stores, and this was the version delivered to participants. To maintain the incentives associated with package size variation in the experiment, we adjusted the final price of each product to reflect the unit price of the option selected by the participant. For example, if a participant chose a 4 oz package displayed at $3.20, but the available real-world product was sold in a 5 oz package, we valued the purchased item at $4.00 (or 80 cents more per package), using the participant's selected unit price of $0.80 per ounce. The resulting adjustment in total expenditure, considering the adjusted price difference multiplied by the number of packages purchased across all products, was then subtracted from, or added to, the

individual's remaining budget. Participants randomized into delivery subsequently received any unused adjusted budget plus a $2 bonus.[12]

A key implication of the experimental incentives is that because the adjustments operated through the unit price, the displayed package price and size did not factor into the incentives separately. Rather, only the price per unit was relevant. Other considerations that may affect ordinary consumption choices, such as preferences over package sizes, were held fixed: respondents could receive only the real-world size of each product. This design therefore yields a rational benchmark that is not available in the observational data: the price and size elasticities should coincide. To ensure that respondents understood the incentive structure, we administered a comprehension check. Respondents who answered incorrectly on the first attempt were shown the incentive explanation again and retested.

Experimental Randomizations. Participants were randomly assigned to one of three groups that differed in the frequency with which product sizes differed from their baseline values. For each participant-product-scenario observation, prices and sizes were independently randomized. The size distribution varied by group. Products appeared at their baseline size with probability 40%, 60%, or 80%, depending on the arm. Within each group, the residual probability mass was allocated equally among size deviations of -20%, -10%, +10%, and +20%. The price distribution was held fixed across arms: products appeared at their baseline price with a 60% probability, and deviated by -20%, -10%, +10%, or +20% with a 10% probability each. This randomization into size-change-frequency groups was independent of all other experimental manipulations.

The shopping task began with a practice scenario, which was excluded from the six scenarios used in the analysis and was not eligible for implementation. The practice scenario was included to familiarize participants with the choice environment. Participants then completed the first three shopping scenarios. Between scenarios, product prices and sizes were re-randomized. The position of each category, whether in the top or bottom row, and the order of products within each row were also randomized across scenarios. In each scenario, participants had a $20 budget and could purchase any bundle of goods satisfying the budget constraint. Panel (a) of Figure 8 shows an example of the initial shopping scenarios.

Participants then proceeded to the final three shopping scenarios. They were randomly assigned with equal probability to one of four conditions: (i) control, which replicated the initial shopping environment; (ii) unit price, in which the price per unit (e.g., per ounce) was displayed; (iii) prominent size display, in which package sizes were displayed as prominently as package prices; and (iv) size change label, in which products with sizes that differed from their baseline value displayed labels alerting participants to the size difference. Participants completed the three final scenarios in their assigned environment. As before, prices and sizes were randomized at the individual-product-scenario level. Panel (b) of Figure 8 shows the treatment manipulations for a single product. Appendix Figures C.1 through C.4 provide screenshots of

[12]The additional $2 was added to minimize the chances that participants would end up with a negative remaining budget after the price adjustments.

the full shopping environment for each treatment arm.

Belief Elicitation. For a randomly selected 20% of participants, we elicited beliefs prior to the shopping scenarios. Specifically, we asked these respondents the local retail prices and package sizes of four products—two from each category—that they would subsequently encounter in the survey. Participants not assigned to this elicitation proceeded directly to the shopping scenarios after the incentive explanation and comprehension checks. Following the first block of shopping scenarios, participants assigned to the initial belief-elicitation module completed a second elicitation. In this elicitation, we asked them to report the price and package size of the same four products in the immediately preceding shopping scenario, Scenario 3.

After all the shopping scenarios, we informed all participants that they would be asked about the prices, unit prices, and package sizes of selected products from the final scenario, Scenario 6. We then elicited beliefs about the likelihood that their responses would fall within 5% of the true value for each product's package price and package size. For a randomly selected 25% of participants, one belief (either package size, package price, or unit price for one product) was selected for bonus payment up to \$1 based on its accuracy.[13] These participants were informed about this incentive, but did not know which belief would be selected for payment. All participants were then asked about the package price and size of four products from Scenario 6 and answered whether, for those products, they had noticed any changes in prices, sizes, or package design across scenarios 4 through 6. On the next screen, participants were asked about the unit prices of the same four products.

Additional Questions. The final section of the survey elicited qualitative responses on participants' preferred shopping environment, familiarity with the products shown in the scenarios, preferences over price versus size changes, and views on shrinkflation and related legislation. We also collected demographic information. At the end of the survey, participants were informed whether they had been selected for delivery implementation.

3.2 Sample Selection and Data

Recruitment and Sample Selection. The experiment was conducted in April 2026 with a sample recruited through Prolific. A total of 2,000 individuals completed the survey. Following the pre-registered sample selection process,[14] we exclude 49 observations flagged by our quality-control procedure, which combines a measure provided by the survey platform Qualtrics and two custom AI-detection questions.[15] These restrictions yield a final sample of

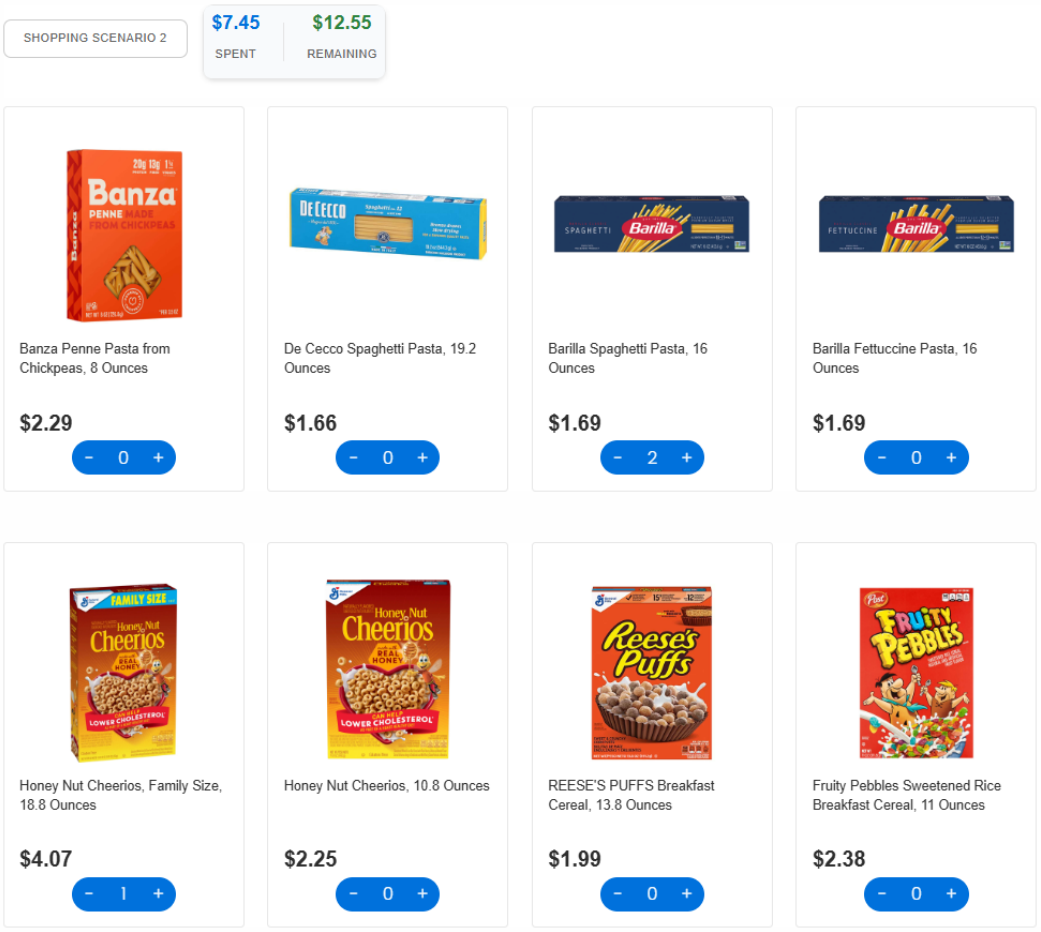
[13] For the selected belief, the bonus was given by $\max\left\{0, 1 - \left(\frac{x-y}{0.2y}\right)^2\right\} \times \$1$, where (x) denotes the participant's response and (y) denotes the true value. A correct response earned \$1, while errors of 5, 10, and 15 percent earned approximately \$0.94, \$0.75, and \$0.44, respectively. Errors of 20 percent or more earned no bonus.

[14] Available at https://doi.org/10.1257/rct.15352-1.2

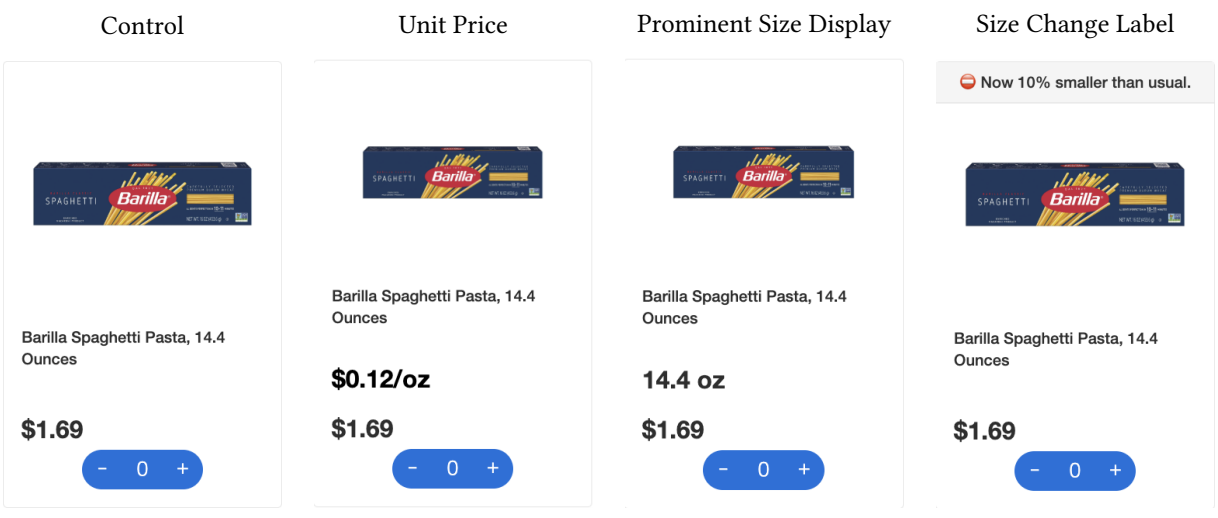
[15] Specifically, we exclude from the main sample participants who: (i) scored below 0.4 on the reCAPTCHA v3 bot-detection measure; (ii) failed to correctly identify at least two out of three letters that visually flashed in a question; or (iii) were unable to draw geometric shapes following our instructions.

Figure 8: Experimental Scenarios

(a) Example Baseline Scenario



(b) Shopping Environment Manipulations



Notes: Panel (a) shows an example of what the shopping environment looked like for everyone in the first three scenarios, and for the control group through all their scenarios. Panel (b) shows how one product in the scenario looked for each of the four treatment arms.

1,951 responses.

The median survey completion time was 17 minutes, and the mean completion time was 20 minutes. In the control group, the survey completion rate was 87.1%. This rate does not differ statistically across shopping environment treatment groups. Individuals who did not finish the survey are excluded from our analyses.

Because the experimental incentive structure is designed to provide a rational benchmark that equates price and size demand elasticities, participants' understanding of the incentives is central to the design. We find that 95% of the sample correctly answered both incentive-comprehension questions. Of these participants, 83% answered both questions correctly on the first attempt, while 12% did so after viewing the information a second time.

Summary Statistics. Appendix Table C.1 presents summary statistics for the main analysis sample. The sample is approximately 75% white and evenly split by gender. Respondents are well distributed across age groups. About 55% have at least a college degree. Household income is fairly spread out across the main income categories: 28% of respondents have household income below $50,000, 39% have income between $50,000 and $100,000, and 30% have income above $100,000. About 31% live in households with four or more members, and 79% report shopping for groceries at least once per week.[16]

The bottom part of this table reports respondents' shopping attitudes to package size changes. A large majority, 92%, report that they are likely to notice a price change, while only 66% report that they are likely to notice a size change. Attitudes toward shrinkflation are unfavorable: 41% classify shrinkflation as very unfair; and when asked what they would prefer firms do after a cost increase, 32% preferred a price increase, while only 15% preferred a size decrease. These attitudes are reflected in strong stated behavioral responses: 77% report that they would be likely to switch brands following shrinkflation, and 79% support a law requiring disclosure of shrinkflation.

As described above, we asked participants which product categories they expected to purchase on their next grocery-shopping trip. The shares selecting each category were 82% for pasta, 81% for chips, 55% for hand soap, and 70% for cereal. Each participant was shown products from exactly two categories, following the survey selection procedure. Category exposure shares were 58% for pasta, 58% for chips, 36% for hand soap, and 47% for cereal. Participants spent an average of $13.11 out of the $20 shopping budget in each scenario, or about 66% of the total budget. The 10th and 90th percentiles of the average spent budget share distribution at the individual level were 25% and 97%, respectively.

Appendix Table C.3 reports the set of products that survey participants could be shown. The set consists of six products in four categories: pasta, chips, hand soap, and cereal. Within each category, we selected products with broad availability at major online retailers. These

[16] Appendix Table C.2 compares the demographic composition of the analysis sample with the U.S. adult population. The sample is more white, younger, and more educated than the U.S. adult population, while the gender composition is similar. Household income is broadly comparable at the lower end, although the sample has fewer people in households earning above $100,000.

categories are also ones which are relatively likely to experience package size changes. For each product, the table first reports the baseline price and package size. Baseline prices are based on prices observed at online retailers in March and April of 2026, discounted by approximately 20 percent. We also report mean packages demanded across the shopping scenarios, conditional on a product being visible to participants as described above, as well as the share of participants who report that they would purchase the product at usual prices. These measures suggest significant interest in the products included in the survey.

To limit the influence of outliers, we trim the top and bottom 1% of beliefs about product price, size, and unit price. Generally, size beliefs are less precise than price beliefs. Consistent with this, across all treatment groups, the mean absolute error in final size beliefs is 33.1%, compared to 21.2% for price beliefs. In our main analysis, we also exclude unit price beliefs that are more than 10 times the actual unit price for that product-scenario. The resulting unit price average absolute belief error is equal to 177%. We find no evidence that the accuracy bonus affected belief errors, giving us confidence in the quality of the belief measures across the full sample and across all elicitations. We report additional descriptive statistics for the belief measures in Appendix C.3.

3.3 Methodology and Main Results

Our baseline experimental results show that subjects respond less to size than to price. Under the experiment's incentive structure, this behavior implies a welfare loss. Using recall and belief measures, as well as our shopping environment manipulations, we provide evidence that consumer inattention drives these results. These shopping environment manipulations also provide evidence on the potential effects of related policies: mandatory disclosure of unit prices and more prominent size labeling can increase welfare, while labels highlighting size changes can induce a bias in the opposite direction.

Demand Responses to Price and Size. We estimate the responsiveness to prices and sizes in two ways.[17] First, by estimating a Poisson regression. Specifically, we estimate:

$$\ln(\mathbb{E}[q_{igt}]) = \beta_C^P \ln(price_{igt}) + \beta_C^S \ln(size_{igt}) + \sum_{m \in \{UP,PSD,SCL\}} \left[\beta_m^P \ln(price_{igt}) \cdot \mathbb{1}_{m,i} + \beta_m^S \ln(size_{igt}) \cdot \mathbb{1}_{m,i}\right] + FEs \quad (6)$$

where i indexes individuals, g products, and t shopping scenarios. The outcome q_{igt} denotes the number of packages of product g selected by individual i in scenario t. The variables $price_{igt}$ and $size_{igt}$ denote, respectively, the displayed package price and package size to individual i for product g in scenario t. The indicator $\mathbb{1}_{m,i}$ equals one if individual i was assigned to treatment group m. The treatment groups UP, PSD, and SCL refer to the unit price, prominent size

[17] We discuss alternative models in Appendix C.4 and show that our main findings are robust to using many other parametric models. Because survey choices are discrete and participants choose zero packages for 56% of product-scenario observations, we rely on specifications that do not use ln quantity as the dependent variable.

display, and size change label treatment groups, respectively. Our main specification uses only choices from scenarios 4–6, after the shopping environment manipulations were introduced. It includes individual fixed effects, scenario-by-treatment-group fixed effects, and product-by-treatment-group fixed effects. This set of fixed effects replicates the results from running separate regressions for each treatment group using only scenarios 4–6. We cluster standard errors at the individual level.

The main parameters of interest are the β^P and β^S. The control group price and size elasticities are β^P_C and β^S_C, respectively. For treatment arm m, the corresponding elasticities are $\beta^P_C+\beta^P_m$ and $\beta^S_C+\beta^S_m$. Thus, the statistical significance of β^P_m and β^S_m test whether treatment group m changes price and size elasticities relative to the control group. We also test whether the treatments affect the ratio of size to price elasticities. For the control group, define this ratio as $R_C = \frac{\beta^S_C}{|\beta^P_C|}$, and for treatment arm m, define it as $R_m = \frac{\beta^S_C+\beta^S_m}{|\beta^P_C+\beta^P_m|}$. We test the null hypotheses $R_m = R_C$ and $R_m = 1$ for each m. These tests are motivated by the rational benchmark that demand should be equally responsive to price and package size in our experiment. We therefore ask whether each treatment changes this relative responsiveness compared with the control group and whether behavior in each group is consistent with the benchmark of equal elasticities in absolute value.

Second, to reduce reliance on functional form assumptions,[18] we estimate an alternative specification that exploits the price and size randomization processes. As described above, for each respondent-product-scenario, price and size were independently set either to their baseline values or to values shifted by $\pm 10\%$ or $\pm 20\%$. This design allows us to estimate the following specification without imposing a particular shape on the demand function:[19]

$$q_{igt} = \sum_{r\in\{-20,-10,10,20\}} \beta^P_r D^P_{r,igt} + \sum_{r\in\{-20,-10,10,20\}} \beta^S_r D^S_{r,igt} + FEs + u_{igt} \tag{7}$$

where $D^P_{r,igt}$ and $D^S_{r,igt}$ are indicators for whether price and size, respectively, were set $r\%$ above (below if negative) the baseline value for individual i, product g, in scenario t. The baseline value is the omitted category. We estimate separate regressions for each treatment group using scenarios 4-6, and an additional regression that pools all control group scenarios with scenarios 1-3 for the treated groups. All specifications include product, individual, and scenario fixed effects, and standard errors are clustered at the individual level. The coefficients β^P_r and β^S_r are the main parameters of interest. They capture the average effect on demand of changing price or size by $r\%$ relative to baseline.

The estimates from the baseline shopping environment, without treatment interventions, show that consumers are substantially more responsive to price than to size. Column (1) of Table 1 reports these results. In the Poisson specification (Equation 6), the estimated price elasticity is 1.62 in absolute value, compared with a size elasticity of 0.55. We reject the equality

[18] For example, the Poisson specification imposes constant price and size elasticities of expected demand.

[19] When interpreting these results we should keep in mind that equal percentage changes in price and size, with opposite signs, induce only approximately comparable changes in unit price. For example, a 10% increase in price raises unit price by 10%, whereas a 10% decrease in size raises unit price by $\frac{1}{0.9} - 1 = 11.1\%$.

of the two elasticities. Their ratio is 0.34, which is well below 1. The estimated difference between these elasticities represents a deviation from optimal behavior.

Table 1: Demand Responses to Price and Size Changes

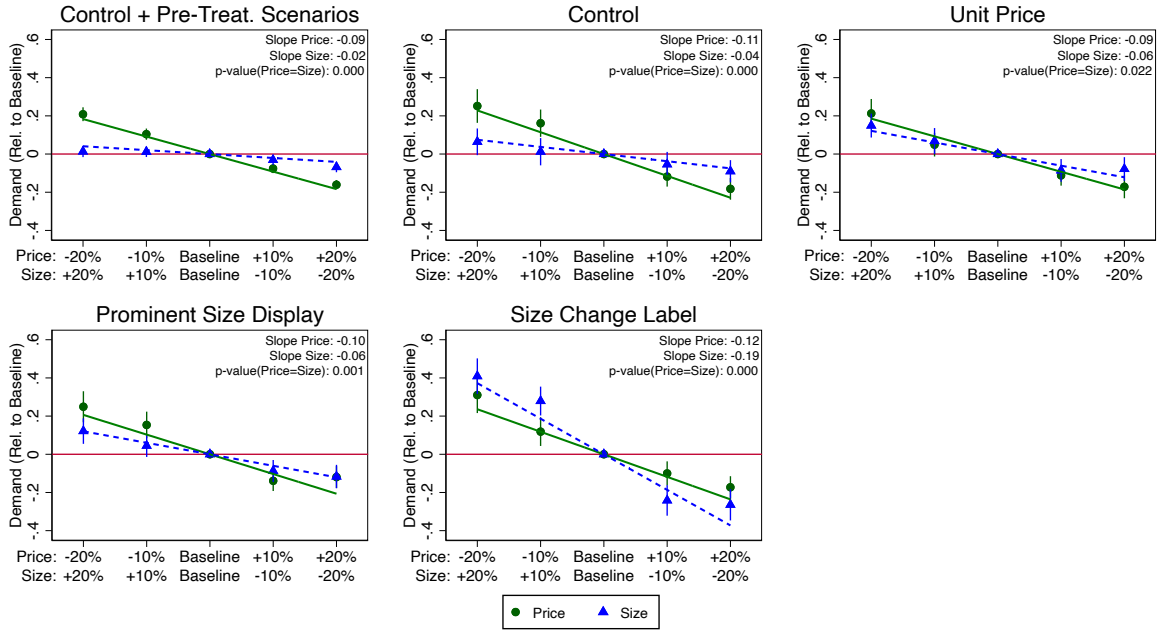
	(1) Quantity	(2) Quantity
Ln(Price)	-1.62***	-1.13***
	(0.17)	(0.12)
Ln(Size)	0.55***	0.37***
	(0.14)	(0.099)
Unit Price × Ln(Price)	0.34	0.22
	(0.23)	(0.17)
Unit Price × Ln(Size)	0.29	0.21
	(0.20)	(0.14)
Prominent Size Display × Ln(Price)	0.15	0.10
	(0.23)	(0.16)
Prominent Size Display × Ln(Size)	0.32	0.22
	(0.20)	(0.13)
Size Change Label × Ln(Price)	-0.023	-0.045
	(0.23)	(0.17)
Size Change Label × Ln(Size)	2.15***	1.44***
	(0.27)	(0.18)
Observations	46488	46824
Size/Price Ratio [P-val vs. Control] {P-val vs. 1}		
Control	0.34 {0.00}	0.33 {0.00}
Unit Price	0.66 [0.04] {0.03}	0.64 [0.04] {0.02}
Prominent Size Display	0.59 [0.06] {0.00}	0.57 [0.05] {0.00}
Size Change Label	1.65 [0.00] {0.00}	1.53 [0.00] {0.00}
Scenarios Included	Treatment Only	Treatment Only
Individual FE	✓	✓
Scenario × Treatment Group FE	✓	✓
Product × Treatment Group FE	✓	✓
Model	Poisson	Linear-Log

Notes: $^{***}p < .01$, $^{**}p < .05$, $^{*}p < .1$. Clustered standard errors at the individual level in parentheses. The samples include scenarios 4–6 for all groups. Column (1) presents the results of estimating a Poisson regression. Column (2) estimates a linear-log model, reproducing Equation 6 but using packages demanded as the dependent variable. We report the (absolute value of the) ratio of the size to price responses of each group, and p-values of the difference in this ratio between the control group and each treatment. We also report the p-values for the equality of size and price coefficients (in absolute value) for each group.

The non-parametric specification (Equation 7) yields the same conclusions. For each of the

four price changes, $\pm 10\%$ and $\pm 20\%$, we compare the effect on demand with the corresponding size-equivalent changes, $\mp 10\%$ and $\mp 20\%$. In the samples without shopping-environment manipulations, demand responds consistently much more to price than to size. The first two panels of Figure 9 present these results. To assist in the visual comparison, these panels also report linear fits of quantity on the within-product price rank (equal to 1 for prices 20% below baseline, 2 for prices 10% below baseline, and so on) and on the inverse within-product size rank, using the same fixed effects. We also reject the equality of the price and size slopes in this specification.

Figure 9: Demand Responses to Price and Size Changes: Non-Parametric



Notes: The figure plots the coefficients obtained from regressing quantity on the indicators for each price-change and size-change level relative to the baseline values ($\pm 10\%$ and $\pm 20\%$), including product, individual and scenario fixed effects. We plot separately the results from using a sample that contains all of the control group and scenarios 1–3 for everyone, and samples restricted to scenarios 4–6 within each treatment arm. We include a linear fit of quantity on the within-product price ranking (i.e., 1 if the price was 20% below baseline, 2 if 10% below baseline, etc.) and the inverse of the within-product size ranking, using the same set of fixed effects. For each plot, we report the slopes of these two lines and the p-value of the test of their equality.

Shopping Environment Interventions. The effects of the shopping environment manipulations on demand elasticities provide a basis for evaluating policy interventions. They also provide evidence on attention as a mechanism for our findings. Both the Unit Price and Prominent Size Display treatments increased the ratio of size to price elasticity, although the ratio remained below the full-attention rational benchmark of one. The Poisson elasticity estimates and treatment-effect tests are again reported in Table 1. Specifically, the ratio rose from 0.34 in the control group to 0.66 and 0.59 in the Unit Price and Prominent Size Display groups, respectively. This suggests that these interventions, perhaps by increasing the attention paid

to size or unit price, improved consumer decision-making. The Size Change Label treatment had a substantially larger effect: the ratio increased to 1.65, indicating that participants in this group became more responsive to size than to price. This also represents a deviation from the benchmark of equal demand responses, but in the opposite direction. Table 1 also reports estimates from a linear-log demand model, which yields semi-elasticities. The qualitative conclusions are unchanged. The corresponding nonparametric demand estimates are shown in Figure 9.

We also elicited participants' stated preferences over the shopping environments corresponding to our treatment interventions, as described in Figure 8, Panel (b). The results, shown in Appendix Figure C.6, indicate a clear preference for environments displaying unit prices: 45 percent of participants selected this option. By contrast, roughly 20% selected the baseline environment or the prominent size display, while 16% preferred size change labels. The low stated preference for size change labels suggests that participants do not view this intervention as especially helpful. One interpretation is that consumers perceive them as drawing attention to a dimension that may induce excessive responses to size changes, consistent with our demand response results.[20] At the same time, the preference for unit price displays provides additional evidence that consumers value the additional information these displays provide.

In Appendix Figure C.7, we examine the effects of the shopping environment manipulations on the mean absolute error of post-final-scenario beliefs, measured in percentage terms. We find no statistically significant effects on the accuracy of price beliefs. In contrast, the treatments significantly affect the accuracy of size beliefs. Relative to a baseline mean absolute error of 33.6% in the control group, the Unit Price treatment decreases accuracy of size beliefs by 3.9 p.p. (p-value = 0.050), the Prominent Size Display improves them by 6.5 p.p. (p-value < 0.001), and the Size Change Label improves them by 3 p.p. (p-value = 0.091). For unit price beliefs, the Unit Price treatment is the only intervention with a statistically significant effect. It reduces mean absolute errors by 35 p.p. (p-value = 0.002) relative to a control group baseline of 145%. This suggests that displaying the unit price redirects consumers' attention from package size toward price per unit.

Attention as a Mechanism. Several pieces of evidence suggest that attention is an important mechanism underlying the underreaction to size relative to price. First, within the control condition, we split individuals according to the accuracy of their beliefs elicited after the final scenario. Respondents were asked to report the price and size of four products shown in that scenario, and we compare their responses with the true values. We then compute, separately for price and size, each respondent's mean absolute belief error. We label individuals as accurate on price if their mean absolute error on price beliefs is below the sample median and as accurate on size if their mean absolute size belief error is below the sample median.

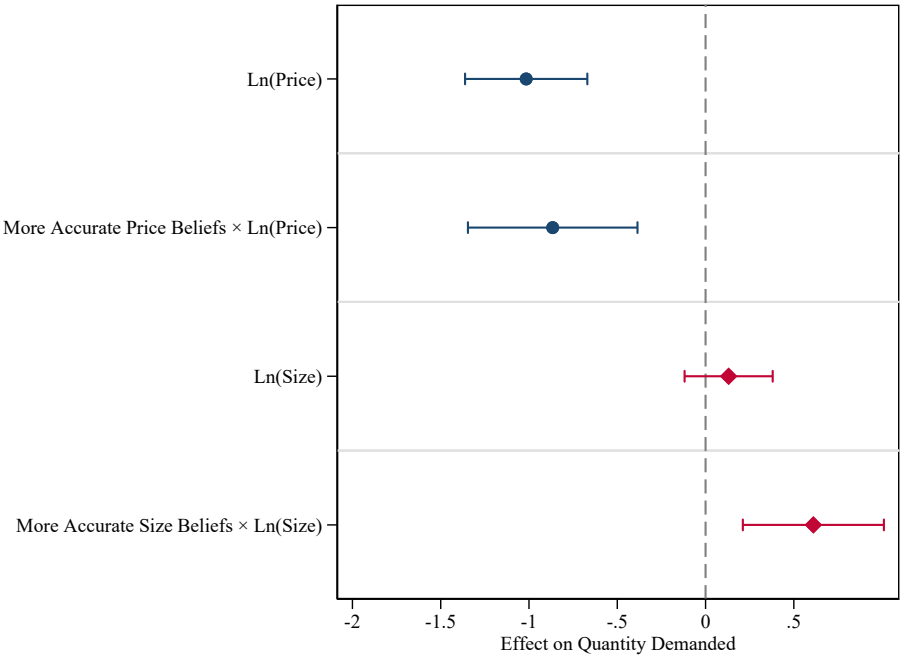
[20] Salience models predict that prominently displayed attributes can receive disproportionate weight in consumer decisions (Bordalo et al., 2013, 2020, 2022). Preferences over such labels may therefore also reflect sophistication about their own susceptibility to salience effects.

Using all control group scenarios, we then estimate the following Poisson model:

$$ln(\mathbb{E}[q_{igt}]) = \beta^P ln(price_{igt}) + \beta^P_A ln(price_{igt}) \mathbb{1}_{AP,i} + \beta^S ln(size_{igt}) + \beta^S_A ln(size_{igt}) \mathbb{1}_{AS,i} + FEs \tag{8}$$

Here, $\mathbb{1}_{AP,i}$ is an indicator equal to one if individual i is classified as accurate on price according to the definition above, and $\mathbb{1}_{AS,i}$ is defined analogously for size accuracy. The specification includes individual fixed effects and allows scenario and product fixed effects to vary across each combination of price- and size-accuracy classifications. The results, reported in Figure 10, show that individuals who more accurately recall past prices are more price elastic, while those who more accurately recall past sizes are more size elastic, relative to individuals with less accurate beliefs. These findings reinforce the interpretation that differences in price and size elasticities partially reflect differences in attention to these attributes.

Figure 10: Price and Size Elasticities by Accuracy of Beliefs (Control Group)



Notes: The figure reports estimates from Equation 8, using all scenarios in the control group. Respondents are classified by belief accuracy using the belief elicitation conducted after their final scenario. More Accurate Price and Size Beliefs are defined as having a mean absolute error below the sample median for price and size beliefs, respectively. The regressions include individual fixed effects, and scenario and product fixed effects are allowed to vary across the four combinations of price- and size-accuracy classifications.

As a second piece of evidence on attention as a mechanism, we use the post-scenarios question asking respondents whether they noticed changes in prices, sizes, or package design between scenarios 4 and 6 for four products they had seen. Package design was included as a placebo, since it did not vary across scenarios. We compare the frequency with which individuals report noticing changes with the actual frequency of changes faced by each individual.

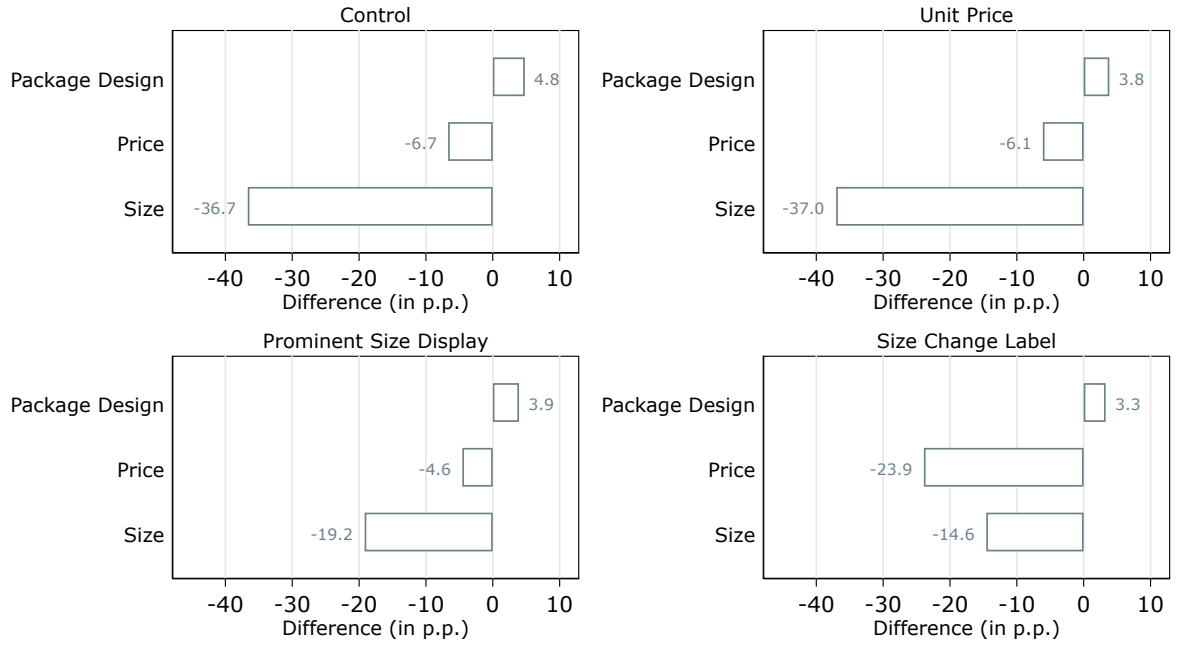
The first panel of Figure 11 shows that control-group individuals underestimate the frequency of price changes by 6.7 p.p. Their performance is substantially worse for size: they underestimate the frequency of size changes by 36.7 p.p.[21] This gap provides direct evidence that consumers are less attentive to size than to price. Because size changes are less likely to be perceived, this should translate into weaker demand responses, providing a rationale for the result that the estimated size elasticity of demand is smaller than the price elasticity.

The treatment effects on demand for the Prominent Size Display and Size Change Label groups are mirrored in the changes' recall patterns reported in Figure 11. When sizes are displayed as prominently as prices, respondents underestimate the occurrence of variation in sizes less than those in the control group, although they continue to miss size changes substantially more often than price changes. When labels identify size changes, the recall patterns reverse: respondents underestimate price changes more than size changes. Relative to the control group, they are substantially more likely to notice size changes, despite still underestimating them, but much worse at noticing price changes. These results indicate that the treatments altered the attention consumers paid to prices and sizes, consistent with their effects on demand elasticities.

The Unit Price treatment does not generate meaningful differences relative to the control group in the perceived changes questions, despite increasing size elasticity. One interpretation is that, in their decision making, participants may rely on the unit prices displayed to them rather than attending directly to package size. Under this interpretation, the treatment should increase responsiveness to size, since size enters unit prices, without necessarily improving people's ability to recall size changes.

[21] Appendix Figure C.5 presents a more detailed version of this analysis, separating correct and incorrect reports of variation in each dimension, and yields the same qualitative insights.

Figure 11: Recall of Changes vs Actual Frequency



Notes: The figure compares respondents' reported noticing of changes in prices, sizes, and package design between scenarios 4 and 6 with the actual variation in each attribute, averaged across the four products for which this was elicited in the post-scenario questions. Panels report these patterns separately for the control group and each treatment group.

This evidence helps us understand the results from the analyses based on the observational scanner data. The dynamic effects for downsized products, where we see a smaller response in the short term than in the long term, can be explained by a lack of attention. As the experiment shows, individuals can underreact to changes if they do not pay attention to a product dimension, and this can lead to a muted short term demand response. Over time, as consumers learn that package size is different than expected, the demand response can increase. Moreover, firms may strategically advertise size increases, but not size decreases, in a manner similar to the environment manipulation in the Size Change Label treatment. Such a strategy can direct consumer attention to package size and thereby increase even short-run demand responses. This can help explain the different dynamics for upsized and downsized products we found in the observational data.

Is Attention Rational? A natural question is whether inattention to size reflects optimal allocation of attention: because attention is costly, individuals may rationally choose not to attend to size.[22] Two pieces of evidence suggest that is not the case in the experiment. First, a strong interpretation of these frameworks implies a certain degree of sophistication in attention. Individuals must know their capabilities and hence be able to adequately choose

[22]This interpretation aligns with models of rational inattention, which treat attention as a scarce resource whose allocation across attributes responds to the expected benefits and costs of information processing (Maćkowiak et al., 2023; Sims, 2003).

the contexts in which to exert more attention. We show evidence to the contrary. After the last shopping scenario, we informed participants that they would be asked to recall the price and size of products shown in that scenario. Then, we asked them what they thought their probability of being within 5% of the actual value was. We compare this value to the actual attributes. Results, presented in Appendix Figure C.8, show that participants are substantially overconfident for both price and size beliefs. Moreover, the relationship between confidence and accuracy is nearly flat: more confident participants perform only marginally better.

Second, we exploit the variation in the frequency of size changes. Individuals were cross-randomized into three groups that varied in the probability of being shown a size that differed from the baseline size. A prediction of many models of rational inattention is that people should pay more attention to dimensions with greater variation. To test this prediction, we estimate fifteen Poisson regressions: one for each of the three levels of size variation in each of five samples. The first sample includes all individuals in the pre-treatment scenarios. The remaining four samples correspond to the shopping environment treatment groups and use only the final three scenarios. All regressions include individual, scenario, and product fixed effects. The results are presented in Appendix Figure C.9. We find no systematic evidence of a positive relationship between the level of size variation and the size elasticity, which does not align with the predictions of these models.

Distribution of Attention. We now use the heterogeneity in individual demand responses to package size to estimate the distribution of attention. Specifically, we impose homogeneous price responsiveness and attribute heterogeneity in size responsiveness to heterogeneity in attention. Because a fully rational and attentive consumer in the experiment should have price and size elasticities that are equal in absolute value, we measure individual attention as the ratio of the individual-level size response to the common price response. To implement this, we estimate a random effects Poisson model:

$$ln(E[q_{igt}]) = \alpha_i + \beta^P ln(price_{igt}) + \beta_i^S ln(size_{igt}) + \text{ProductFE}_g + \text{ScenarioFE}_t \quad (9)$$

Where α_i is a random intercept and β_i^S is a random size slope, assumed to be independently normally distributed. ProductFE$_g$ and ScenarioFE$_t$ are fixed effects by product and scenario, respectively. The remaining notation is as defined above. Estimating this model yields the mean and variance of the distribution of individual-level size elasticity, β_i^S. The attention parameter for individual i is then defined as $\theta_i = -\beta_i^S/\beta^P$. Its distribution is derived from the distribution of the size elasticity. That is, θ_i follows a normal distribution with mean $-E(\beta_i^S)/\beta^P$ and standard deviation $\frac{\sqrt{Var(\beta_i^S)}}{|\beta^P|}$. The estimated standard deviations also enter the welfare calculations in subsection 4.2.

We estimate these models separately by treatment arm using scenarios 4–6. We also estimate a pooled baseline model that combines scenarios 1–3 for all respondents with scenarios 4–6 for the control group. The results for the average attention, presented in Table 2, are comparable to the ratio of elasticities from our previous Poisson regressions (Table 1). The estimates again indicate substantial inattention to package size in the control group: the mean

attention parameter is 0.32, well below the benchmark value of one. All three treatments increase estimated attention. The unit-price and prominent-size treatments raise mean attention to 0.66 and 0.57, respectively, while the size-change label raises it to 1.628, making consumers more sensitive to size than implied by the full-attention benchmark.

Table 2: Attention Estimates

Sample	β^P	$E[\beta_i^S]$	$SD(\beta_i^S)$	$E[\theta_i]$	$SD(\theta_i)$	Obs.
Control + Pre-Treat	-1.311	0.265	0.140	0.202	0.107	58,704
Control	-1.592	0.507	0.134	0.318	0.084	11,880
Unit Price	-1.243	0.818	0.116	0.658	0.094	11,616
Prominent Size Display	-1.464	0.831	0.126	0.568	0.086	11,544
Size Change Label	-1.639	2.669	0.104	1.628	0.064	11,784

Notes: This table presents the estimates of Equation 9. The first row includes all the control group and scenarios 1–3 for everyone. The remaining rows use only scenarios 4–6 for the corresponding treatment group.

Additional Results. Appendix C.5 reports additional results from the survey experiment. First, the asymmetry between price and size responses is robust across alternative demand specifications, including specifications based on unit prices: demand is substantially more responsive to unit-price variation induced by price changes than to comparable variation induced by size changes. We also find evidence that consumers substitute toward a product when close alternatives become more expensive or smaller in display size. Second, the pattern of underreaction to product size versus price is broad-based across products and respondent groups, with little evidence of meaningful heterogeneity by demographics, shopping behavior, or views about shrinkflation. Finally, heterogeneity in belief errors is limited. Belief accuracy is somewhat related to stated attention to product attributes, but not to demographic characteristics such as education or income.

4. Models and Welfare

4.1 Demand Model with Inattentive Consumers

The preceding empirical results point to systematic inattention to package size changes. We now provide a demand model that can explain these patterns, and we show how the same framework can be used to quantify their welfare consequences. The model yields sufficient statistics for consumer-surplus losses from inattention to size changes, which can be estimated using combined evidence from our observational and experimental analyses.

Setup. Assume there is a unit mass of consumers. Each consumer decides whether to purchase a single package of a product with size s and package price p. We assume homogeneous income y, as in Taubinsky and Rees-Jones (2018). The flow utility for individual i in

period t can be expressed as:

$$u_{it} = \begin{cases} v(s) + y - p + \varepsilon_{it} & \text{if purchased} \\ y & \text{otherwise} \end{cases} \tag{10}$$

In addition, ε_{it} is a shock to the utility from purchasing the product, or equivalently, the difference between the purchase and no-purchase utility shocks. We assume that $\varepsilon_{it} \sim F$, distributed independently across periods, that the resulting demand function is smooth, and that $v'(s) > 0$.

Consumers might misperceive package sizes after size changes. Perceived size, $\hat{s}_{it}$, evolves according to:

$$\hat{s}_{it} = \theta_i s + (1 - \theta_i)\hat{s}_{i,t-1}, \qquad \hat{s}_{i0} = s_0$$

$$\hat{s}_{it} = \left[1 - (1 - \theta_i)^t\right] s + (1 - \theta_i)^t s_0 = \phi_{it} s + (1 - \phi_{it}) s_0, \qquad \phi_{it} \equiv 1 - (1 - \theta_i)^t \tag{11}$$

θ_i is a time-invariant individual characteristic, which we interpret as individual i's attention parameter. In the first period after the size change, θ_i is the weight placed on the true size, while $1 - \theta_i$ is the weight placed on the old size, s_0. In later periods, the weight on the true size is ϕ_{it}, which is determined by θ_i through the updating process above.

When taking expectations or integrating the distribution of θ_i, denoted by H, we write ϕ_t as shorthand for the generic function $\phi_t(\theta) \equiv 1 - (1 - \theta)^t$. We assume $E_H(\phi_t) \neq 0$ and $\varepsilon_{it} \perp\!\!\!\perp \theta_i$; that is, the average weight placed on the true size is nonzero, and the utility shock is independent of the attention parameter. We otherwise leave the distribution of θ unrestricted. For example, H may allow for either homogeneous or heterogeneous attention across consumers, and may include consumers who under-adjust their perceived size, $\theta_i < 1$, as well as consumers who overshoot, $\theta_i > 1$.

The consumption decision depends only on variables observed in the current period. In each period, individuals purchase based on perceived utility, which depends on perceived size. Formally, individual i buys in period t if $v(\hat{s}_{it}) + \varepsilon_{it} \geq p$. We define $\varepsilon_t^*(p, s, \phi_t)$ as the utility shock of the marginal consumer, which is a function of (p, s, ϕ_t):

$$\varepsilon_t^*(p, s, \phi_t) = p - v\left(\phi_t s + (1 - \phi_t) s_0\right) = p - v\left(\left[1 - (1 - \theta)^t\right] s + (1 - \theta)^t s_0\right) \tag{12}$$

We can write period-t demand by computing, for each value of the attention parameter, the share of consumers whose utility shock exceeds the marginal one:

$$D_t(p, s) = \int_\Theta \left(\int_{\varepsilon_t^*(p,s,\phi_t)}^{\infty} f(\varepsilon) d\varepsilon \right) dH(\theta) \tag{13}$$

Note that, at $s = s_0$, demand is independent of t and θ because all consumers correctly perceive s_0 regardless of ϕ_t. So, $D(p, s_0) = D_t(p, s_0) = \int_{\varepsilon^*(p,s_0)}^{\infty} f(\varepsilon) d\varepsilon$.

Elasticities. We can use this setup to derive demand elasticities. We evaluate them at $s = s_0$. The price elasticity does not vary over time and is independent of the distribution of attention. Specifically, it is given by:

$$\eta^p(p, s_0) = \eta^p_t(p, s_0) = D'_{p,t}(p, s_0)\frac{p}{D(p, s_0)} = -f(\varepsilon^*(p, s_0))\frac{p}{D(p, s_0)} \tag{14}$$

By contrast, the size elasticity depends on the distribution of attention and changes over time. Specifically, it takes the following form:

$$\begin{aligned}\eta^s_t(p, s_0) &= D'_{s,t}(p, s_0)\frac{s_0}{D(p, s_0)} \\ &= f(\varepsilon^*(p, s_0))v'(s_0)\frac{s_0}{D(p, s_0)}E_H(\phi_t) \\ &= f(\varepsilon^*(p, s_0))v'(s_0)\frac{s_0}{D(p, s_0)}\left(1 - E_H[(1-\theta)^t]\right)\end{aligned} \tag{15}$$

Thus, the size elasticity is time-varying through the term $1 - E_H[(1-\theta)^t]$. This implies that:

$$\lim_{t\to\infty} E_H[(1-\theta)^t] = 0 \;\Rightarrow$$
$$\lim_{t\to\infty} \eta^s_t(p, s_0) = f(\varepsilon^*(p, s_0))v'(s_0)\frac{s_0}{D(p, s_0)} = \eta^s_t(p, s_0|\theta_i = 1\ \forall i) \tag{16}$$

Thus, another result is that, if $\lim_{t\to\infty} E_H[(1-\theta)^t] = 0$, then the size elasticity at $s = s_0$ converges to the full attention size elasticity as time goes to infinity. A sufficient condition for this convergence is that $\theta_i \in (0, 2)$ for all i. Intuitively, this condition says each consumer updates their perceived size in the correct direction each period, moving from the prior toward the true size, and that they end each update closer to the real size than their previous perception.[23] This condition also implies that perceived size converges to the true size for everyone: $\lim_{t\to\infty} \hat{s}_{it} = s$ for all i.

Another implication is that $E_H[\phi_t]$ determines whether demand under- or overreacts to changes in size relative to the full attention benchmark (obtained if $E_H[\phi_t] = 1$). If $E_H[\phi_t] \in (0, 1)$, the size elasticity of demand is smaller than under full attention. A sufficient, though not necessary, condition for this is $\theta_i \in (0, 1)$ for all i. In contrast, if $E_H[\phi_t] > 1$, the size elasticity of demand is greater than under full attention.

Welfare formulas. To derive sufficient statistics for consumer welfare, we start from the period-t aggregate consumer surplus expression:

$$CS_t(p, s) = \int_\Theta \left[\int_{\varepsilon^*_t(p,s,\phi_t)}^{\infty} (v(s) - p + \varepsilon)f(\varepsilon)d\varepsilon\right] dH(\theta) \tag{17}$$

[23] Furthermore, the assumption $\theta_i \in (0, 2)$ is economically reasonable, because values outside this range imply unrealistic learning dynamics: if $\theta_i < 0$, perception moves explosively in the wrong direction; if $\theta_i = 0$, perception never adjusts; if $\theta_i = 2$, perception oscillates indefinitely around the true size with constant amplitude; and if $\theta_i > 2$, perception oscillates with an increasing and explosive amplitude.

Using a second-order Taylor approximation around the old size, s_0, we obtain expressions for three important objects: the period-t change in consumer surplus induced by a size change, the welfare loss from inattention to sizes, and the tax-induced percent price change that would generate the same demand-side deadweight loss as size inattention. Here, we present the formulas and economic intuitions, while the derivations are left for Appendix subsection D.1.

First, we can show that the change in consumer surplus from a change in size can be approximated by:

$$\Delta CS_t \approx D(p, s_0)(\Delta s)v'(s_0) + \frac{1}{2}(\Delta s)^2 D(p, s_0)v''(s_0) + \frac{1}{2}(\Delta s)^2 \underbrace{\left(-D'_p(p, s_0)v'(s_0)\right)}_{D'_s(p,s_0|\theta_i=1\ \forall i)} v'(s_0)\left[1 - (1 - E_H(\phi_t))^2 - Var_H(\phi_t)\right] \quad (18)$$

This expression has a familiar shape. The first two terms capture the welfare effects for inframarginal consumers. The first term is the standard rectangle in the (D, s) plane: the initial mass of buyers, $D(p, s_0)$, is exposed to a size change Δs, valued at the marginal value of size, $v'(s_0)$. The second term is an inframarginal curvature correction: it accounts for the fact that the marginal value of size may change as size moves away from s_0.

The third term captures the effects on marginal consumers. To interpret it, consider the full attention benchmark, under which the bracketed term equals one because full attention implies $E_H(\phi_t) = 1$ and $Var_H(\phi_t) = 0$. The term then corresponds to the standard triangle: for a size change Δs (the triangle height), the mass of buyers under the assumption of fully attentive consumers increases by $D'_s(p, s_0|\theta_i = 1\ \forall i)\Delta s$ (the triangle base), with $v'(s_0)$ converting the triangle area into dollar-equivalent surplus.

The decomposition also shows how attention affects consumer surplus. Attention enters only through the marginal-consumer term. Furthermore, its effect depends not only on the average bias in cumulative attention, $1 - E_H(\phi_t)$, but also on its variance, $Var_H(\phi_t)$. This is in line with the findings of Taubinsky and Rees-Jones (2018) and Allcott et al. (2025), which emphasize that heterogeneity in biases, rather than only their average level, can be crucial for understanding the welfare implications of mistakes. As $\frac{1}{2}(\Delta s)^2\left(-D'_p(p, s_0)v'(s_0)\right)v'(s_0) > 0$, consumer surplus after a size change will be lower if the average cumulative attention is further away from the full attention benchmark, captured by a larger $(1 - E_H(\phi_t))^2$, and if it exhibits more heterogeneity across individuals, so that $Var_H(\phi_t)$ is larger.

The intuition for the last part is that heterogeneity leads to more extreme mistakes. This is clearest when $E_H(\phi_t) = 1$ but $Var_H(\phi_t) > 0$: even though attention is correct on average, some consumers under-react while others over-react, generating welfare losses. More generally, for any given average attention level, greater variance moves some consumers farther from full attention. Even if it moves others closer to the benchmark, the welfare cost rises because larger deviations are weighted more heavily.

To obtain the consumer surplus loss from inattention, we take the difference between consumer surplus when all consumers are fully attentive and the unrestricted expression. From

Equation 18, we know this difference comes entirely from the last term of that expression. To express the loss in terms of estimable objects, we use $v'(s_0) = -\frac{D'_{s,t}(p,s_0)}{E_H(\phi_t)D'_p(p,s_0)}$ (which follows from combining Equations 14 and 15). After some algebraic manipulations, this yields a sufficient statistic formula for the consumer surplus loss from inattention as a function of product spending, the percent magnitude of the size change, the price elasticity, the period-t size elasticity, and the distribution of attention:

$$CS_t(p,s|\phi_t = 1\ \forall i) - CS_t(p,s) \approx \frac{1}{2}pD(p,s_0)\left(\frac{\Delta s}{s_0}\right)^2 \frac{\eta^s_t(p,s_0)^2}{|\eta^p(p,s_0)|}\frac{(1-E_H(\phi_t))^2 + Var_H(\phi_t)}{E_H(\phi_t)^2}$$

$$\approx \frac{1}{2}pD(p,s_0)\left(\frac{\Delta s}{s_0}\right)^2 \frac{\eta^s_t(p,s_0)^2}{|\eta^p(p,s_0)|}\frac{E_H[(1-\theta)^{2t}]}{(1-E_H[(1-\theta)^t])^2} \quad (19)$$

To benchmark these welfare losses with a familiar metric, we compare them to the consumer's share of the deadweight loss from a tax that increases the price by Δp. That is:

$$-\frac{1}{2}D'_p(p,s_0)(\Delta p)^2 = \frac{1}{2}|\eta^p(p,s_0)|\frac{D(p,s_0)}{p}(\Delta p)^2 \quad (20)$$

To obtain this metric, we equate equations 19 and 20 and solve for the proportional price change:

$$\frac{\Delta p_t}{p} \approx \left|\frac{\Delta s}{s_0}\right| \frac{|\eta^s_t(p,s_0)|}{|\eta^p(p,s_0)|}\frac{\sqrt{(1-E_H(\phi_t))^2 + Var_H(\phi_t)}}{|E_H(\phi_t)|}$$

$$\approx \left|\frac{\Delta s}{s_0}\right| \frac{|\eta^s_t(p,s_0)|}{|\eta^p(p,s_0)|}\frac{\sqrt{E_H[(1-\theta)^{2t}]}}{|1-E_H[(1-\theta)^t]|} \quad (21)$$

4.2 Welfare Estimates

We now quantify Equation 19 by combining evidence from the distribution of size changes, observational estimates of demand elasticities, and experimental estimates of attention.

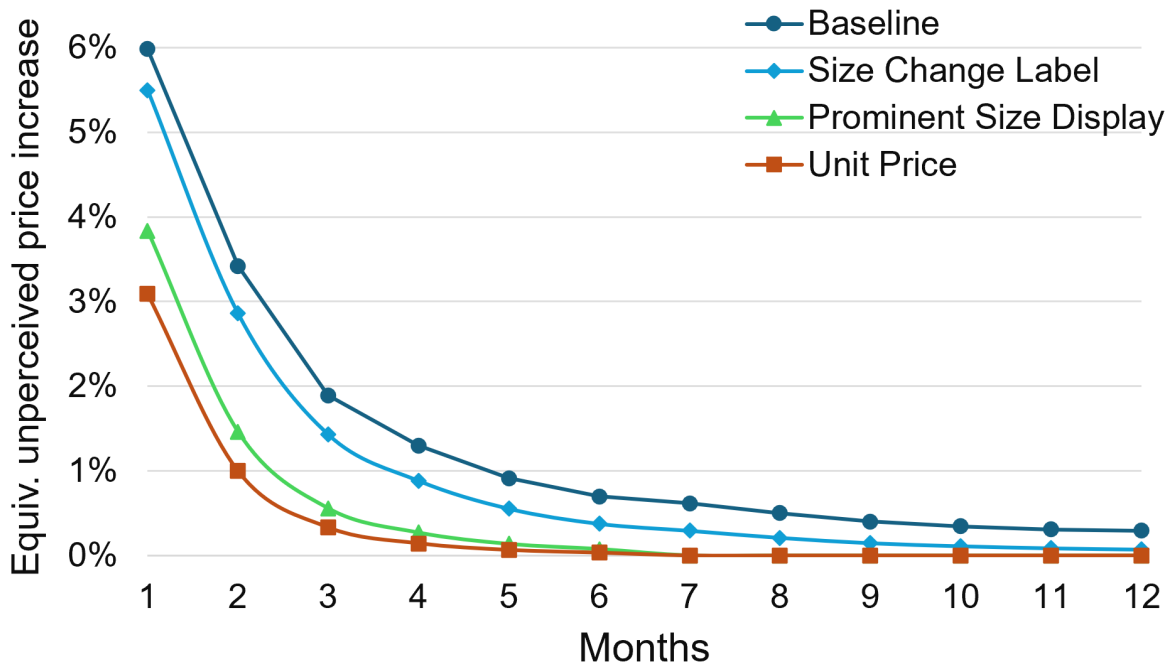
For the main estimates, we use the median size decrease in the observational data, 12.5%. The elasticities come from our retail-scanner data specification which uses the natural logarithm of packages demanded as the outcome variable and is reported in Appendix Table B.2. To compute the moments of the attention distribution, we rely on the experimental estimates of the distribution of θ. In the base estimation, we use the control group estimates reported in Table 2.

We also evaluate the welfare losses from inattention under the three interventions studied in the experiment. For each intervention, we use the corresponding attention distribution estimated from the experimental data. The model also predicts that the size elasticity changes with the attention distribution. In particular, Equation 15 implies that the elasticity under a given treatment can be obtained by scaling the observational estimate by the ratio of cumulative attention under that treatment to cumulative attention under the control condition.

Formally: $\eta_t^{S,Treatment} = \eta_t^{S,Observed} E[\phi_t^{Treatment}]/E[\phi_t^{Control}]$. The percent size change and the price elasticity used are the same as in baseline.

The estimates are presented in Figure 12. At baseline, consumer surplus losses from inattention due to package size changes in the first period are equivalent to the consumer-attributable deadweight loss from a tax-induced 6% price increase. The loss decreases monotonically over time, as consumers become more likely to notice a size change. The interventions evaluated in the experiment reduce welfare losses, but do not eliminate them. Size change labels generate only modest welfare gains, reflecting substantial overreaction to size changes in that treatment. The prominent display of sizes and that of unit prices have similar welfare implications, with the Unit Price intervention being slightly better in welfare terms. In the first month, the tax-induced price change required to generate the same consumer-side deadweight loss is half the baseline value. By the second month, it is at the level of what the baseline group achieves only on month five. These patterns imply that prominent displays of size, and especially unit prices, meaningfully reduce consumer surplus losses arising from inattention to size changes.

Figure 12: Welfare Losses From Inattention – Equivalent Price Change



Notes: This figure presents the estimates of Equation 21, the percent price change induced by a tax that would generate a deadweight loss attributable to consumers equivalent to the consumer surplus losses from inattention to size changes. We quantify it at the median size decrease, 12.5%. The distribution of attention for the baseline case is the one for the control group from Table 2. The other groups use their respective estimates from that same table.

4.3 Supply Model

We consider a firm that faces the demand described in the previous section, with the added feature that it can influence attention by advertising size changes. The firm receives a cost shock and chooses a price, a package size, and whether to advertise size changes. Changing the size requires paying a one-time sunk cost. We show that under general conditions, the firm has an incentive to advertise size increases but not decreases, that after a cost decrease there is an incentive to increase package sizes, while after cost increases the firm has an incentive to downsize.

Setup. Before period $t = 1$, the firm sells packages of size s_0 at price p_0. The marginal cost of a unit of size is initially c_0, so the production cost of a package is $c_0 s_0$. At $t = 1$ the firm receives a permanent cost shock. The marginal cost of a unit changes from c_0 to c_1. We assume that the cost shock is not large enough to make the company completely stop production. After observing c_1, the firm chooses (p, s, A), where p is the package price, s is the package size, and $A \in \{0, 1\}$ indicates whether the firm advertises the size change when $s \neq s_0$. Advertising affects consumer attention, and we consider different alternatives for its effect below. If the firm chooses to adjust the size, it pays a one-time adjustment cost Γ. There are no additional costs associated with changing prices or advertising size changes (e.g., the firm can print a prominent label advertising the size change directly on the package).

The firm makes a durable decision and discounts the future at rate $\delta \in [0, 1)$. This discount factor captures both standard time discounting and the probability that the chosen (p, s, A) remains in place in future periods. For tractability, we assume that if these choices are no longer in place, it is because the firm made a new decision for reasons independent of the original (p, s, A). Since this continuation path is common across choices, we do not need to model it explicitly.

The firm's problem is:

$$\max_{p,s,A} -\Gamma \mathbb{1}\{s \neq s_0\} + \sum_{t=1}^{\infty} \delta^{t-1}(p - c_1 s) D_t(p, s, A) \tag{22}$$

Demand retains the structure described in subsection 4.1, but we now allow consumer attention to be affected by a firm decision. Throughout this section, we assume $\theta_i(A) \in (0, 2)\ \forall(i, A)$. As explained in the previous section, values outside that range have unrealistic implications for size perception.

We consider three alternative options for the effect of advertisement on attention:

Assumption 1.1. Advertisement leads to full attention: $\theta_i(1) = 1\ \forall i$.

Assumption 1.2. Advertisement (weakly) increases attention: $\theta_i(1) \geq \theta_i(0)\ \forall i$.

Assumption 1.3. Advertisement (weakly) increases attention. Everyone ends up with at least full attention. Over-reaction is allowed. Formally: $\theta_i(1) \geq \theta_i(0)\ \wedge\ \theta_i(1) \geq 1\ \forall i$.[24]

[24]Note that Assumption 1.3 implies Assumption 1.2.

PROPOSITION 1. **Advertisement of Size Changes.** In this setting, under Assumption 1.2, the firm has a local incentive to advertise size increases and not to advertise size decreases. Since Assumption 1.3 implies Assumption 1.2, the same conclusion also holds under Assumption 1.3.

The economic intuition is straightforward. Ceteris paribus, demand is increasing in perceived package size. Because advertising increases the attention parameter θ_i, it increases the present discounted value of the attention weights ϕ_{it} that consumers apply to the new size. Therefore, when a company upsizes a product, it has a local incentive to increase attention so that the larger size carries a higher weight, thus profiting on the resulting higher demand. Conversely, when reducing the product size, increasing attention would place a higher weight on the smaller size, resulting in sharper drops in demand. Consequently, the firm has a local incentive to withhold advertisement of size decreases. The proof of the proposition is presented in the Appendix subsection D.2.

Size Changes After Cost Changes. We now present results on size changes after cost shocks. To solve the problem and be able to compare the new optimal size with the old one, we need information on what the old size was. Here, we adopt the following:

ASSUMPTION 2. The old size, s_0, was optimal under cost c_0 and full attention; and $v(s)$ is differentiable and strictly concave. This implies $v'(s_0) = c_0$.

PROPOSITION 2. **Cost Decrease.** Under this setup and Assumption 2, if the marginal cost falls ($c_1 < c_0$), then: 1. Under Assumption 1.3, the firm has a strict local incentive to implement an advertised size increase. That is, abstracting from the fixed adjustment cost, a marginal increase in size (from s_0) raises variable profits when the size change is advertised (which, by Proposition 1, we know is locally better than an unadvertised one). 2. Under Assumption 1.1, we can globally characterize the optimal adjusted size under advertising. It solves: $v'(s) = c_1$. By Assumption 2, this implies $s > s_0$, that is, the adjusted advertised optimal size is larger than the old one.

It is worth noting what this proposition does not imply. First, in all cases, a firm might choose to retain the old size (s_0) if the sunk cost of changing size (Γ) is larger than the additional profits from adjusting the size. Second, we cannot guarantee under Assumption 1.1 that the company would want to advertise even after size increases. Intuitively, this is because consumers might overreact even to unadvertised size increases. On one end, if everyone has $\theta_i(0) > 1$, the firm would not want to advertise size increases, as making them fully attentive ($\theta_i = 1$) would indeed reduce demand in that case. On the other end, if everyone has $\theta_i(0) < 1$, we can guarantee that the firm would have a local incentive to advertise when upsizing (this would be a special case of Assumption 1.2, covered by Proposition 1). Third, in all of these cases, downsizing could still be better than upsizing. This force is (at least locally) weaker when unadvertised attention is higher; and if unadvertised consumers mostly underreact, a higher δ also weakens this incentive. See the Appendix subsection D.2 for proofs.

ASSUMPTION 3. Under advertising regime A, the expected value of the normalized discounted sum of cumulative attention is below one: $E_H\left[\frac{\sum_{t=1}^{\infty}\delta^{t-1}\phi_t(A)}{\sum_{t=1}^{\infty}\delta^{t-1}}\right] = E_H\left[\frac{\theta(A)}{1-\delta+\delta\theta(A)}\right] \leq 1$. A

sufficient (but not necessary) condition for this is the absence of over-reactors, i.e., $\theta_i(A) \leq 1 \; \forall i$.

Proposition 3. **Cost Increase.** Under this setup and Assumptions 2 and 3, fix an advertising regime A. If the marginal cost increases ($c_1 > c_0$), then the firm has a local incentive to reduce the package size under advertising regime A.

Assumption 3 is particularly natural for $A = 0$, as we expect consumers to be more likely to underreact to size changes when they are not advertised. Moreover, under Assumption 1.2, Proposition 1 implies that, conditional on decreasing size, the firm has a local incentive not to advertise the size change. The proof of the proposition is presented in the Appendix subsection D.2.

As before, this local incentive does not imply that adjusting size is optimal; that depends on the relationship between the sunk cost of changing size, Γ, and the additional variable profits from the adjustment.

5. Conclusion

Although package sizes are a central dimension of consumer goods, as well as a determinant of unit prices, there is still much to learn about their causes and consequences. This paper addresses important gaps in the literature by shedding light on an economically and policy-relevant phenomenon. Our contributions span solving the data challenge of linking product versions to credibly identify size variation, estimating consumer responses, providing evidence of mechanisms, evaluating policies, rationalizing results through the lens of a model, and conducting welfare analyses.

We started by tackling a central data challenge: developing and validating an algorithm to identify size changes. Beyond our study, this algorithm lays the groundwork for further research on this issue. We applied the algorithm to the NielsenIQ retail scanner data, and provided novel descriptive statistics on the key patterns related to size changes.

Next, we estimated the price and dynamic size elasticities of demand. We found a clear initial low reaction to size changes for downsized products, with the responsiveness growing over time. The size elasticities for upsized products have a different dynamic: there is an immediate response which remains stable over time.

With an online shopping experiment, we show that attention plays a central role in explaining the relationship between price and size elasticities, and that consumer behavior in the control group deviated from the optimal benchmark. Through a series of shopping environment manipulations, we tested interventions that are closely related to policies being implemented around the world.

Finally, we showed that the empirical results can be explained by a dynamic model of inattention on the demand side, paired with strategic choices by firms to advertise size increases but not decreases. Based on the model, we derived sufficient statistics to analyze the

impact of inattention and related policies on consumer welfare, and we quantified them by pooling estimates from the different sections of the paper.

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Appendices

A. Size Change Examples

Figure A.1: Examples of Size Changes

(a) Cake: Size Reduction



(b) Toothpaste: Size Reduction



(c) Paper Towels: Size Reduction



(d) Cereal: Size Increase



Notes: Sub-figures (a)-(c) provide examples of size reductions. Sub-figure (d) is an example of a size increase.

B. Additional Details on the Retail Scanner Data Analyses

B.1 Details on the Algorithm to Identify Product Transitions

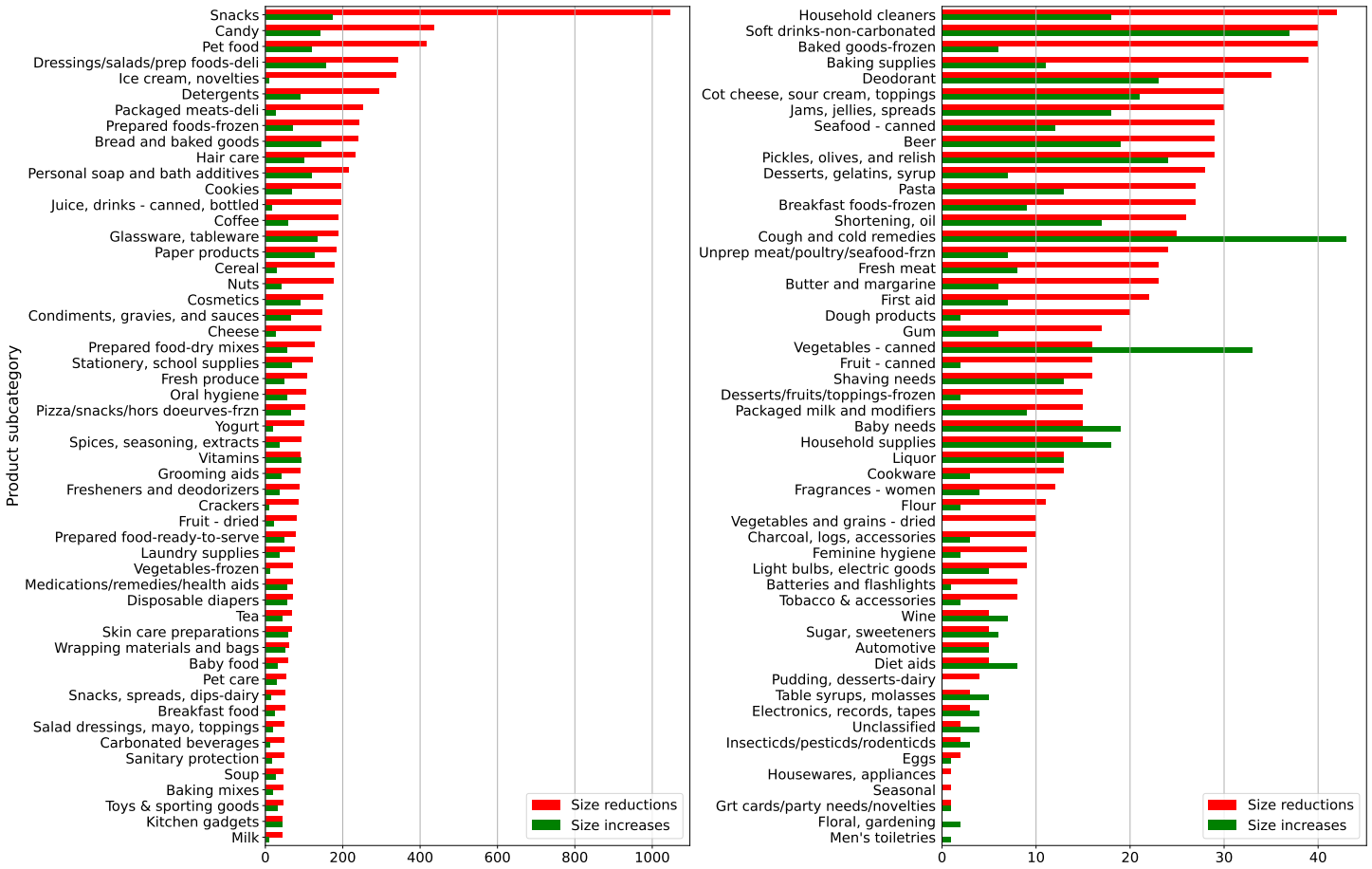
Here we provide details on the algorithm described in subsection 2.2, specifying precisely how we determine when a product transition occurred in the data.

Formally, we take each pair of UPC-versions identified in the previous step as potential instances of size changes. For each store that sold either version at any point in time, we calculate three key metrics used as criteria to determine whether a transition occurred. These metrics are: (i) The proportion of months with sales of the candidate for the old version that happened strictly before the month in which the candidate for the new version was first sold. If that proportion is above 70% we say this criterion is satisfied for this pair of UPC-versions at this store. (ii) The proportion of months with sales of the candidate for the new version that happened strictly after the month the candidate for the old version was last sold. Again, if that proportion is above 70% we say this criterion is satisfied. (iii) The number of months between the month in which the candidate for the old version was last sold and the month in which the candidate for the new version was first sold. We say this criterion is satisfied if the number is weakly below 5 in absolute value. Next, we aggregate across stores for each UPC-version pair, checking the percentage of stores for which each of the three criteria are satisfied. We classify a candidate pair as a product transition if all three measures are above 65%.

After applying these filters, multiple candidates may still remain for the new version of a discontinued size. We resolve these cases by first keeping the candidate with the most similar product description. If ties remain, we retain the candidate whose sales pattern most closely resembles a transition according to the criteria defined above. We drop the small number of cases in which more than one candidate remains after these steps. We also address cases in which a given UPC-version is identified as the new version for more than one discontinued product. In these situations, we keep the match with the most similar product description. If ties remain, we drop all matches.

B.2 Additional Descriptive Statistics

Figure B.1: Instances of Up and Downsizing Identified, by Product Subcategory



Notes: This figure shows the number of instances of size increases and of size reductions identified, grouped by product subcategory.

Figure B.2: Distribution of Size Changes by Time Period

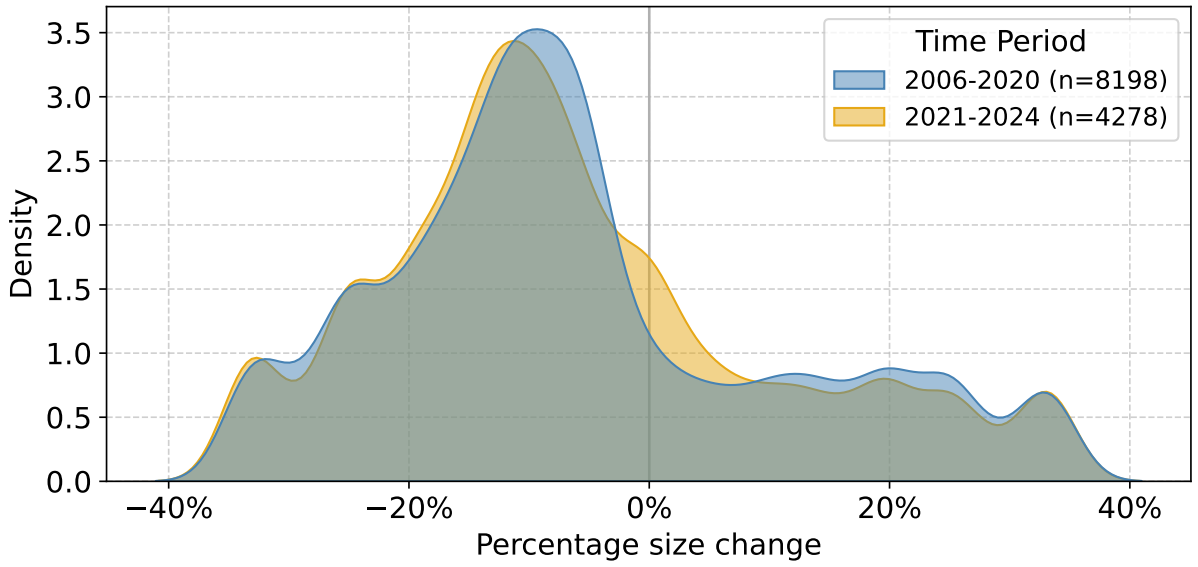
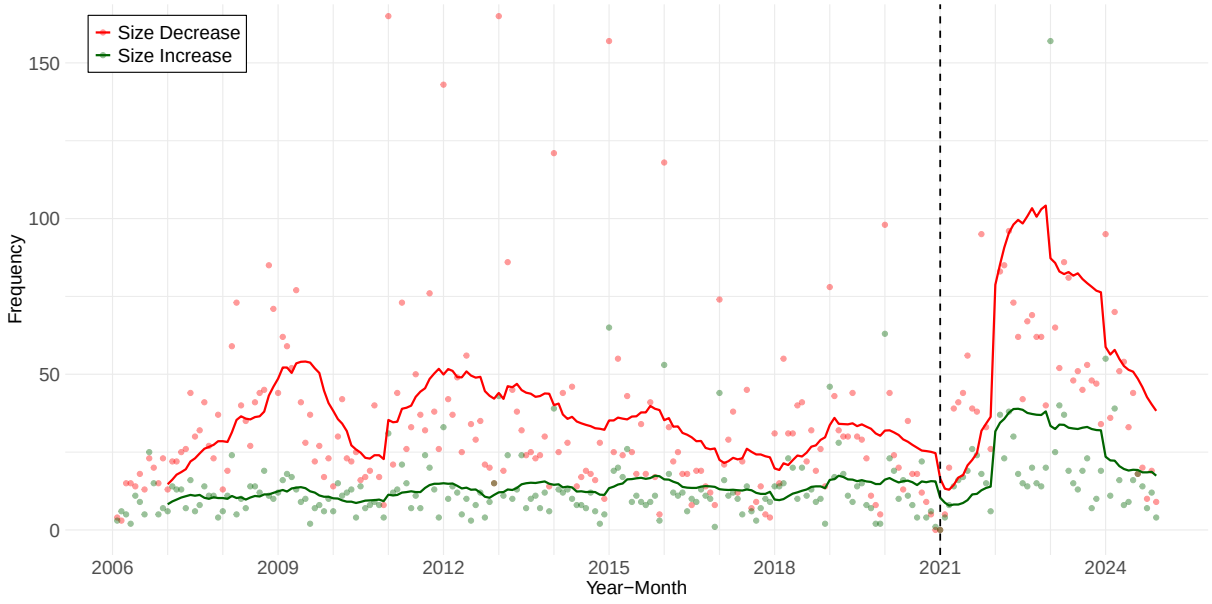


Figure B.3: Number of Size Changes Per Month: Up and Downsizing



Notes: This plot shows instances of size changes (circles) and the 12-month moving average (lines) over time. Size reductions are plotted in red, size increases in green. The time of a size change is defined as the median of the introduction time of the new version across stores. The top 2% of circles are not plotted because they are out of the y-axis range.

B.3 Additional Results and Robustness Checks

Table B.1: Sample Descriptions

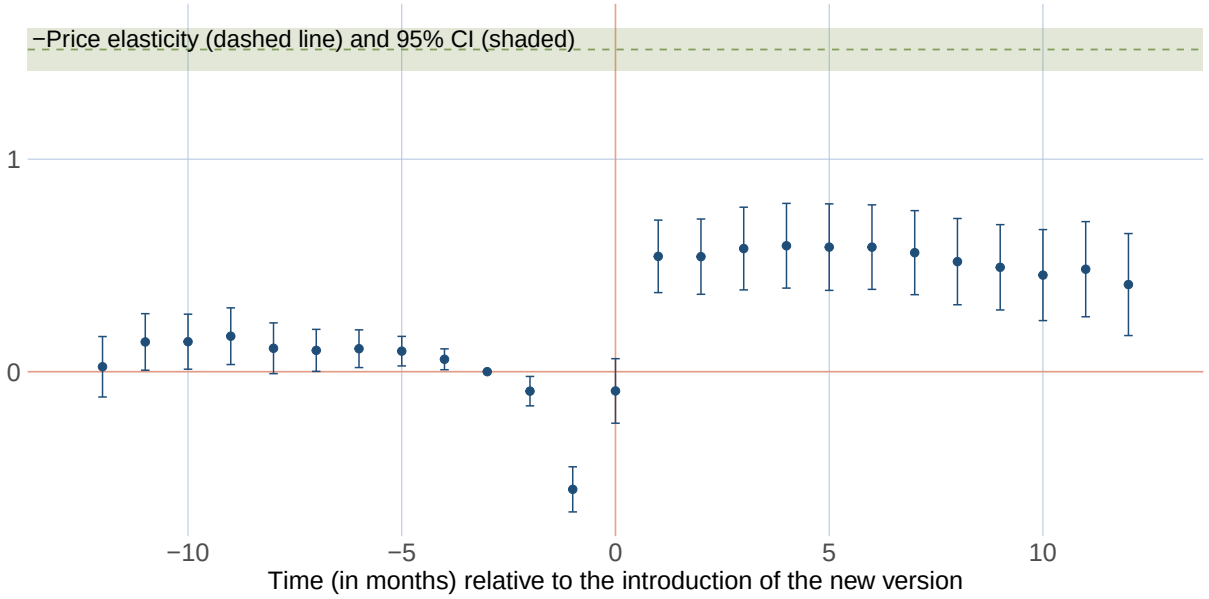
Full Samples		
Direction of Size Change	**Products**	**Rows**
Up	3,237	556,684,233
Both	203	108,925,524
Down	8,368	2,725,640,255

Samples Used for Estimation		
Direction of Size Change	**Products**	**Rows**
Up	994	53,921,427
Down	3,459	363,249,455

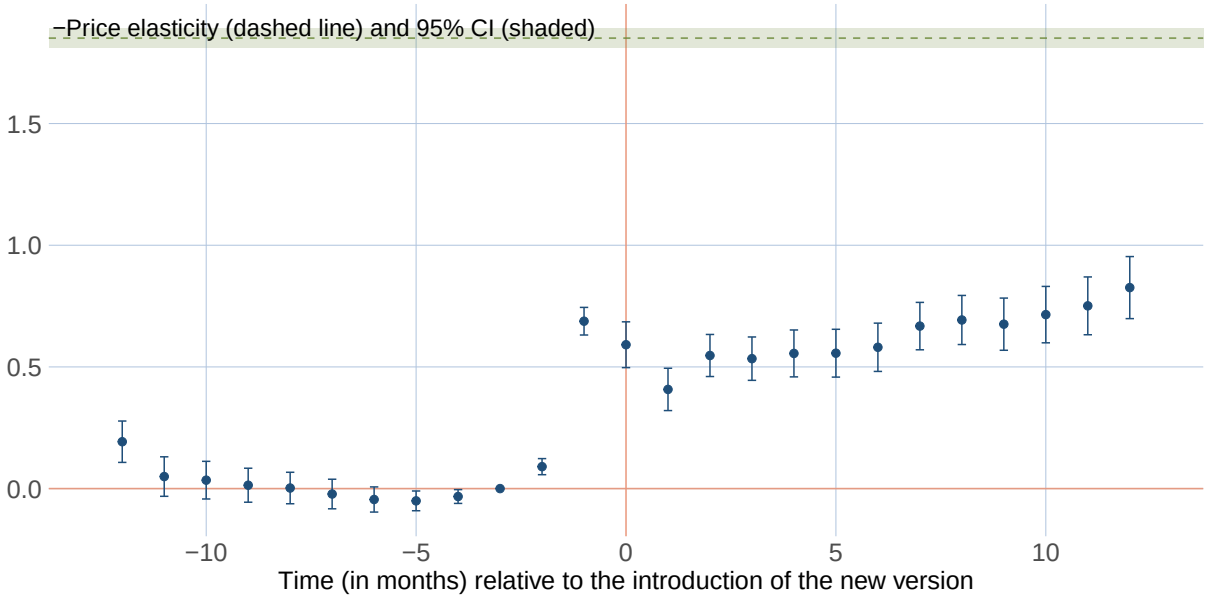
Notes: The full samples include all the sales at the monthly level of products that had size changes. The samples used for estimation have several restrictions. First, they include only products that experienced one and only one size change between 2006–2020 or 2021–2024 and that changed UPCs after the size change. Second, they include only stores that have the same non-missing DMA region for all years. Third, they exclude all observations for which the price instrument cannot be calculated. Fourth, for each product, they contain only stores that had at most one month difference between the month in which they last sold the old version and the month in which they first sold the new version. Fifth, for each product, they have only stores with positive sales in the month of the first sale of the new version, the 12 months before it, and the 12 months after it. Finally, they include only stores that have at least 2 packages sold in each month in the 12 months before the size change.

Figure B.4: Size and Price Elasticities of Packages Demanded

(a) Upsized Products



(b) Downsized Products



Notes: Estimated coefficients and 95% confidence intervals. The coefficients of the binned endpoints are not plotted. The dashed line plots the negative of the price elasticity, $-\widehat{\beta}^{P}$, with the shaded area showing its 95% confidence interval. The sample in the top panel (a) includes only upsized products, while the sample in the bottom panel (b) includes only downsized products. The outcome variable is the natural logarithm of packages sold. Separate regressions were estimated for each product category in each time period following Equation 1, with standard errors clustered at the product level. These aggregate results use the number of products in each regression as weights.

Table B.2: Regression Coefficients of Size Change Event Study Results (Packages)

	Upsized		Downsized	
	ln(Packages) (RHS Binary)	ln(Packages) (RHS Interacted)	ln(Packages) (RHS Binary)	ln(Packages) (RHS Interacted)
	(1)	(2)	(3)	(4)
ln(package price)	-1.52***	-1.52***	-1.85***	-1.85***
	(0.05)	(0.05)	(0.02)	(0.02)
$k \leq -13$	-0.09***	-0.37***	-0.09***	0.63***
	(0.02)	(0.10)	(0.01)	(0.06)
$k = -12$	-0.01	0.02	-0.03***	0.19***
	(0.01)	(0.07)	(0.01)	(0.04)
$k = -11$	0.02*	0.14**	-0.01	0.05
	(0.01)	(0.07)	(0.01)	(0.04)
$k = -10$	0.03**	0.14**	-0.00	0.03
	(0.01)	(0.07)	(0.01)	(0.04)
$k = -9$	0.04***	0.17**	-0.00	0.01
	(0.01)	(0.07)	(0.00)	(0.04)
$k = -8$	0.03**	0.11*	0.00	0.00
	(0.01)	(0.06)	(0.00)	(0.03)
$k = -7$	0.02**	0.10**	0.00	-0.02
	(0.01)	(0.05)	(0.00)	(0.03)
$k = -6$	0.02**	0.11**	0.00	-0.04*
	(0.01)	(0.05)	(0.00)	(0.03)
$k = -5$	0.02***	0.10***	0.00	-0.05**
	(0.01)	(0.04)	(0.00)	(0.02)
$k = -4$	0.01***	0.06**	0.00	-0.03**
	(0.00)	(0.03)	(0.00)	(0.01)
$k = -2$	-0.03***	-0.09***	-0.01***	0.09***
	(0.01)	(0.04)	(0.00)	(0.02)
$k = -1$	-0.11***	-0.55***	-0.09***	0.69***
	(0.01)	(0.05)	(0.00)	(0.03)
$k = 0$	-0.03*	-0.09	-0.07***	0.59***
	(0.01)	(0.08)	(0.01)	(0.05)
$k = 1$	0.10***	0.54***	-0.04***	0.41***
	(0.02)	(0.09)	(0.01)	(0.04)
$k = 2$	0.10***	0.54***	-0.06***	0.55***
	(0.02)	(0.09)	(0.01)	(0.04)
$k = 3$	0.11***	0.58***	-0.06***	0.53***
	(0.02)	(0.10)	(0.01)	(0.05)
$k = 4$	0.10***	0.59***	-0.06***	0.56***
	(0.02)	(0.10)	(0.01)	(0.05)
$k = 5$	0.10***	0.59***	-0.07***	0.56***
	(0.02)	(0.10)	(0.01)	(0.05)
$k = 6$	0.10***	0.59***	-0.07***	0.58***
	(0.02)	(0.10)	(0.01)	(0.05)
$k = 7$	0.09***	0.56***	-0.08***	0.67***
	(0.02)	(0.10)	(0.01)	(0.05)
$k = 8$	0.09***	0.52***	-0.08***	0.69***
	(0.02)	(0.10)	(0.01)	(0.05)
$k = 9$	0.09***	0.49***	-0.08***	0.68***
	(0.02)	(0.10)	(0.01)	(0.05)
$k = 10$	0.07***	0.45***	-0.09***	0.71***
	(0.02)	(0.11)	(0.01)	(0.06)
$k = 11$	0.08***	0.48***	-0.09***	0.75***
	(0.03)	(0.11)	(0.01)	(0.06)
$k = 12$	0.07**	0.41***	-0.11***	0.83***
	(0.03)	(0.12)	(0.01)	(0.07)
$k \geq 13$	0.05*	0.41***	-0.16***	1.17***
	(0.03)	(0.12)	(0.01)	(0.08)
Year x month	✓	✓	✓	✓
Sub-subcategory x year	✓	✓	✓	✓
Sub-subcategory x month	✓	✓	✓	✓
Store x product	✓	✓	✓	✓
Display	✓	✓	✓	✓
Feature	✓	✓	✓	✓
Products	994	994	3,451	3,451
Observations	53,921,427	53,921,427	363,246,613	363,246,613

Notes: $^{***}p < .01$, $^{**}p < .05$, $^{*}p < .1$. These aggregate results use the number of products in each regression as weights. Separate regressions were estimated for each product category in each time period, with standard errors clustered at the product level. Columns (1) and (3) estimate the event study without rescaling by the proportion of the size change. Columns (2) and (4) estimate our main specification, which rescaled the event dummies by the proportion of the size change. The outcome variable for all specifications is the natural logarithm of packages sold.

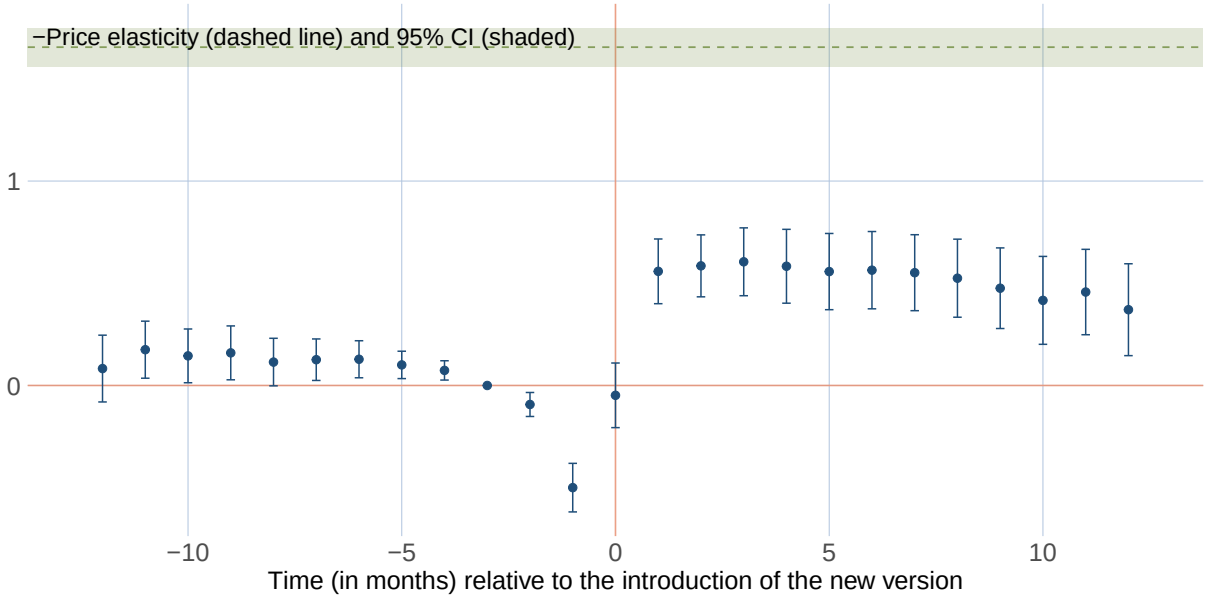
Table B.3: Regression Coefficients of Price and Size Change Event Study Results

	Upsized			Downsized		
	ln(Package Price)	ln(Unit Price)	ln(Units Demanded)	ln(Package Price)	ln(Unit Price)	ln(Units Demanded)
	(1)	(2)	(3)	(4)	(5)	(6)
ln(package price)			-1.51***			-1.85***
			(0.05)			(0.02)
$k \leq -13$	-0.02**	-0.01	-0.37***	0.01***	0.02***	0.62***
	(0.01)	(0.01)	(0.10)	(0.00)	(0.00)	(0.06)
$k = -12$	-0.01	-0.00	0.03	0.01***	0.01***	0.19***
	(0.01)	(0.01)	(0.07)	(0.00)	(0.00)	(0.04)
$k = -11$	-0.01	-0.00	0.14**	0.01***	0.01***	0.04
	(0.00)	(0.00)	(0.07)	(0.00)	(0.00)	(0.04)
$k = -10$	-0.01*	-0.01	0.14**	0.01***	0.01***	0.03
	(0.00)	(0.00)	(0.07)	(0.00)	(0.00)	(0.04)
$k = -9$	-0.01	-0.00	0.17**	0.01***	0.01***	0.01
	(0.00)	(0.00)	(0.07)	(0.00)	(0.00)	(0.04)
$k = -8$	-0.01**	-0.01	0.11*	0.01***	0.01***	0.00
	(0.00)	(0.00)	(0.06)	(0.00)	(0.00)	(0.03)
$k = -7$	-0.01**	-0.01	0.10**	0.00*	0.00**	-0.02
	(0.00)	(0.00)	(0.05)	(0.00)	(0.00)	(0.03)
$k = -6$	-0.00	-0.00	0.11**	0.00	0.00	-0.05*
	(0.00)	(0.00)	(0.05)	(0.00)	(0.00)	(0.03)
$k = -5$	-0.00	-0.00	0.10***	0.00	0.00	-0.05**
	(0.00)	(0.00)	(0.04)	(0.00)	(0.00)	(0.02)
$k = -4$	-0.00	-0.00	0.06**	-0.00	-0.00	-0.03**
	(0.00)	(0.00)	(0.03)	(0.00)	(0.00)	(0.01)
$k = -2$	-0.01***	-0.01***	-0.09***	0.00	0.00	0.09***
	(0.00)	(0.00)	(0.04)	(0.00)	(0.00)	(0.02)
$k = -1$	-0.03***	-0.04***	-0.55***	-0.00***	-0.00**	0.69***
	(0.00)	(0.01)	(0.05)	(0.00)	(0.00)	(0.03)
$k = 0$	0.03***	-0.07***	0.48***	-0.02***	0.06***	1.23***
	(0.01)	(0.01)	(0.08)	(0.00)	(0.00)	(0.05)
$k = 1$	0.08***	-0.08***	1.39***	-0.02***	0.11***	1.46***
	(0.01)	(0.01)	(0.09)	(0.00)	(0.00)	(0.05)
$k = 2$	0.08***	-0.08***	1.42***	-0.02***	0.11***	1.65***
	(0.01)	(0.01)	(0.09)	(0.00)	(0.00)	(0.04)
$k = 3$	0.08***	-0.08***	1.46***	-0.02***	0.11***	1.64***
	(0.01)	(0.01)	(0.10)	(0.00)	(0.00)	(0.05)
$k = 4$	0.08***	-0.08***	1.48***	-0.02***	0.11***	1.67***
	(0.01)	(0.01)	(0.10)	(0.00)	(0.00)	(0.05)
$k = 5$	0.08***	-0.08***	1.47***	-0.02***	0.11***	1.67***
	(0.01)	(0.01)	(0.10)	(0.00)	(0.00)	(0.05)
$k = 6$	0.08***	-0.08***	1.47***	-0.02***	0.11***	1.69***
	(0.01)	(0.01)	(0.10)	(0.00)	(0.00)	(0.05)
$k = 7$	0.08***	-0.08***	1.44***	-0.02***	0.11***	1.78***
	(0.01)	(0.01)	(0.10)	(0.00)	(0.00)	(0.05)
$k = 8$	0.08***	-0.08***	1.40***	-0.02***	0.11***	1.80***
	(0.01)	(0.01)	(0.10)	(0.00)	(0.00)	(0.05)
$k = 9$	0.08***	-0.07***	1.37***	-0.02***	0.11***	1.79***
	(0.01)	(0.01)	(0.10)	(0.00)	(0.00)	(0.06)
$k = 10$	0.09***	-0.07***	1.34***	-0.02***	0.11***	1.83***
	(0.01)	(0.01)	(0.11)	(0.00)	(0.00)	(0.06)
$k = 11$	0.09***	-0.07***	1.37***	-0.02***	0.11***	1.86***
	(0.01)	(0.01)	(0.11)	(0.00)	(0.00)	(0.06)
$k = 12$	0.09***	-0.07***	1.29***	-0.03***	0.11***	1.94***
	(0.01)	(0.01)	(0.12)	(0.00)	(0.00)	(0.07)
$k \geq 13$	0.09***	-0.07***	1.29***	-0.02***	0.11***	2.29***
	(0.01)	(0.01)	(0.12)	(0.00)	(0.00)	(0.08)
Year x month	✓	✓	✓	✓	✓	✓
Sub-subcategory x year	✓	✓	✓	✓	✓	✓
Sub-subcategory x month	✓	✓	✓	✓	✓	✓
Store x product	✓	✓	✓	✓	✓	✓
Display	✓	✓	✓	✓	✓	✓
Feature	✓	✓	✓	✓	✓	✓
Products	994	994	994	3,451	3,451	3,459
Observations	53,921,427	53,921,427	53,921,427	363,246,613	363,246,613	363,249,455

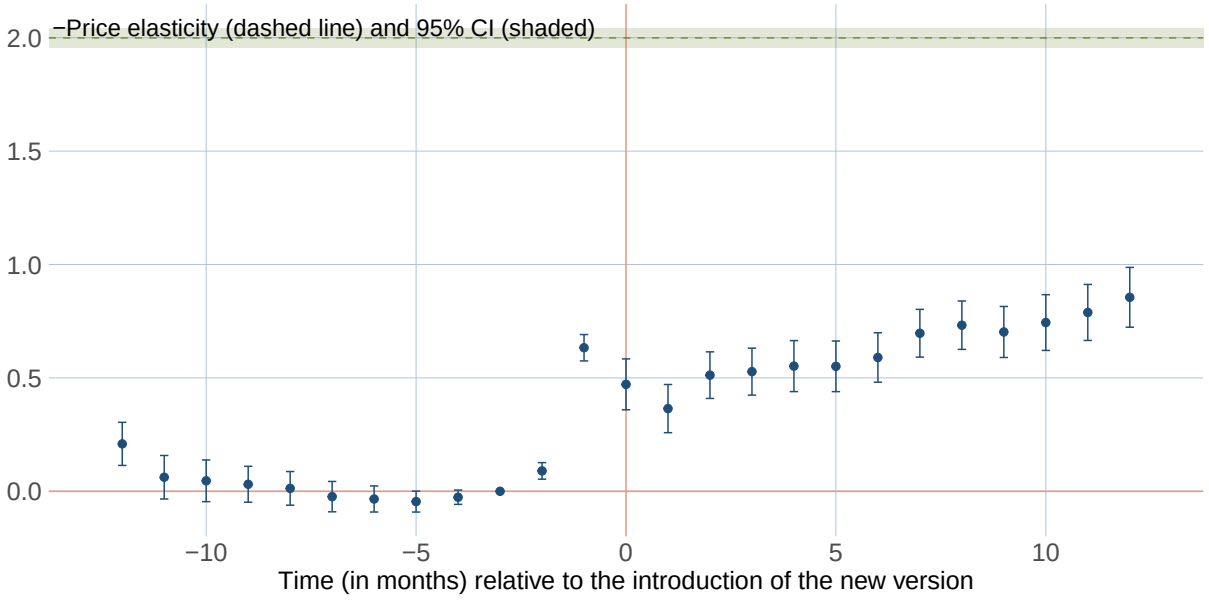
Notes: ***$p < .01$, **$p < .05$, *$p < .1$. Standard errors in parentheses. These aggregate results use the number of products in each regression as weights. Separate regressions were estimated for each product category in each time period, with standard errors clustered at the product level.

Figure B.5: Size and Price Elasticities of Packages Demanded: Observations as Weights

(a) Upsized Products



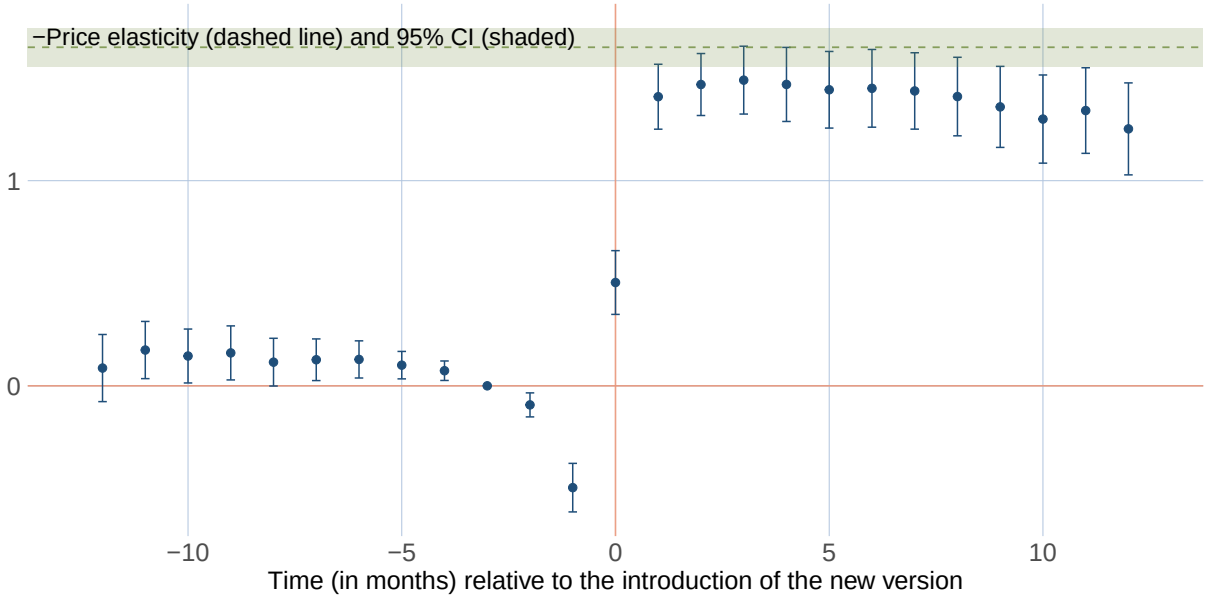
(b) Downsized Products



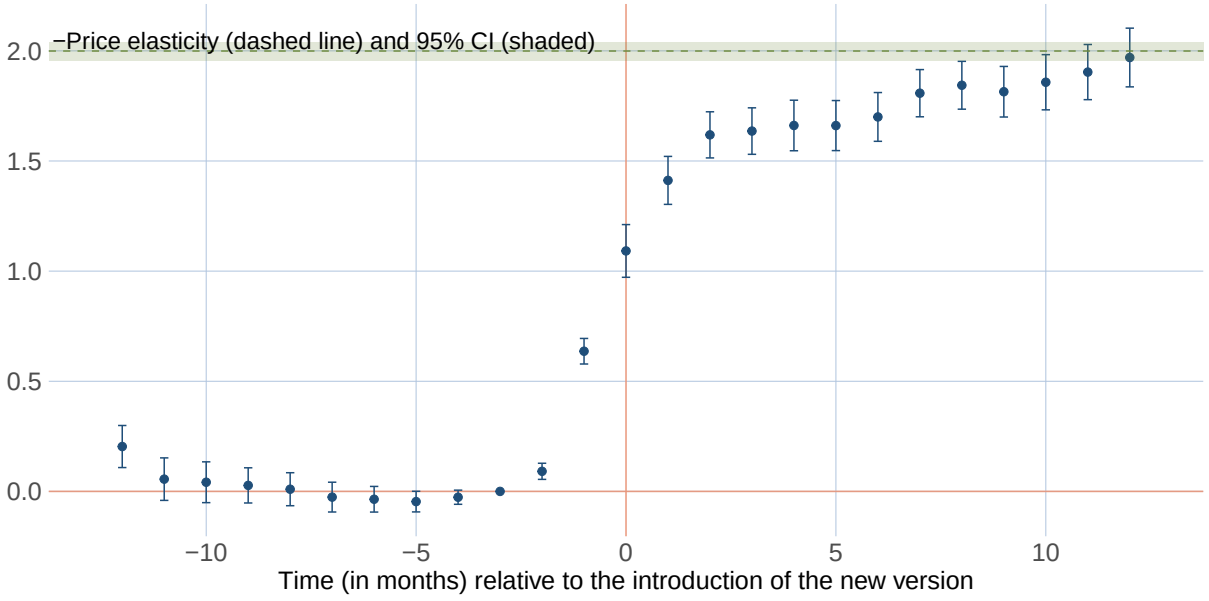
Notes: Estimated coefficients and 95% confidence intervals. The coefficients of the binned endpoints are not plotted. The dashed line plots the negative of the price elasticity, $-\widehat{\beta}^{P}$, with the shaded area showing its 95% confidence interval. The sample in the top panel (a) includes only upsized products, while the sample in the bottom panel (b) includes only downsized products. The outcome variable is the natural logarithm of packages sold. Separate regressions were estimated for each product category in each time period following Equation 1, with standard errors clustered at the product level. These aggregate results use the number of observations in each regression as weights.

Figure B.6: Size and Price Elasticities of Units Demanded: Observations as Weights

(a) Upsized Products



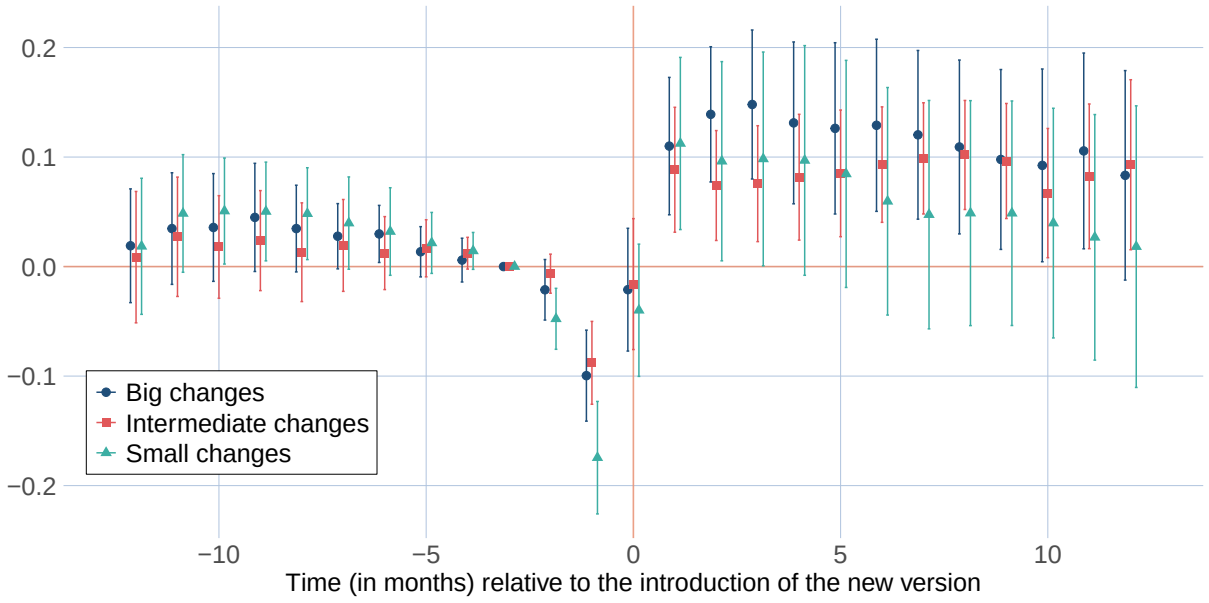
(b) Downsized Products



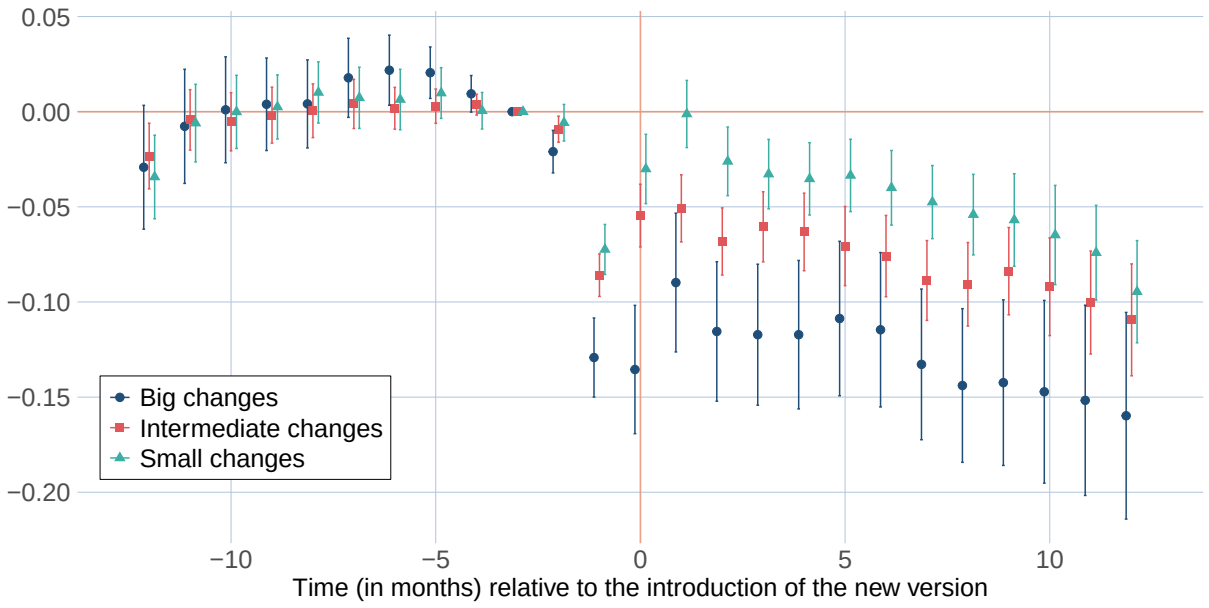
Notes: Estimated coefficients and 95% confidence intervals. The coefficients of the binned endpoints are not plotted. The dashed line plots the negative of the price elasticity, $-\widehat{\beta}^{P}$, with the shaded area showing its 95% confidence interval. The sample in the top panel (a) includes only upsized products, while the sample in the bottom panel (b) includes only downsized products. The outcome variable is the natural logarithm of units sold. Separate regressions were estimated for each product category in each time period following Equation 1, with standard errors clustered at the product level. These aggregate results use the number of observations in each regression as weights.

Figure B.7: Size Change Event Studies: Sample Split by Percent of Size Change

(a) Upsized Products



(b) Downsized Products



Notes: Estimated coefficients and 95% confidence intervals. Standard errors were clustered at the product level. The sample in the top panel (a) includes only upsized products, while the sample in the bottom panel (b) includes only downsized products. The outcome variable is the natural logarithm of packages sold. We split the upsized and downsized products into three groups each based on the magnitude of the size change. Then, we estimate Equation 2 separately for each group. The coefficients of the binned endpoints are not plotted.

C. Additional Details on the Online Shopping Experiment

C.1 Survey Experiment Screenshots

This section reports the shopping-decision screens shown to participants. We first present the baseline version, which is the one experienced by everyone in the first three scenarios, and by the control group in all scenarios. Then, we present the shopping environments of the three treatment groups. The participants in those groups saw the baseline environment for their first three scenarios, and the modified one in scenarios four through six.

Figure C.1: Example of Control Shopping Environment

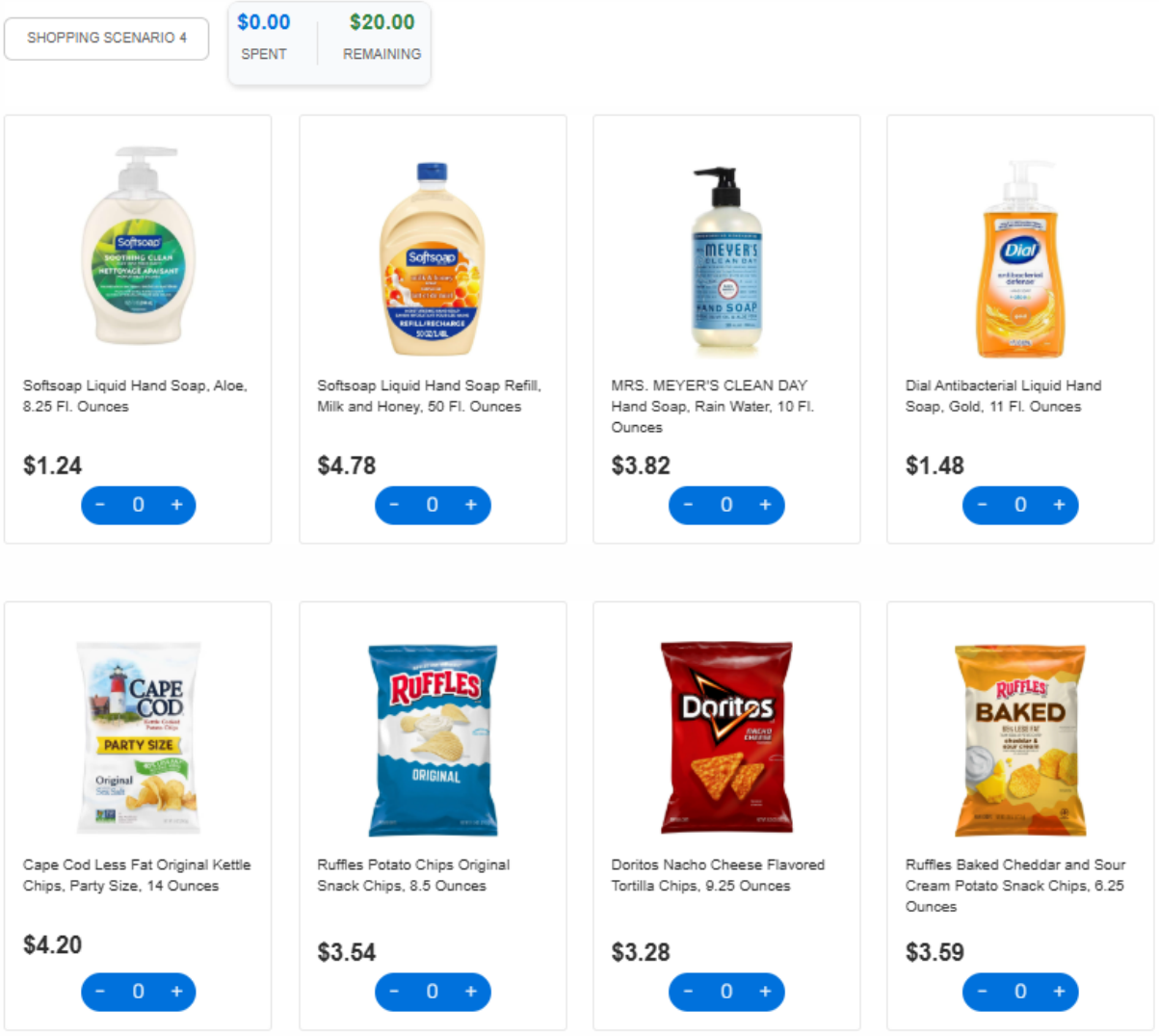


Figure C.2: Example of Unit Price Shopping Environment

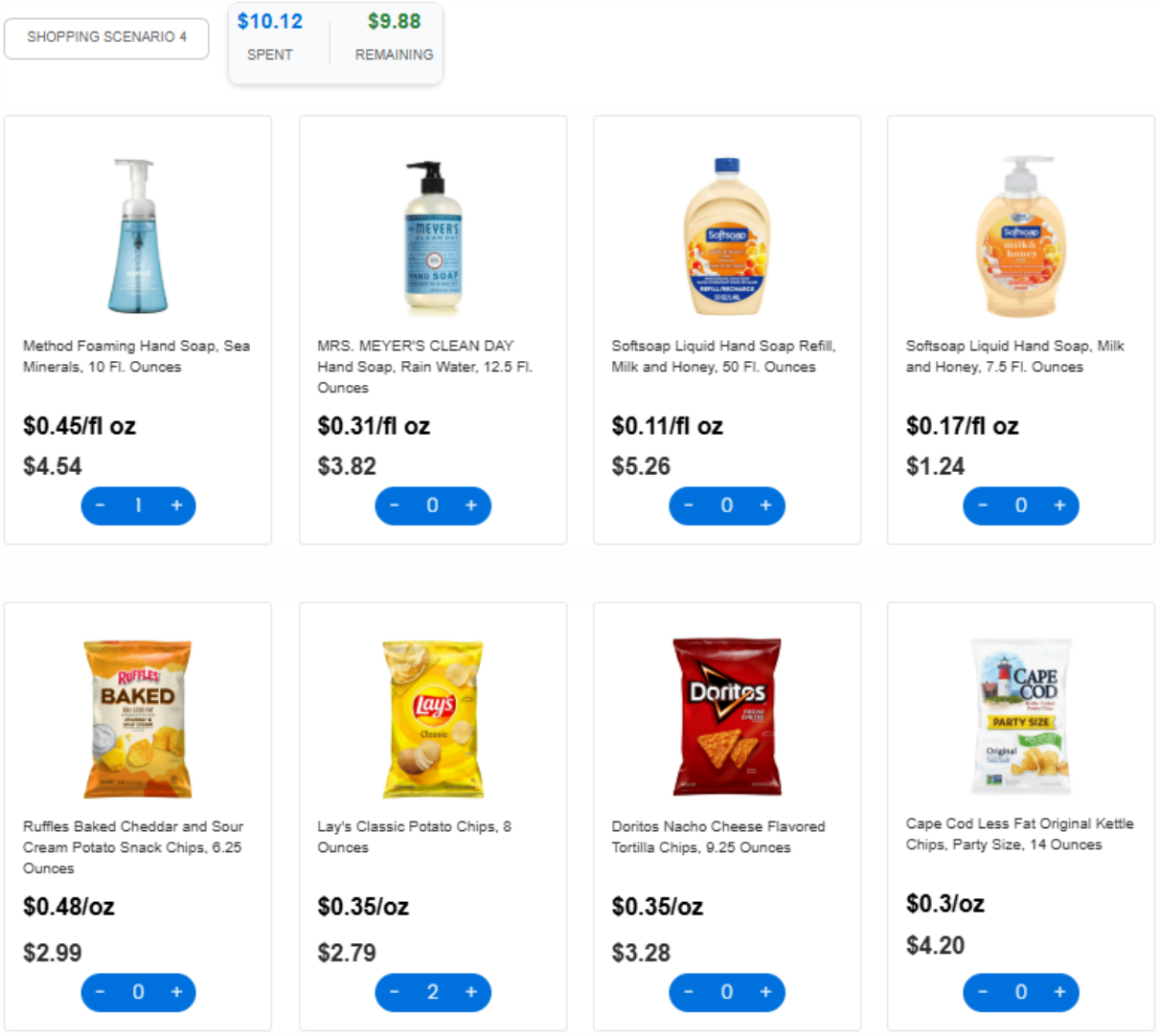


Figure C.3: Example of Prominent Size Display Shopping Environment

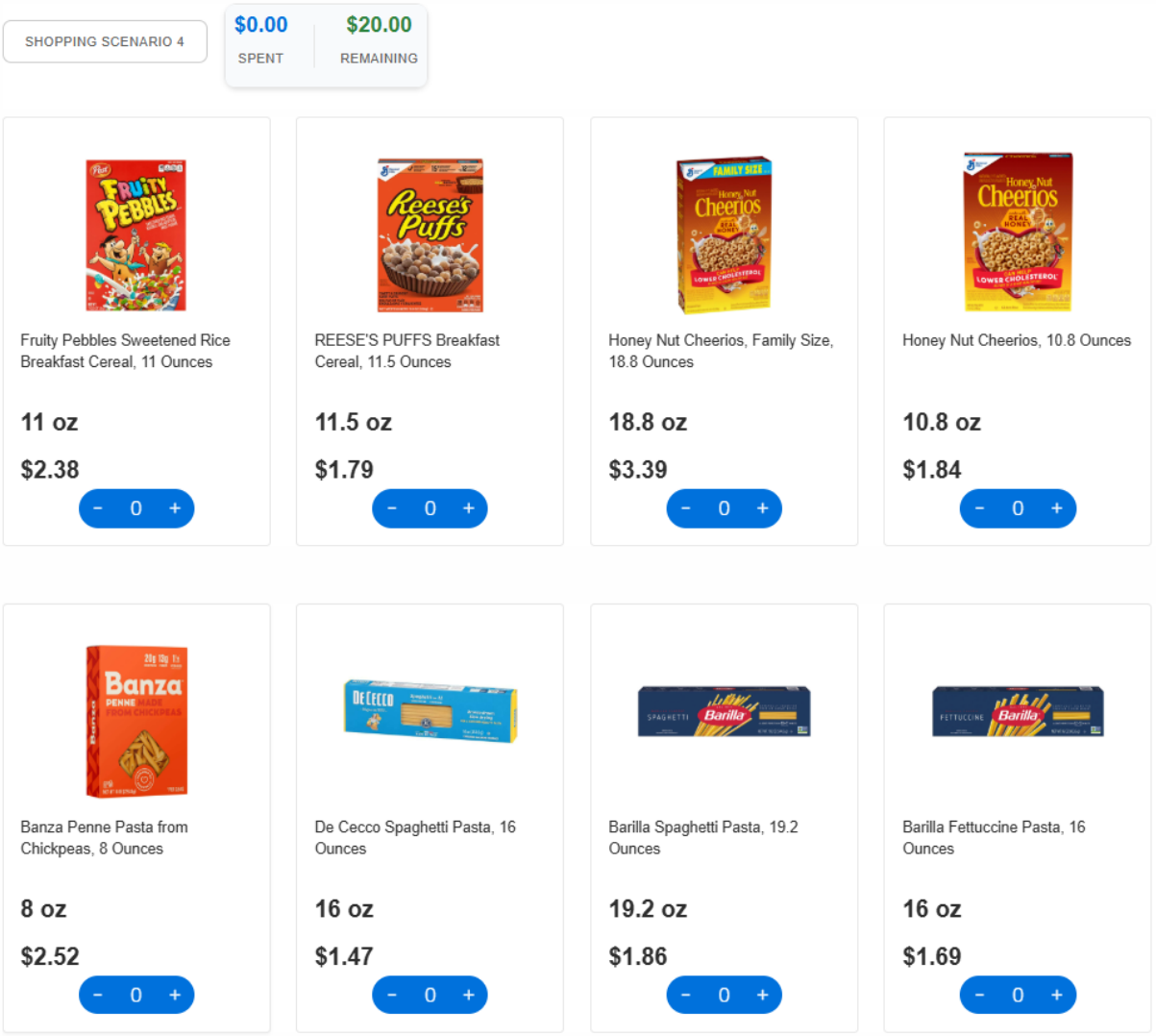
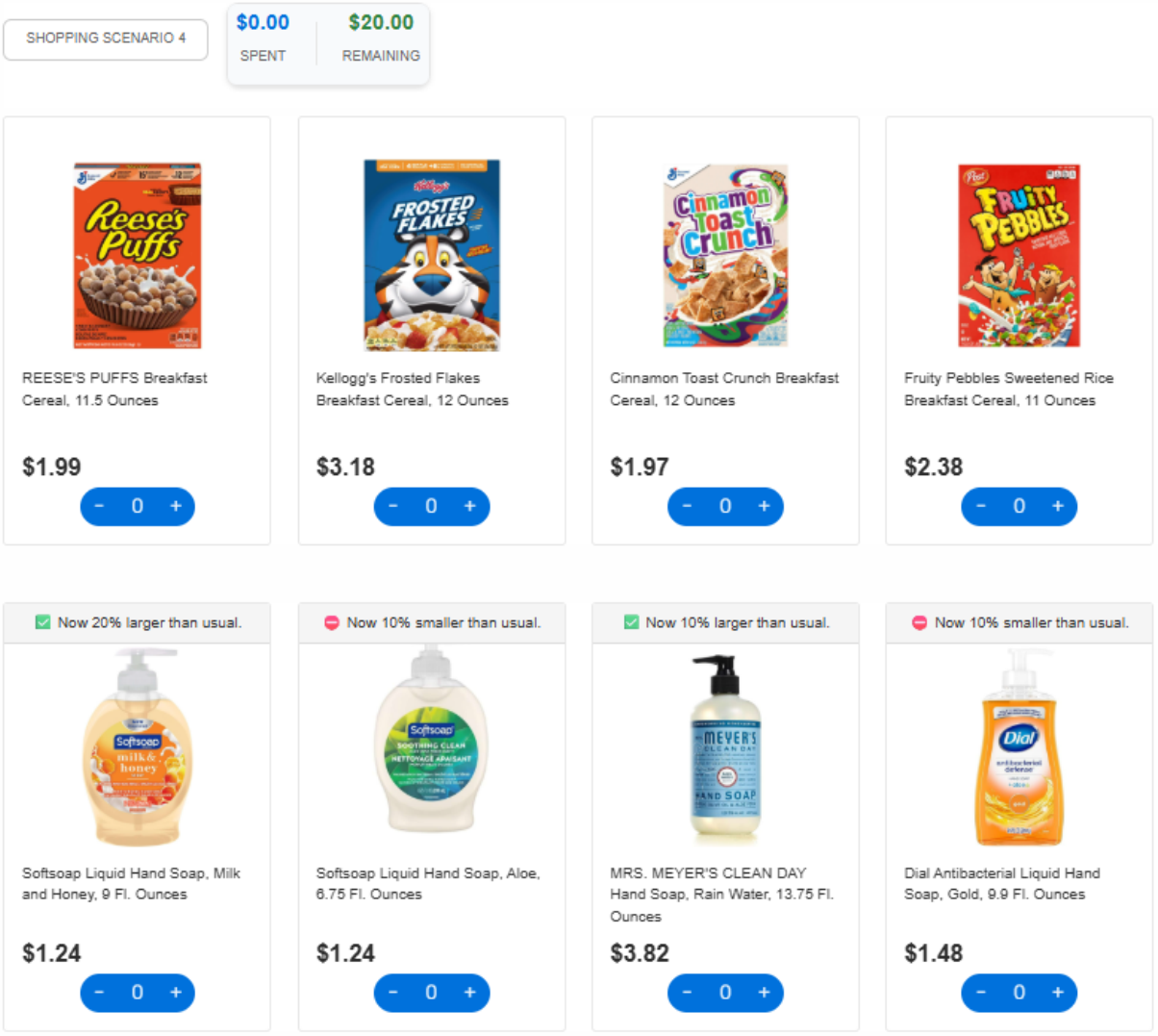


Figure C.4: Example of Size Change Labels Shopping Environment



C.2 Additional Descriptive Statistics and Results

Table C.1: Summary Statistics

	Percent	Std. Dev.
Demographics		
White	74.58	(43.55)
Female	53.05	(49.92)
Age Below 34 Years	40.39	(49.08)
Age 35-54 Years	45.21	(49.78)
Age 55+ Years	14.40	(35.12)
College Degree or Above	55.41	(49.72)
HHold Income Below 50k	28.60	(45.20)
HHold Income 50-100k	39.16	(48.82)
HHold Income Above 100k	30.04	(45.85)
Household Size 4 and Above	31.06	(46.29)
Shops Weekly or More	79.24	(40.57)
Shopping Attitudes and Behaviors		
Says Likely to Notice Size Change	66.53	(47.20)
Says Likely to Notice Price Change	92.16	(26.89)
Thinks Shrinkflation is Very Unfair	40.54	(49.11)
Dislikes Shrinkflation After Cost Incr.	32.14	(46.71)
Dislikes Price Hike After Cost Incr.	15.22	(35.93)
Likely to Switch Brand After Shrinkflation	76.93	(42.14)
Supports Shrinkflation Law	79.45	(40.42)
Observations	1951	

Notes: This table reports summary statistics for the main analysis sample. Entries are reported as percentages, with standard deviations in parentheses. The first panel reports demographic characteristics. The second panel reports shopping attitudes and responses to package size changes.

Table C.2: Sample Demographics and U.S. Population Comparison

	Sample	U.S.	Difference
White	74.58	62.07	12.51
Female	53.05	50.99	2.06
Age Below 34 Years	40.39	29.00	11.39
Age 35-54 Years	45.21	32.48	12.73
Age 55+ Years	14.40	38.53	-24.13
College Degree or Above	55.41	34.14	21.27
HHold Income Below 50k	28.60	30.57	-1.97
HHold Income 50-100k	39.16	28.27	10.89
HHold Income Above 100k	30.04	41.16	-11.12
Household Size 4 and Above	31.06	21.71	9.35
Observations	1951		

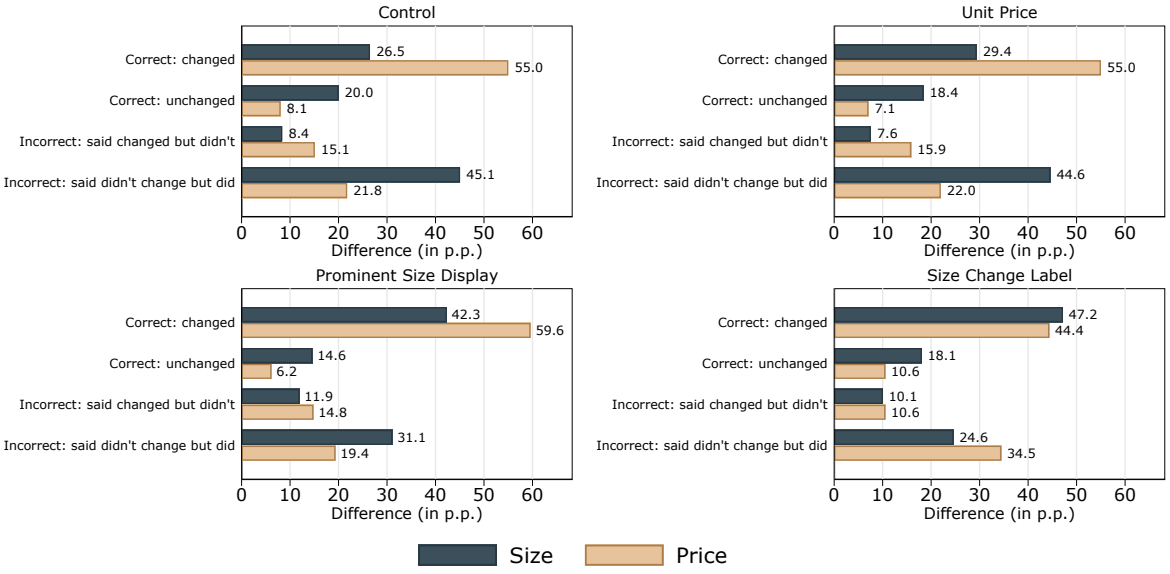
Notes: This table compares the demographic composition of the main analysis sample with the U.S. adult population. Entries are reported as percentages. The first column reports sample shares. The second column reports U.S. shares from the 2024 American Community Survey (ACS) 1-year estimates published by the U.S. Census Bureau. The third column reports the difference between the sample and U.S. shares in percentage points. U.S. shares for race, sex, age, and education are computed over the population aged 18 and over. White refers to respondents reporting White alone. College degree or above refers to a bachelor's degree or higher. U.S. shares for household income and household size are computed over households.

Table C.3: Products in the Survey Experiment

	Size (oz./fl. oz.)	Price ($)	Mean Packs Demanded	Would Buy (%)	Obs.
Pasta					
De Cecco Spaghetti Pasta	16.0	1.84	0.57	34.8	4,044
Banza Penne Pasta from Chickpeas	8.0	2.29	0.91	22.6	3,126
Barilla Spaghetti Pasta	16.0	1.69	0.95	51.2	5,532
Barilla Farfalle Pasta	16.0	1.69	0.96	47.3	5,334
De Cecco Penne Rigate Pasta	16.0	1.84	0.76	34.1	3,972
Barilla Fettuccine Pasta	16.0	1.69	0.79	47.1	5,184
Chips					
SunChips Original Flavor Whole Grain Snack Chips	7.0	2.99	0.58	29.0	4,032
Cape Cod Less Fat Original Kettle Chips, Party Size	14.0	4.20	0.53	29.7	4,188
Lay's Classic Potato Chips	8.0	2.79	0.61	35.3	4,866
Doritos Nacho Cheese Flavored Tortilla Chips	9.2	3.28	0.76	36.9	4,842
Ruffles Potato Chips Original Snack Chips	8.5	3.22	0.48	33.8	4,608
Ruffles Baked Cheddar and Sour Cream Potato Snack Chips	6.2	2.99	0.60	30.8	4,392
Soaps					
Dial Antibacterial Liquid Hand Soap, Gold	11.0	1.48	0.62	23.5	2,898
Softsoap Liquid Hand Soap, Milk and Honey	7.5	1.24	0.71	26.5	3,240
Softsoap Liquid Hand Soap, Aloe	7.5	1.24	0.56	26.9	3,162
MRS. MEYER'S CLEAN DAY Hand Soap, Rain Water	12.5	3.82	0.70	21.2	2,472
Method Foaming Hand Soap, Sea Minerals	10.0	3.78	0.46	21.4	2,454
Softsoap Liquid Hand Soap Refill, Milk and Honey	50.0	4.78	0.61	24.1	2,958
Cereals					
Honey Nut Cheerios	10.8	2.05	0.53	29.7	3,990
Honey Nut Cheerios, Family Size	18.8	3.39	0.68	29.7	4,068
Cinnamon Toast Crunch Breakfast Cereal	12.0	1.97	0.91	27.7	3,930
Kellogg's Frosted Flakes Breakfast Cereal	12.0	3.18	0.49	25.4	3,774
Fruity Pebbles Sweetened Rice Breakfast Cereal	11.0	2.38	0.55	20.7	3,426
REESE'S PUFFS Breakfast Cereal	11.5	1.99	0.76	19.3	3,156

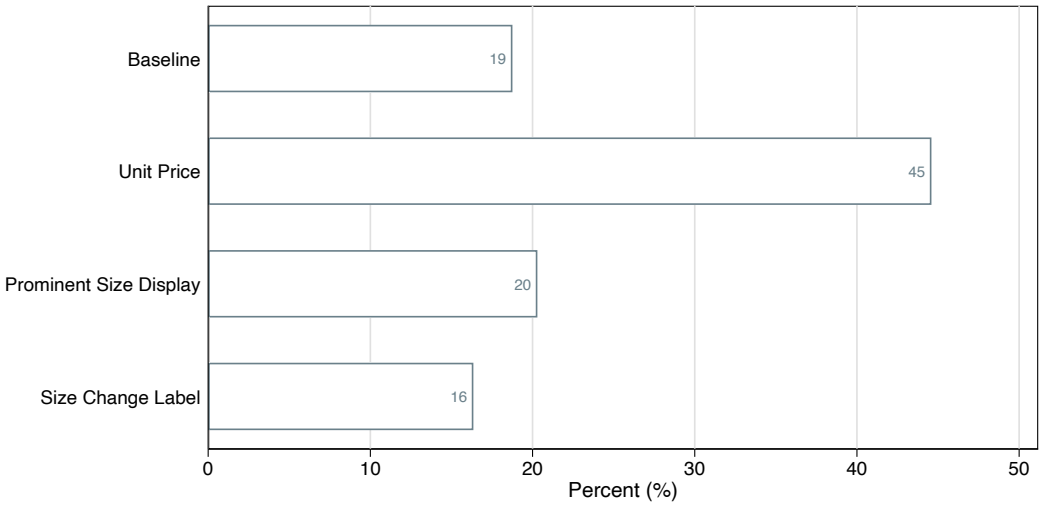
Notes: This table lists the products used in the survey experiment. For each product, the table reports the baseline package size, baseline price, mean number of packages demanded across shopping scenarios, the share of respondents who report that they would buy the product at regular prices in their area, and the number of observations. Mean packages demanded is calculated conditional on the product being visible to the participant.

Figure C.5: Recall of Changes vs Actual Frequency: Details



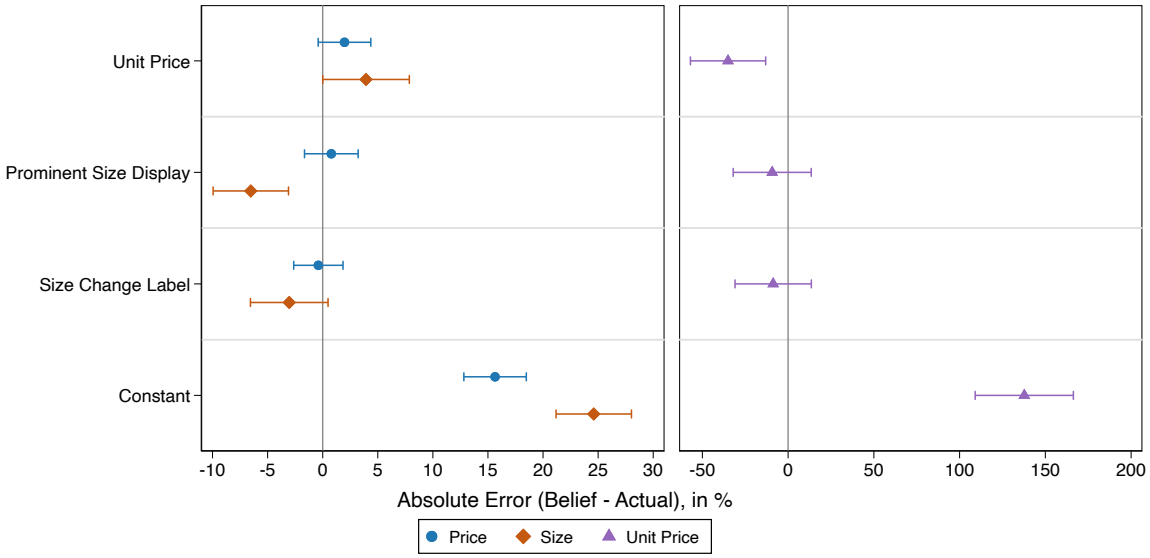
Notes: The figure compares respondents' reported noticing of changes in prices, sizes, and package design between scenarios 4 and 6 with the actual variation in each attribute, averaged across the four products for which this was elicited in the post-scenario questions. We split the responses by whether they correctly or incorrectly identified variation in each dimension. Panels report these patterns separately for the control group and each treatment group.

Figure C.6: Stated Preference for Shopping Environment



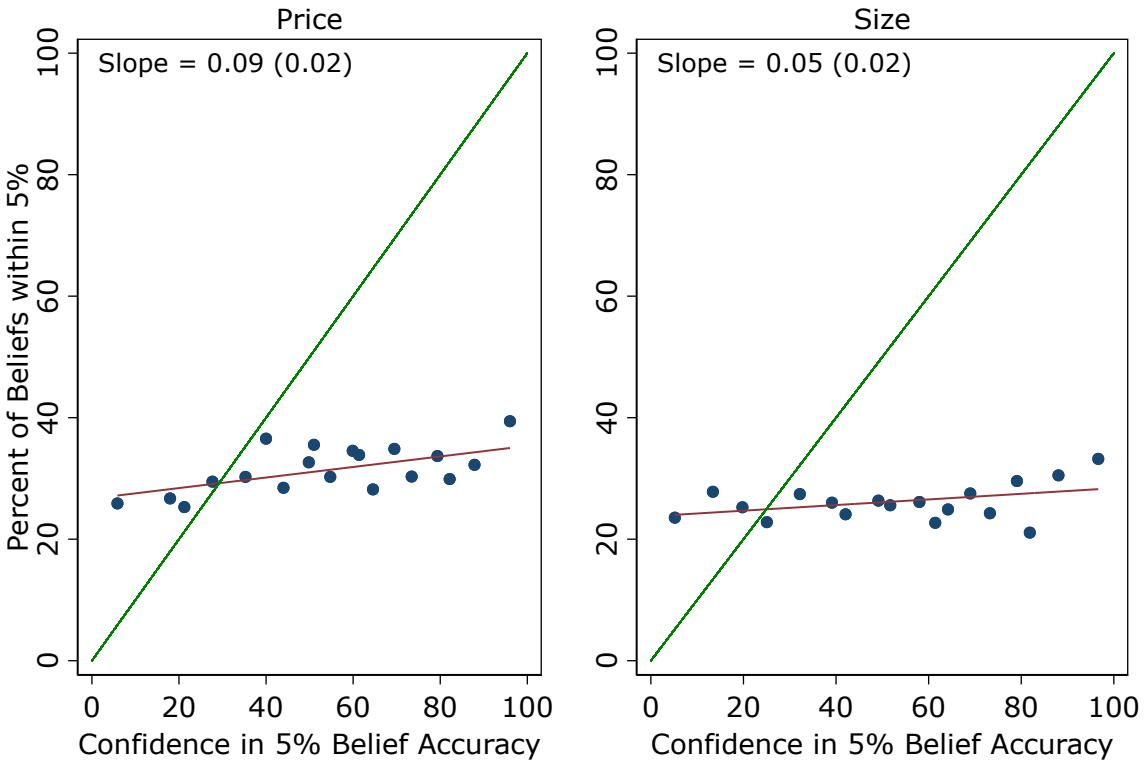
Notes: This figure reports participants' stated preferences over the shopping environments corresponding to the treatment interventions, as shown in Figure 8, Panel (b). Participants were asked: "Which way of showing products would you prefer to see when shopping for any type of product in the real world?"

Figure C.7: Treatment Effects on Belief Accuracy



Notes: The figure reports treatment effects on the mean absolute error of beliefs elicited after the final scenario, expressed in percentage terms. Outcomes are reported separately for price beliefs, size beliefs, and unit price beliefs. Treatment effects are estimated relative to the control group for the Unit Price, Prominent Size Display, and Size Change Label treatments.

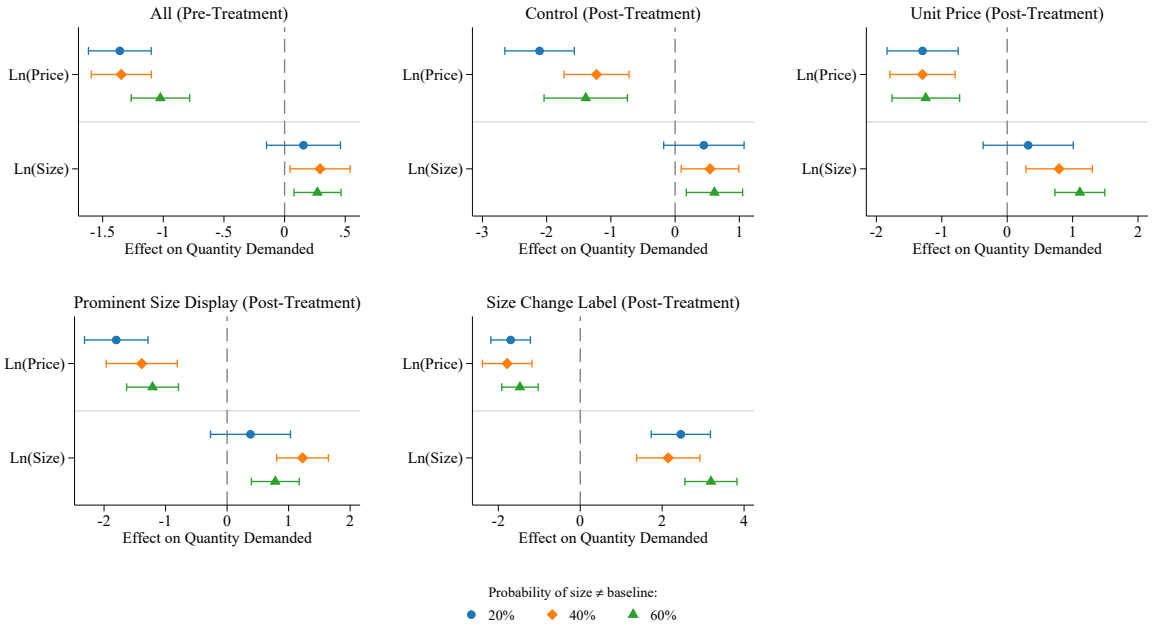
Figure C.8: Belief Accuracy and Sophistication



Mean (Confidence - Accuracy): Price: 23.0 p.p. and Size: 27.0 p.p.

Notes: The figure presents binned scatterplots relating respondents' stated confidence in the accuracy of their beliefs to realized belief accuracy. We include a 45-degree line. Results are shown separately for price and size beliefs. All treatment groups and the control group are included.

Figure C.9: Elasticities by Level of Size Variation



Notes: The figure reports estimates from Poisson regressions that exploit cross-randomized variation in the frequency of size changes. Respondents were assigned to one of three groups that differed in the probability of seeing a product size different from baseline. Estimates are reported separately by size-variation group and sample. The first sample includes all respondents in the pre-treatment scenarios; the remaining samples correspond to each shopping-environment manipulation group and use only the final three scenarios. All regressions include individual, scenario, and product fixed effects.

C.3 Details on Product Attribute Beliefs

Appendix Table C.4 reports percent belief errors, defined as the respondent's belief minus the actual product attribute, divided by the actual value. Beliefs about prices before the shopping scenarios are relatively inaccurate, when compared with the baseline prices used in the survey. This reflects both that the prices shown in the survey are discounted relative to typical retail prices and that real-world prices vary more across locations than package sizes. Once price beliefs are elicited later in the survey, the distribution is tighter and centered much closer to the truth. The median percent error on price beliefs is nearly zero for both the elicitation between scenarios 3 and 4 and the one after the last scenario, scenario 6.

Beliefs about package size are centered on the truth: the median size error is 0% for the three elicitation moments. However, the distribution of size errors is more spread out than the distribution of price errors for both the second and third belief elicitations. This suggests that errors in size beliefs are relatively more widespread across respondents.

Unit price beliefs are less accurate and more biased than either price or size beliefs. The untrimmed unit price belief errors reported in this table have a median of 50%, and the 75th percentile exceeds 600%. Due to this, in our main analysis, we also exclude unit price beliefs that are more than 10 times the actual unit price for that product-scenario. This restriction excludes about 20 percent of unit price beliefs. The resulting unit price belief error, also reported in this table, has a median of 20%. We also construct an implied unit price measure by dividing price beliefs by size beliefs. This measure is more accurate, with a median error of 3%.

Appendix Figure C.10 plots binscatters of price and package size beliefs against the corresponding actual product attributes. Appendix Figure C.11 presents analogous binscatters for unit price beliefs and the implied unit price belief. In each panel, beliefs are plotted on the y-axis and actual values (or implied unit price beliefs) on the x-axis, together with a 45-degree line. The sample is restricted to the control group. Each panel residualizes the relationship between beliefs and actual attributes using individual fixed effects and plots their linear fit. The figures show the same patterns as the summary statistics. Price beliefs are more strongly related to actual prices than size beliefs are to actual sizes, and both are centered around the truth. Unit prices, by contrast, are generally overestimated, with most beliefs lying above the 45-degree line. Even so, these beliefs remain strongly related to actual unit prices. Unit price beliefs are also closely related to unit prices implied by respondents' stated price and size beliefs. However, this relationship is less than one-for-one, with a slope of 0.67.

One concern is whether belief elicitation was properly incentivized. As discussed above, for a randomly selected subgroup of respondents, one randomly selected post-treatment belief was incentivized with a monetary bonus for accuracy. We find no evidence that this accuracy bonus of up to $1 affected absolute errors. The average bonus payment was 36.9 cents. The estimated effects of the incentive on absolute errors are -2 p.p. for size beliefs (p-value=0.22), -0.6 p.p. for price beliefs (p-value=0.55), and -3.9 p.p. for unit price beliefs (p-value=0.76). These results show that the unincentivized belief elicitations are not less accurate than the incentivized ones, giving us confidence in the quality of the belief measures across the full

sample and across all elicitations.

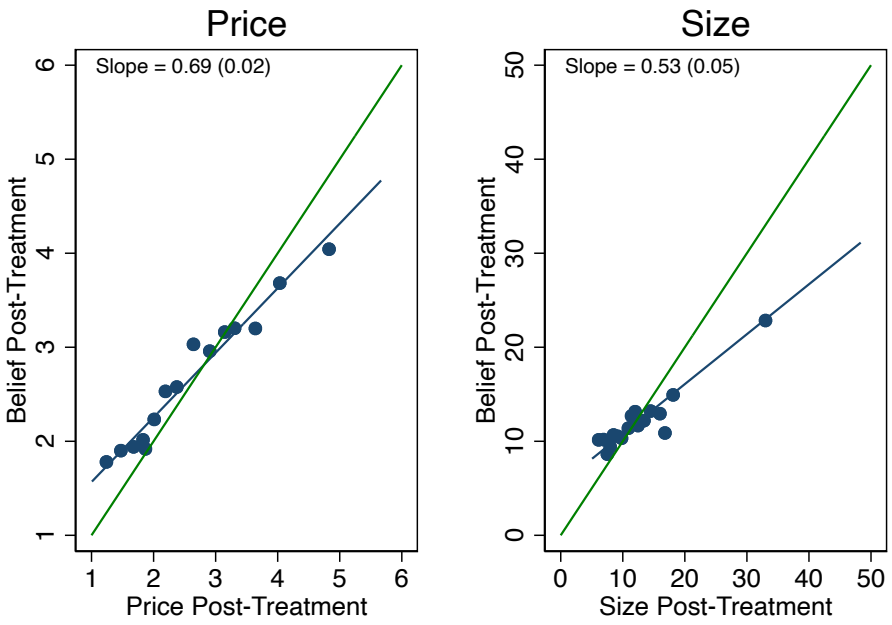
For most participants, we elicited beliefs about prices, sizes, and unit prices only after the shopping scenarios, rather than also eliciting them before and halfway through the scenarios. We find that this post-scenarios only elicitation is associated with larger absolute belief errors. The estimated effects are 16.5 p.p. for size beliefs (p-value$<$0.001), 5.3 p.p. for price beliefs (p-value$<$0.001), and 43.4 p.p. for unit-price beliefs (p-value=0.002). These results indicate that repeated belief elicitations of product attributes make respondents track product attributes more closely. In our main analyses, as well as in the tables and plots below, we pool both of these elicitation samples.

Table C.4: Percent Belief Errors

	Difference (Belief − Actual) in Product Attribute (%)						
	Mean	10th	25th	Median	75th	90th	N
Price							
Prior	81.2	4.5	23.9	61.3	110.1	175.3	1,530
Pre-Treatment	15.6	-19.6	-4.9	0.6	21.3	59.4	1,521
Post-Treatment	8.6	-21.7	-8.7	0.0	16.8	45.1	7,668
Size							
Prior	20.7	-40.0	-12.5	0.0	42.0	100.0	1,520
Pre-Treatment	6.3	-41.7	-16.7	0.0	14.3	60.0	1,525
Post-Treatment	2.3	-52.0	-22.1	0.0	14.3	56.5	7,667
Unit Price							
Post-Treatment (Untrimmed)	941.6	-26.3	-1.8	52.7	610.7	1862.6	7,661
Post-Treatment	116.4	-31.0	-7.1	20.0	127.2	485.3	6,181
Post-Treatment (Implied)	33.6	-37.0	-15.4	2.6	38.9	122.4	7,426

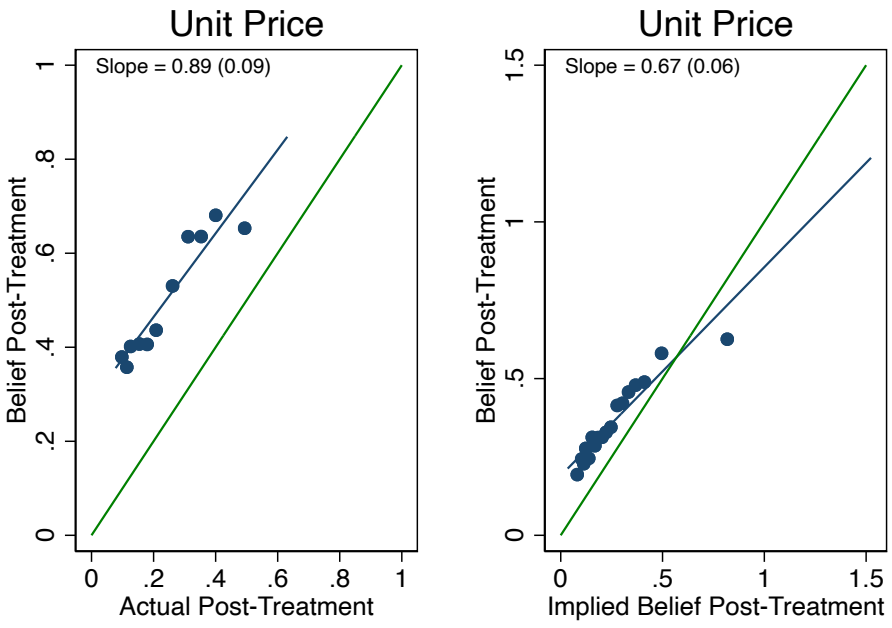
Notes: This table reports summary statistics for percent belief errors in product attributes, defined as $100 \times (\text{belief} - \text{actual})/\text{actual}$. Prior beliefs are elicited before the shopping scenarios, pre-treatment beliefs are elicited before the treatment intervention (between scenarios 3 and 4), and post-treatment beliefs are elicited after the intervention (after the final scenario). We trim the top and bottom 1% of beliefs about price, size, and unit price. The untrimmed unit price row includes all available unit price beliefs. The unit price row excludes observations in which the unit price belief exceeds ten times the actual unit price for that product-scenario. Implied unit price beliefs are constructed by dividing price beliefs by size beliefs.

Figure C.10: Relating Belief and Actual Attributes: Price and Size



Notes: This figure plots binscatters relating respondents' post-treatment beliefs about price and package size to the corresponding actual product attributes. Beliefs are plotted on the vertical axis and actual values on the horizontal axis. The sample is restricted to the control group. Each panel residualizes the relationship using individual fixed effects and plots a linear fit. The displayed slope is estimated from a regression of beliefs on the corresponding actual attribute with individual fixed effects. Standard errors, clustered at the individual level, are reported in parentheses. The 45-degree line indicates equality between beliefs and actual values.

Figure C.11: Relating Belief and Actual Attributes: Unit Price and Implied Unit Price



Notes: This figure plots binscatters relating respondents' post-treatment unit price beliefs to actual and implied unit price measures. In the left panel, the vertical axis reports unit price beliefs and the horizontal axis reports actual unit price. In the right panel, the vertical axis reports unit price beliefs and the horizontal axis reports implied unit price beliefs, constructed from respondents' price and size beliefs. The sample is restricted to the control group. The right panel further excludes observations with stated or implied unit price beliefs above 2. Each panel residualizes the relationship using individual fixed effects and overlays a linear fit. The displayed slope is estimated from a regression with individual fixed effects. Standard errors, clustered at the individual level, are reported in parentheses. The 45-degree line indicates equality between the two measures.

C.4 Elasticities Under Alternative Models

Appendix Table C.5 reports price and size elasticities for quantity demanded, ε_p and ε_s, recovered under seven estimators. The final two columns report their absolute ratio, $|\varepsilon_s/\varepsilon_p|$ and its delta-method standard error, which summarize how much consumers respond to variation in product size relative to price. Unless otherwise noted, all specifications include participant, scenario, and product fixed effects, and cluster standard errors at the participant level. The sample includes all scenarios for control individuals, as well as everyone's first three scenarios (before the shopping environment manipulations). We now describe how we calculate both elasticities for each model.

Linear–Log. We regress quantity demanded q on ln price and ln size, plus fixed effects. The coefficients on $\log p$ and $\log s$ are semi-elasticities $\partial q/\partial \log x$. We convert to point elasticities by dividing by the in-sample mean of q, and obtain standard errors by the delta method.

Two-Stage (Linear Probability Model/Log-Log). We decompose demand into an extensive margin (linear probability model for a dummy of any positive quantity on $\log p$ and $\log s$) and an intensive margin (log-log regression of quantity on $\log p$ and $\log s$ on the subsample with $q > 0$). The total elasticity is $\hat{\beta}_x^{\text{ext}}/\bar{P}(q > 0) + \hat{\beta}_x^{\text{int}}$, following the standard hurdle decomposition. We obtain standard errors using cluster bootstrapping at the participant level.

Poisson and Negative Binomial. The coefficients on the log regressors ($\ln p$ and $\ln s$) are themselves the elasticities. The negative binomial does not include participant fixed effects.

Zero-Inflated Poisson. Because a substantial number of individual-product-scenario chosen quantities are zero (55.5%), we also estimate a zero inflated Poisson model. This model includes only scenario and product fixed effects.

Nonlinear GMM. We estimate the constant-elasticity demand system $q = p^a\, s^b\, \exp(\mathbf{x}'\boldsymbol{\gamma}) + u$ directly by GMM, so a and b are by construction ε_p and ε_s. We use p, s, and product fixed effects as instruments and report one-step GMM estimates.

Ordered Logit. To exploit the discreteness of quantity, we estimate an ordered logit on price and size in levels, after top-coding quantity at three units ($\tilde{q} \equiv \min\{q, 3\}$). We then form the conditional expectation of the censored outcome, $\mathbb{E}[\tilde{q} \mid x] = \sum_{k=0}^{3} k \cdot \Pr(\tilde{q} = k \mid x)$, and report average elasticities of $\mathbb{E}[\tilde{q} \mid x]$ with respect to price and size, averaged over the estimation sample. We do not include participant fixed effects.

Results. The results broadly reproduce our base estimation of price and size elasticities. Despite the very different functional form and distributional assumptions, the estimates line up around the Poisson benchmark. Across all seven methods, the price elasticity is negative, statistically different from zero and around -1. The size elasticity is positive in every specification. The implied ratio $|\varepsilon_s/\varepsilon_p|$ is well below one in all rows, confirming that price responsiveness dominates by a wide margin. Taken together, the table indicates that our Poisson estimates are not an artifact of count-data assumptions or of how zeros are handled, and the magnitudes we discuss in the main text are robust to a wide range of alternative assumptions.

Table C.5: Calculating Price and Size Elasticities with Different Models

	Outcome: Quantity Demanded					
	ε_p	ε_p SE	ε_s	ε_s SE	$\lvert\varepsilon_s/\varepsilon_p\rvert$	$\varepsilon_s/\varepsilon_p$ SE
Linear-Log	-1.345	0.058	0.310	0.058	0.230	0.044
Two-Stage (LPM/Log-Log)	-1.155	0.024	0.234	0.025	0.202	0.022
Poisson	-1.312	0.072	0.310	0.063	0.237	0.048
Negative Binomial	-1.305	0.070	0.281	0.063	0.215	0.048
Zero-Inflated Poisson	-1.302	0.070	0.281	0.063	0.216	0.048
Nonlinear GMM	-1.278	0.066	0.087	0.024	0.068	0.017
Ordered Logit	-0.925	0.050	0.155	0.044	0.168	0.049

Notes: This table reports price and size elasticities of quantity demanded, ε_p and ε_s, estimated using alternative functional-form and distributional assumptions. The final two columns report the absolute ratio $|\varepsilon_s/\varepsilon_p|$ and its standard error, summarizing the size response relative to the price response. Unless otherwise noted, all specifications include participant, scenario, and product fixed effects, and cluster standard errors at the participant level. The sample includes all scenarios for control individuals, as well as everyone's first three scenarios (before the shopping environment manipulations). In the linear–log model, log-price and log-size coefficients are converted from semi-elasticities to point elasticities by dividing by mean quantity. The two-stage specification combines an extensive-margin linear probability model with an intensive-margin log-log model among observations with positive quantity; standard errors are obtained by participant-level cluster bootstrap. In the Poisson and negative binomial models, coefficients on ln price and ln size are the reported elasticities. The negative binomial specification omits participant fixed effects. For the zero-inflated Poisson model, elasticities are average elasticities of the predicted unconditional mean and the specification includes only scenario and product fixed effects. The nonlinear GMM model estimates a constant-elasticity demand equation directly, using price, size, and product fixed effects as instruments. The ordered logit model top-codes quantity at three units and reports average elasticities of the predicted conditional expectation of the censored outcome. It omits participant fixed effects. Standard errors for elasticity ratios are computed by the delta method.

C.5 Supplementary Experimental Analyses

Additional Demand Specifications. Appendix Table C.6 reports demand responses to price and size changes for a sample that pools the control group with everyone's shopping scenarios 1–3. We present results in the aggregate and separately for downsizing and upsizing cases, using both the linear-log and Poisson specifications. The upsizing sample includes product-scenario-individual observations in which package size is weakly larger than in the preceding scenario. The downsizing sample is defined analogously for weakly smaller package sizes. The estimates show asymmetric demand responses to price and size changes in both subsamples, with somewhat stronger responses in the downsizing sample. This pattern contrasts with the short-term retail scanner data results reported above, where demand responded more strongly to size increases. The difference may reflect the absence of marketing effects in the experiment, or the fact that, by design, our experimental setting abstracts from preferences over package sizes.

As we mentioned before, in the experiment, demand should respond equivalently to changes in prices and sizes. In particular, size changes should affect demand only through their induced effect on unit prices. Considering this, Appendix Figure C.12 compares demand responses to unit price variation generated by price versus size changes. We start by regressing quantity on ln unit price and our usual set of fixed effects, reporting a baseline OLS specification. This is followed by three complementary approaches aiming to isolate the contribution of one of the two components: (i) controlling for the other component directly in the regression, (ii) restricting the sample to scenarios in which the other component is held fixed at its baseline value, and (iii) instrumenting unit price with the experimentally varied price or size. The sample pools all scenarios from the control group with scenarios 1–3 from the other groups. Across specifications, demand is substantially more responsive to unit price variation generated by price changes than by size changes. The implied semi-elasticity is roughly -0.9 when identified from price variation, compared with about -0.2 when identified from size variation. Thus, consistent with our previous results, consumers do not treat price and size changes symmetrically: they react about 4 to 5 times as much to prices as to sizes.

Substitution. Appendix Table C.7 reports Poisson demand estimates that allow for substitution patterns across products. In addition to own price and own size, we include leave-one-out average prices and sizes for products in the same category and parent brand, products in the same category but under a different parent brand, and products in other categories. This specification is restricted to observations with a complete set of product alternatives, and therefore differs from the samples used in other demand regressions.[25] The results for the own price and size elasticities are in line with the evidence we previously presented. Individuals are more responsive to the product's price than to its size, except in the Size Change Label group. The cross-price elasticities are generally positive, especially for products in the same category and parent brand, suggesting that consumers substitute toward the product when close alter-

[25]For example, observations are excluded when a product has no other products in the same category and parent brand, since the corresponding leave-one-out average price and size cannot be computed.

natives become more expensive. Conversely, increases in the sizes of close substitutes tend to reduce demand for the focal product. Results are broadly similar across treatment groups.

Heterogeneity in Demand Responses and Beliefs. Appendix Figure C.13 presents product level baseline price and size elasticities. The figure shows that the larger response to price than to size is not driven by a small set of products, but instead holds broadly at the product level. This pattern is indeed present for nearly every product. Elasticities appear larger for pasta than for the other product categories, while the responses are somewhat smaller for hand soaps.

We examine heterogeneity in baseline demand responses by separately interacting our main specification with respondent characteristics. Appendix Figure C.14 reports these estimates for views about shrinkflation, race, gender, household income, shopping frequency, education, product familiarity, and survey comprehension. We find little evidence of heterogeneity in responses to size, with interaction estimates generally small and statistically indistinguishable from zero. There is somewhat more heterogeneity in price responses: women are less price sensitive, while White respondents and those very familiar with the products are more elastic to price.

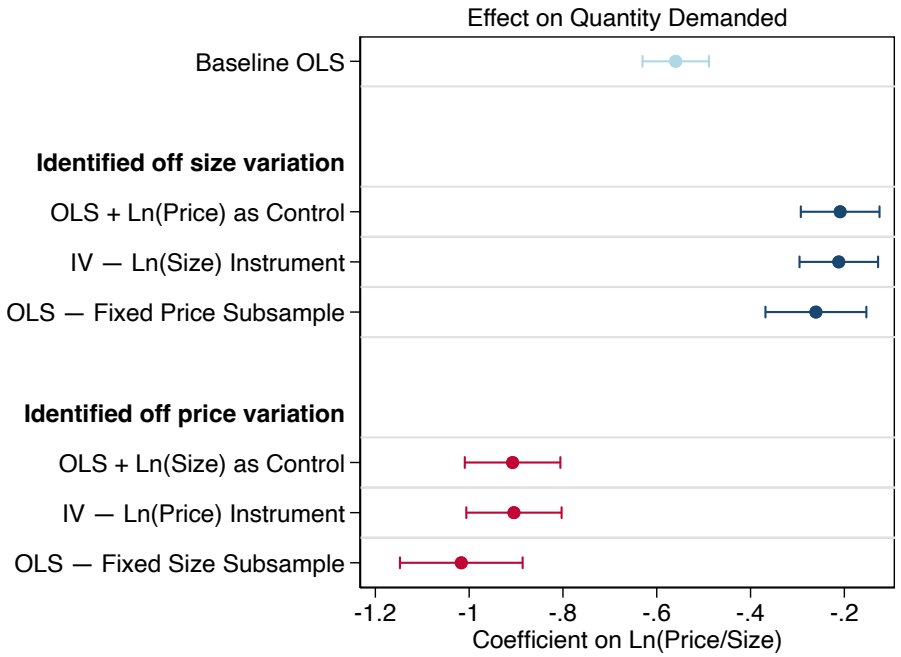
Appendix Table C.8 considers, in turn, heterogeneity in belief errors by respondent characteristics. Panel A reports regressions using respondents' views on shrinkflation and stated attention paid to product attributes, while Panel B reports regressions using demographic characteristics. The dependent variable is an indicator for whether the belief error exceeds 10% (using the final belief elicitation), and standard errors are clustered at the individual level. First, belief accuracy varies with stated attentiveness. For instance, respondents who say they are likely to notice size changes are less likely to make above 10% size errors (-2.8 p.p., significant at 10%) and above 10% unit price errors (-4.0 p.p., significant at 1%). Shrinkflation views are related to errors less systematically, though respondents who dislike shrinkflation after cost increases or would switch brands after it are less likely to make above 10% unit price errors. Demographics are mostly uncorrelated with belief errors, except that weekly or more frequent shoppers make lower unit price errors. This indicates that belief accuracy is not larger for particular demographic groups, such as those in wealthier or poorer households.

Table C.6: Demand Responses Divided by Upsizing and Downsizing Cases

	(1)	(2)	(3)	(4)	(5)	(6)
	Quantity	Quantity	Quantity	Quantity	Quantity	Quantity
Ln(Price)	-0.91***	-0.95***	-0.90***	-1.31***	-1.35***	-1.31***
	(0.052)	(0.061)	(0.055)	(0.072)	(0.082)	(0.078)
Ln(Size)	0.21***	0.099	0.28***	0.31***	0.13	0.43***
	(0.043)	(0.072)	(0.060)	(0.063)	(0.10)	(0.092)
Observations	58704	43472	43354	58344	43149	43050
P-value (Size = -Price)	0.000	0.000	0.000	0.000	0.000	0.000
Size Changes		Upsizing	Downsizing		Upsizing	Downsizing
Individual FE	✓	✓	✓	✓	✓	✓
Scenario FE	✓	✓	✓	✓	✓	✓
Product FE	✓	✓	✓	✓	✓	✓
Model	Linear-Log	Linear-Log	Linear-Log	Poisson	Poisson	Poisson

Notes: This table reports demand responses to price and size changes for control observations, pooling the control group and pre-treatment shopping scenarios, divided by upsizing and downsizing cases. The dependent variable is quantity demanded. Columns present estimates using linear–log and Poisson specifications. The upsizing sample includes product-scenario-individual observations in which package size is weakly larger than in the preceding scenario. The downsizing sample is defined analogously for observations in which package size is weakly smaller than in the preceding scenario. All specifications include individual, scenario, and product fixed effects. Standard errors are clustered at the individual level. The reported p-value tests equality of the size coefficient and the negative of the price coefficient.

Figure C.12: Isolating Price versus Size Variation in Unit Price Demand Responses



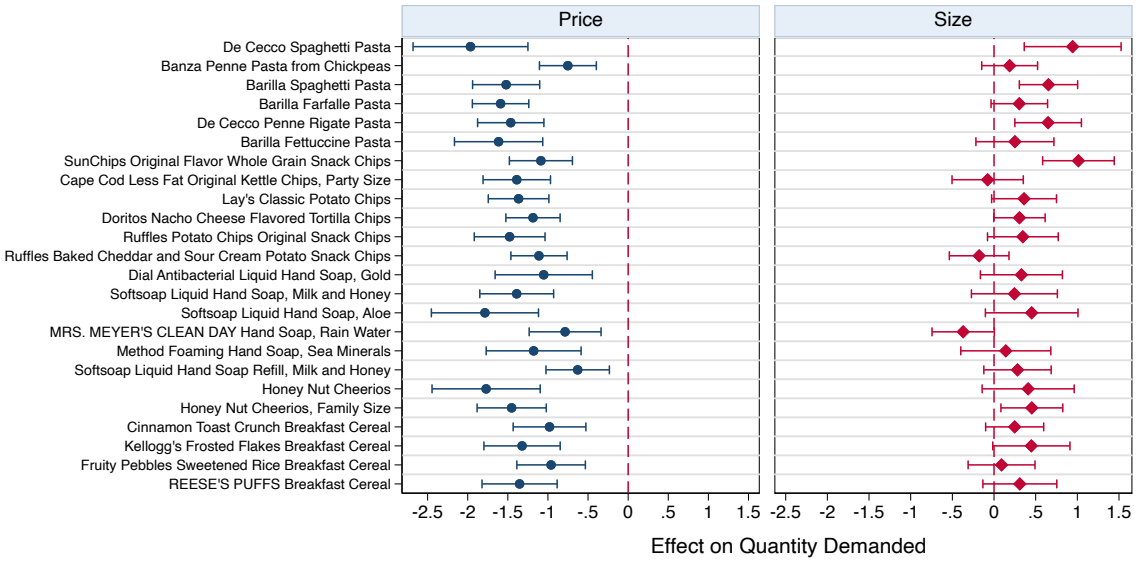
Notes: This figure reports estimates of demand responses to unit price variation generated by price and size changes. The dependent variable is quantity demanded, and the horizontal axis reports the coefficient on log unit price, or $\ln(\text{price}/\text{size})$. The baseline specification regresses quantity demanded on ln unit price and the standard set of fixed effects (scenario, individual and product). The remaining specifications isolate variation in unit price generated by price or size using three approaches: controlling directly for the other component, restricting the sample to scenarios in which the other component is held fixed at its baseline value, and instrumenting ln unit price with the experimentally varied price or size. The sample pools all scenarios from the control group with scenarios 1–3 from the other groups. Standard errors are clustered at the participant level.

Table C.7: Demand Responses Considering Substitution Patterns

	Control + Pre-Treat	Control	Unit Price	Prominent Size Display	Size Change Label
Ln(Price)	-1.41***	-1.94***	-1.48***	-1.54***	-1.55***
	(0.083)	(0.21)	(0.17)	(0.17)	(0.17)
Ln(Size)	0.42***	0.68***	1.06***	0.91***	2.69***
	(0.078)	(0.18)	(0.17)	(0.17)	(0.26)
Avg. Ln(Price) (Same Category and Parent Brand)	0.32***	0.39**	0.68***	0.66***	0.38**
	(0.081)	(0.18)	(0.15)	(0.19)	(0.16)
Avg. Ln(Price) (Same Category, Other Parent Brand)	0.16**	0.27*	0.087	0.15	-0.035
	(0.077)	(0.15)	(0.14)	(0.13)	(0.21)
Avg. Ln(Price) (Other Category and Parent Brand)	0.36**	0.53*	0.16	0.64**	-0.066
	(0.14)	(0.31)	(0.25)	(0.28)	(0.33)
Avg. Ln(Size) (Same Category and Parent Brand)	-0.20***	-0.17	-0.63***	-0.44***	-0.54***
	(0.072)	(0.15)	(0.13)	(0.15)	(0.17)
Avg. Ln(Size) (Same Category, Other Parent Brand)	0.074	0.24	-0.19	-0.26*	-0.23
	(0.083)	(0.18)	(0.14)	(0.14)	(0.15)
Avg. Ln(Size) (Other Category and Parent Brand)	0.13	0.10	-0.12	0.41	-0.71**
	(0.14)	(0.30)	(0.27)	(0.27)	(0.33)
Observations	35157	6990	6987	6882	7023
P-value (Size = -Price)	0.000	0.000	0.063	0.003	0.000
Individual FE	✓	✓	✓	✓	✓
Scenario FE	✓	✓	✓	✓	✓
Product FE	✓	✓	✓	✓	✓

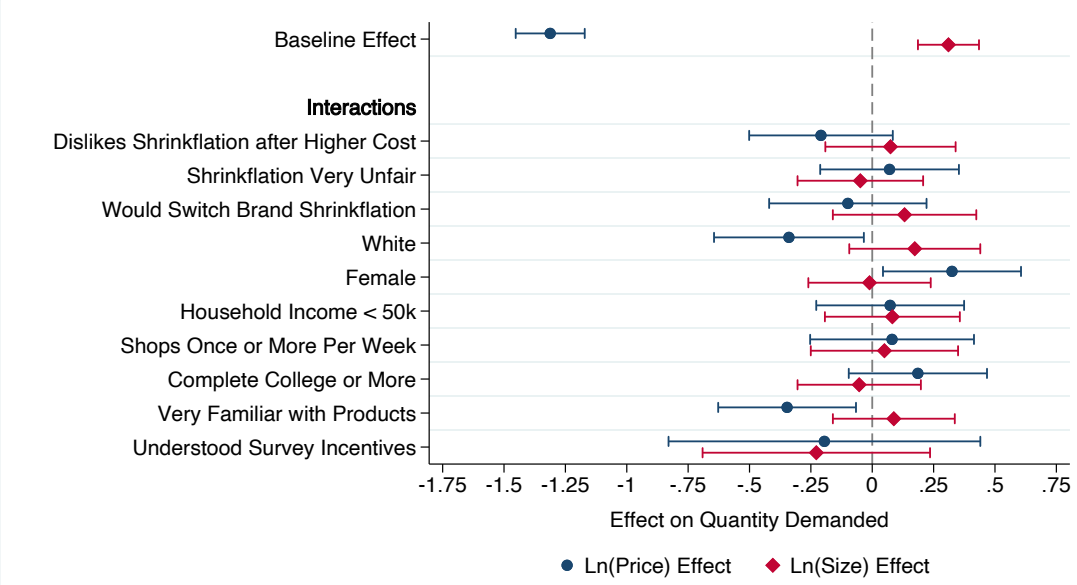
Notes: This table reports Poisson demand estimates allowing for substitution patterns across products. The dependent variable is quantity demanded. In addition to own log price and own log size, the regressions include leave-one-out average log prices and log sizes for products in the same category and parent brand, products in the same category but with a different parent brand, and products in other categories. Columns report estimates separately for the pooled control and pre-treatment sample, and for scenarios 4–6 in the control group, the unit price treatment, the prominent size display treatment, and the size change label treatment. The sample is restricted to observations for which the relevant within-scenario product alternatives are observed. All specifications include individual, scenario, and product fixed effects. Standard errors are clustered at the individual level. The reported p-value tests equality of the own size coefficient and the negative of the own price coefficient.

Figure C.13: Heterogeneity of Demand Responses by Product



Notes: This figure reports product-specific estimates of demand responses to price and size changes. For each of the 24 products in the experiment, we estimate a separate Poisson model of quantity demanded on ln price and ln size. The sample includes all control group scenarios and all pre-treatment scenarios. Each specification includes participant and scenario fixed effects. Standard errors are clustered at the participant level.

Figure C.14: Heterogeneity of Demand Responses by Characteristics



Notes: This figure reports coefficient estimates from Poisson models of quantity demanded. The baseline estimates are from a specification including ln price and ln size without interactions. The remaining estimates report interaction coefficients from separate regressions that interact ln price and ln size with one respondent characteristic at a time. Characteristics include views about shrinkflation, race, gender, household income, shopping frequency, education, product familiarity, and survey comprehension. The sample includes control-group observations and all pre-treatment scenarios. The baseline specification absorbs participant, scenario, and product fixed effects; interaction specifications absorb participant, scenario-by-characteristic, and product-by-characteristic fixed effects. Standard errors are clustered at the participant level. The horizontal axis reports the estimated effect on quantity demanded.

Table C.8: Heterogeneity of Belief Errors by Characteristics

Panel A: Views on Shrinkflation

	Belief has >10% Error		
	Price	Size	Unit Price
Says Likely to Notice Price Change	-0.090***		-0.042*
	(0.024)		(0.023)
Dislikes Shrinkflation After Cost Incr.	-0.0096	-0.023	-0.052***
	(0.014)	(0.016)	(0.015)
Thinks Shrinkflation is Very Unfair	0.0098	0.0061	0.025*
	(0.014)	(0.015)	(0.014)
Likely to Switch Brand After Shrinkflation	-0.015	-0.045**	-0.029*
	(0.016)	(0.018)	(0.016)
Says Likely to Notice Size Change		-0.028*	-0.040***
		(0.015)	(0.014)
Observations	7668	7667	6181

Panel B: Demographics

	Belief has >10% Error		
	Price	Size	Unit Price
White	-0.013	0.024	0.021
	(0.016)	(0.017)	(0.017)
Female	-0.019	-0.0088	0.0079
	(0.013)	(0.014)	(0.014)
College Degree or Above	-0.014	-0.0058	-0.024*
	(0.014)	(0.015)	(0.014)
HHold Income Below 50k	-0.020	-0.0013	0.0067
	(0.015)	(0.016)	(0.016)
Shops Weekly or More	-0.011	-0.013	-0.053***
	(0.017)	(0.017)	(0.016)
Observations	7668	7667	6181

Notes: This table reports heterogeneity in belief errors by respondent characteristics. The dependent variable is an indicator for whether the respondent's belief error exceeds 10 percent (using the final belief elicitation). Panel A reports specifications using respondents' views on shrinkflation and stated attention to product attributes. Panel B reports specifications using demographic and shopping-frequency characteristics. Columns report belief errors for price, size, and unit price, respectively. These samples include everyone. Standard errors, clustered at the individual level, are reported in parentheses.

D. Models: Proofs and Additional Details

D.1 Demand Model

This subsection provides more details on the derivation of our demand model. We start by deriving all the formulas for a static version of the model. Then, we explain how the dynamic formulas follow from the static version and the results presented in the main body of the paper.

Static Version. We start with a model with only one period. There is a unit mass of consumers subindexed with i. Each decides whether to buy a single package of size s at package price p. We assume the following utility function:

$$u_i = \begin{cases} v(s) + y - p + \varepsilon_i & \text{if purchased} \\ y & \text{otherwise} \end{cases} \tag{D.1}$$

ε_i can be equivalently interpreted as the difference in the shocks to the utility of purchasing and not purchasing. We assume $\varepsilon_i \sim F$, that it gives a smooth demand function, and that $v'(s) > 0$.

After a size change, consumers can misperceive the size. We specifically assume:

$$\hat{s}_i = \theta_i s + (1 - \theta_i) s_0 \tag{D.2}$$

With $\hat{s}_i$ being the perceived size for consumer i, and s_0 representing the old size of the package. We interpret θ_i as individual i's attention parameter. Assume $\theta \sim H \ \wedge E(\theta) \neq 0 \ \wedge \ \theta \perp\!\!\!\perp \varepsilon$.

Consumption decisions are made based on the perceived size. Hence, a consumer buys if $v(\hat{s}_i)+\varepsilon_i \geq p$. We can characterize marginal consumers by: $\varepsilon_i^*(p, s, \theta) = p - v(\theta_i s + (1-\theta_i)s_0)$.

We can express aggregate demand and consumer surplus as:

$$D(p, s) = \int_\Theta \left(\int_{\varepsilon^*(p,s,\theta)}^{\infty} f(\varepsilon) d\varepsilon \right) dH(\theta) \tag{D.3}$$

$$CS(p, s) = \int_\Theta \left(\int_{\varepsilon^*(p,s,\theta)}^{\infty} (v(s) - p + \varepsilon) f(\varepsilon) d\varepsilon \right) dH(\theta) \tag{D.4}$$

Note that utility depends on the real size, even if individuals have misperceptions.

To approximate the welfare effects of a size change, we take a second order Taylor approximation around the initial size, s_0:

$$CS(p, s) \approx CS(p, s_0) + CS'_s(p, s_0)\Delta s + \frac{1}{2} CS''_{ss}(p, s_0)(\Delta s)^2 \tag{D.5}$$

$$\Delta CS(p, s) \approx CS'_s(p, s_0)\Delta s + \frac{1}{2} CS''_{ss}(p, s_0)(\Delta s)^2 \tag{D.6}$$

Before jumping to the first and second derivatives of CS, we explore the demand, as it will provide a simplifying formula that will be later used.

$$D'_p(p,s) = -\int_\Theta f(\varepsilon^*(p,s,\theta))dH(\theta) \tag{D.7}$$

$$D'_s(p,s) = \int_\Theta \left[f(\varepsilon^*(p,s,\theta))v'(\hat{s}(s,\theta))\theta\right] dH(\theta) \tag{D.8}$$

At $s = s_0$, we have $\varepsilon^* = p - v(s_0)$. This threshold does not depend on θ, meaning $f(\varepsilon^*)$ can be factored out of the integral.

$$D'_p(p,s_0) = -f(\varepsilon^*(p,s_0)) \tag{D.9}$$

$$D'_s(p,s_0) = f(\varepsilon^*(p,s_0))v'(s_0)\int_\Theta \theta dH(\theta) = f(\varepsilon^*(p,s_0))v'(s_0)E(\theta) \tag{D.10}$$

$$\frac{D'_s(p,s_0)}{D'_p(p,s_0)} = -v'(s_0)E(\theta) \implies v'(s_0) = -\frac{D'_s(p,s_0)}{E(\theta)D'_p(p,s_0)} \tag{D.11}$$

From Equation D.10 we see that the size elasticity $\eta^s(p,s_0) = \frac{s_0}{D(p,s_0)}f(\varepsilon^*)v'(s_0)E(\theta)$ is increasing in the average attention. If individuals are inattentive on average ($E(\theta) < 1$), we will get a smaller response than in the full attention benchmark ($\theta_i = 1\ \forall i$).

We now move to the derivatives of CS:

$$\begin{aligned} CS'_s(p,s) &= \int_\Theta \left[\int_{\varepsilon^*(p,s,\theta)}^\infty v'(s)f(\varepsilon)d\varepsilon - (v(s) - p + \varepsilon^*)f(\varepsilon^*)\frac{\partial \varepsilon^*}{\partial s}\right] dH(\theta) \\ &= \int_\Theta \left[\left(\int_{\varepsilon^*}^\infty v'(s)f(\varepsilon)d\varepsilon\right) + (v(s) - v(\hat{s}))f(\varepsilon^*)v'(\hat{s})\theta\right] dH(\theta) \end{aligned} \tag{D.12}$$

Differentiating again, with some algebraic manipulations and using the $D'_s(p,s)$ expression previously obtained, we get to:

$$\begin{aligned} &CS''_{ss}(p,s) = v''(s)D(p,s) + v'(s)D'_s(p,s) + \\ &+\int_\Theta \left[(v'(s) - v'(\hat{s})\theta)f(\varepsilon^*)v'(\hat{s})\theta + (v(s) - v(\hat{s}))(f'(\varepsilon^*)(-v'(\hat{s})\theta)v'(\hat{s})\theta + f(\varepsilon^*)v''(\hat{s})\theta^2)\right] dH(\theta) \end{aligned} \tag{D.13}$$

Evaluating CS'_s at $s = s_0$:

$$\begin{aligned} CS'_s(p,s_0) &= \int_\Theta \left[\int_{\varepsilon^*}^\infty v'(s_0)f(\varepsilon)d\varepsilon + \underbrace{(v(s_0) - v(\hat{s}(s_0,\theta)))}_{=0} f(\varepsilon^*)v'(\hat{s})\theta\right] dH(\theta) \\ &= v'(s_0)\int_\Theta \left(\int_{\varepsilon^*(p,s_0)}^\infty f(\varepsilon)d\varepsilon\right) dH(\theta) \\ &= v'(s_0)D(p,s_0) \end{aligned} \tag{D.14}$$

Using the expression for $v'(s_0)$ obtained in Equation D.11:

$$CS'_s(p, s_0) = -\frac{D'_s(p, s_0)}{E(\theta)D'_p(p, s_0)}D(p, s_0) \tag{D.15}$$

Evaluating CS''_{ss} at $s = s_0$:

$$\begin{aligned}
CS''_{ss}(p, s_0) &= v''(s_0)D(p, s_0) + v'(s_0)D'_s(p, s_0) \\
&\quad + \int_\Theta (v'(s_0) - v'(s_0)\theta)\underbrace{f(\varepsilon^*(p, s_0))}_{=-D'_p(p,s_0)}v'(s_0)\theta\, dH(\theta) \\
&\quad + \int_\Theta \underbrace{(v(s_0) - v(\hat{s}(s_0, \theta)))}_{=0}\left[f'(\varepsilon^*)(-v'(\hat{s})\theta)v'(\hat{s})\theta + f(\varepsilon^*)v''(\hat{s})\theta^2\right] dH(\theta) \\
&= v''(s_0)D(p, s_0) + v'(s_0)D'_s(p, s_0) - D'_p(p, s_0)v'(s_0)^2 \int_\Theta (1-\theta)\theta\, dH(\theta) \\
&= v''(s_0)D(p, s_0) + v'(s_0)D'_s(p, s_0) - D'_p(p, s_0)v'(s_0)^2\left(E(\theta) - E(\theta^2)\right)
\end{aligned} \tag{D.16}$$

Using the expression for $v'(s_0)$ obtained in Equation D.11 and simplifying:

$$\begin{aligned}
CS''_{ss}(p, s_0) &= v''(s_0)D(p, s_0) - \frac{D'_s(p, s_0)^2}{E(\theta)D'_p(p, s_0)} - \frac{D'_s(p, s_0)^2}{E(\theta)^2 D'_p(p, s_0)}\left(E(\theta) - E(\theta^2)\right) \\
&= v''(s_0)D(p, s_0) - \frac{D'_s(p, s_0)^2}{D'_p(p, s_0)}\frac{2E(\theta) - E(\theta^2)}{E(\theta)^2}.
\end{aligned} \tag{D.17}$$

Going back to the expression for changes in consumer surplus (Equation D.6), we substitute the expressions in Equation D.14 and Equation D.17, and with some algebraic manipulations we obtain:

$$\begin{aligned}
\Delta CS \approx\ & D(p, s_0)(\Delta s)v'(s_0) + \frac{1}{2}(\Delta s)^2 D(p, s_0)v''(s_0) + \\
& \frac{1}{2}(\Delta s)^2 \underbrace{\left(-D'_p(p, s_0)v'(s_0)\right)}_{D'_s(p,s_0|\theta_i=1\ \forall i)} v'(s_0)\left[1 - (1 - E(\theta))^2 - Var(\theta)\right]
\end{aligned} \tag{D.18}$$

This is the $t = 1$ case of Equation 18, which is presented in the main text.

To get the consumer surplus loss from inattention, we compare consumer surplus in the full attention benchmark with the expression with inattention. Then we use Equation D.11

and some algebraic manipulations to get to our final expression:

$$
\begin{aligned}
CS(p,s|\theta=1\ \forall i) - CS(p,s) &\approx -\frac{1}{2}(\Delta s)^2 D'_p(p,s_0)v'(s_0)^2\left[(1-E(\theta))^2 + Var(\theta)\right] \\
&= -\frac{1}{2}(\Delta s)^2 D'_p(p,s_0)\left(\frac{-D'_s(p,s_0)}{E(\theta)D'_p(p,s_0)}\right)^2\left[(1-E(\theta))^2 + Var(\theta)\right] \\
&= \frac{1}{2}(\Delta s)^2 \frac{D'_s(p,s_0)^2}{|D'_p(p,s_0)|}\frac{(1-E(\theta))^2 + Var(\theta)}{E(\theta)^2} \\
&= \frac{1}{2}(\Delta s)^2 \frac{D'_s(p,s_0)^2(s_0^2/s_0^2)(1/D(p,s_0)^2)}{|D'_p(p,s_0)|(p/p)(1/D(p,s_0)^2)}\frac{(1-E(\theta))^2 + Var(\theta)}{E(\theta)^2} \\
&= \frac{1}{2}pD(p,s_0)\left(\frac{\Delta s}{s_0}\right)^2 \frac{\eta^s(p,s_0)^2}{|\eta^p(p,s_0)|}\frac{(1-E(\theta))^2 + Var(\theta)}{E(\theta)^2}
\end{aligned}
\tag{D.19}
$$

Where η^p and η^s represent the price and size elasticities of demand, respectively. This is equivalent to the main body Equation 19 evaluated at $t = 1$.

Finally, we compare this loss in consumer surplus from inattention with the share of the deadweight loss attributable to consumers from a tax that increases the price by Δp:

$$
-\frac{1}{2}D'_p(p,s_0)(\Delta p)^2 = \frac{1}{2}|\eta^p(p,s_0)|\frac{D(p,s_0)}{p}(\Delta p)^2 \tag{D.20}
$$

We set Equations D.19 and D.20 equal to each other and solve for $\Delta p/p$ to obtain:

$$
\frac{\Delta p}{p} = \left|\frac{\Delta s}{s_0}\right|\frac{|\eta^s(p,s_0)|}{|\eta^p(p,s_0)|}\frac{\sqrt{(1-E(\theta))^2 + Var(\theta)}}{|E(\theta)|} \tag{D.21}
$$

This expression corresponds to the main body Equation 21 when considering $t = 1$.

Dynamic Version. The dynamic setup and the period-t sufficient-statistic formulas are presented in the main body of the paper (subsection 4.1). Here, we explain how they follow from the static derivation above. Fixing a period t, the period-t problem is equivalent to the static model after replacing the static attention parameter θ with the cumulative attention parameter $\phi_t(\theta) = 1 - (1-\theta)^t$. Therefore, the static derivations apply period by period if we replace θ with ϕ_t in the static formulas, and correctly adjust for the time varying nature of the components that depend on ϕ_t. We showed in subsection 4.1 that the demand and its price elasticity do not depend on ϕ_t when evaluated at $s = s_0$. The size elasticity does depend on ϕ_t. So, the resulting period-t welfare formulas in the dynamic model are identical to the static ones after changing θ to ϕ_t and η^s to the one corresponding to period t, η^s_t.

D.2 Supply Model

Here we provide the proofs of the propositions presented in the main body of the text.

Proof of Proposition 1. The disclosure decision can first be studied conditional on a candidate package size s. For a fixed $s \neq s_0$, the adjustment cost Γ is the same under $A = 0$ and $A = 1$, and there is no direct cost of advertising. Define

$$V(s, A) = \max_p \sum_{t=1}^{\infty} \delta^{t-1}(p - c_1 s) D_t(p, s, A). \tag{D.22}$$

Conditional on size s, the firm advertises the size change if and only if $V(s, 1) \geq V(s, 0)$.

A local comparison around s_0 gives a simple characterization. Let $\Delta s = s - s_0$, and let the conditionally optimal price be defined as $p^{opt}(s, A) \equiv \arg\max_p \sum_{t=1}^{\infty} \delta^{t-1}(p - c_1 s) D_t(p, s, A)$ (going forward, we do not make explicit the dependency of p^{opt} on (s, A) to simplify the notation). At $s = s_0$, both advertising choices generate the same perceived size for all consumers and periods, so they imply the same optimal price. A first-order approximation around s_0 gives:

$$\begin{aligned} V(s_0 + \Delta s, 1) - V(s_0 + \Delta s, 0) &\approx V(s_0, 1) - V(s_0, 0) + \Delta s \left. \frac{\partial}{\partial s} \left[V(s, 1) - V(s, 0) \right] \right|_{s=s_0} \\ &\approx \Delta s \left. \frac{\partial}{\partial s} \left[V(s, 1) - V(s, 0) \right] \right|_{s=s_0} \end{aligned} \tag{D.23}$$

By the envelope theorem:

$$\begin{aligned} \left. \frac{\partial V(s, A)}{\partial s} \right|_{s=s_0} &= \sum_{t=1}^{\infty} \delta^{t-1} \left[-c_1 D_t(p^{opt}, s_0, A) + (p^{opt} - c_1 s_0) \left. \frac{\partial D_t(p^{opt}, s, A)}{\partial s} \right|_{s=s_0} \right] \\ &= \sum_{t=1}^{\infty} \delta^{t-1} \left[-c_1 D_t(p^{opt}, s_0, A) + (p^{opt} - c_1 s_0) f(p^{opt} - v(s_0)) v'(s_0) E[\phi_{it}(A)] \right] \end{aligned} \tag{D.24}$$

Considering that $D_t(p, s_0, 0) = D_t(p, s_0, 1)$, we get:

$$V(s_0 + \Delta s, 1) - V(s_0 + \Delta s, 0) \approx \Delta s (p^{opt} - c_1 s_0) f(p^{opt} - v(s_0)) v'(s_0) \sum_{t=1}^{\infty} \delta^{t-1} \left(E[\phi_{it}(1)] - E[\phi_{it}(0)] \right) \tag{D.25}$$

We now connect this expression to the underlying attention parameter. Since $\phi_{it}(A) = 1 - (1 - \theta_i(A))^t$ we have:

$$\begin{aligned} \sum_{t=1}^{\infty} \delta^{t-1} E[\phi_{it}(1) - \phi_{it}(0)] &= \sum_{t=1}^{\infty} \delta^{t-1} E\left[(1 - \theta_i(0))^t - (1 - \theta_i(1))^t \right] \\ &= \frac{1}{1 - \delta} E\left[\frac{\theta_i(1)}{1 - \delta + \delta\theta_i(1)} - \frac{\theta_i(0)}{1 - \delta + \delta\theta_i(0)} \right] \end{aligned} \tag{D.26}$$

Under Assumption 1.2, $\theta_i(1) \geq \theta_i(0)\ \forall i$. Since $\frac{\theta}{1-\delta+\delta\theta}$ is increasing in θ (with $\delta \in [0,1)$), the discounted sum above is weakly positive, and strictly positive if advertising increases attention for a positive mass of consumers. Therefore, with $p^{opt} - c_1 s_0 > 0$ and $v'(s_0) > 0$, the sign of $V(s_0 + \Delta s, 1) - V(s_0 + \Delta s, 0)$ is locally the sign of Δs.

Hence, under Assumption 1.2, the firm has a local incentive to advertise size increases and not to advertise size decreases. Since Assumption 1.3 implies Assumption 1.2, the same conclusion also holds under Assumption 1.3.

Proof of Proposition 2. Consider a negative cost shock, $c_1 < c_0$, and adopt Assumption 2, so $v'(s_0) = c_0$. Fix an advertising choice A. Ignoring the fixed adjustment cost, which does not affect the first-order conditions conditional on changing size, the firm solves:

$$\max_{p,s} \sum_{t=1}^{\infty} \delta^{t-1}(p - c_1 s) D_t(p, s, A). \tag{D.27}$$

The first-order conditions are:

$$\sum_{t=1}^{\infty} \delta^{t-1} \left[D_t(p, s, A) + (p - c_1 s) D'_{p,t}(p, s, A) \right] = 0 \tag{D.28}$$

$$\sum_{t=1}^{\infty} \delta^{t-1} \left[-c_1 D_t(p, s, A) + (p - c_1 s) D'_{s,t}(p, s, A) \right] = 0. \tag{D.29}$$

Combining them gives:

$$c_1 = \frac{\sum_{t=1}^{\infty} \delta^{t-1} D'_{s,t}(p, s, A)}{\sum_{t=1}^{\infty} \delta^{t-1} \left[-D'_{p,t}(p, s, A) \right]} = \frac{\sum_{t=1}^{\infty} \delta^{t-1} E_H \left[f(\varepsilon^*_{it}) v'(\hat{s}_{it}) \phi_{it}(A) \right]}{\sum_{t=1}^{\infty} \delta^{t-1} E_H \left[f(\varepsilon^*_{it}) \right]} \tag{D.30}$$

At $s = s_0$, $\hat{s}_{it} = s_0$ and $\varepsilon^*_{it} = p - v(s_0)$ for all i, t; they do not depend on θ. Hence the density terms cancel, and the right-hand side becomes:

$$v'(s_0) \frac{\sum_{t=1}^{\infty} \delta^{t-1} E_H[\phi_{it}(A)]}{\sum_{t=1}^{\infty} \delta^{t-1}} = c_0 E_H \left[\frac{\theta_i(A)}{1 - \delta + \delta \theta_i(A)} \right] \tag{D.31}$$

Where we also used $v'(s_0) = c_0$ (from Assumption 2) and the formula for a convergent geometric sequence. If the equality from Equation D.30 does not hold at $s = s_0$, we know that there is a local incentive to change sizes. To establish in which direction, we look at the marginal value of size. Let $V(s, A) = \max_p \sum_{t=1}^{\infty} \delta^{t-1}(p - c_1 s) D_t(p, s, A)$. By the envelope theorem:

$$V'_s(s, A) = -c_1 \sum_{t=1}^{\infty} \delta^{t-1} D_t(p^{opt}, s, A) + (p^{opt} - c_1 s) \sum_{t=1}^{\infty} \delta^{t-1} D'_{s,t}(p^{opt}, s, A) \tag{D.32}$$

From the firm's first-order condition for optimal price (Equation D.28), we can isolate the markup as $(p^{opt} - c_1 s) = \frac{\sum_{t=1}^{\infty} \delta^{t-1} D_t(p^{opt}, s, A)}{\sum_{t=1}^{\infty} \delta^{t-1} [-D'_{p,t}(p^{opt}, s, A)]}$. Substituting this into the envelope condition and factoring out total demand yields:

$$V'_s(s, A) = \left(\sum_{t=1}^{\infty} \delta^{t-1} D_t(p^{opt}, s, A) \right) \left[\frac{\sum_{t=1}^{\infty} \delta^{t-1} D'_{s,t}(p^{opt}, s, A)}{\sum_{t=1}^{\infty} \delta^{t-1} [-D'_{p,t}(p^{opt}, s, A)]} - c_1 \right] \tag{D.33}$$

Because total discounted demand is strictly positive, the sign of V_s' depends entirely on the bracketed term. Evaluating this bracketed term at $s = s_0$ simplifies the demand derivative ratio to $c_0 E_H\left[\frac{\theta_i(A)}{1-\delta+\delta\theta_i(A)}\right]$ (as shown in Equation D.31). Thus, under advertising choice A, the firm has a strict local incentive to increase size ($V_s'(s_0, A) > 0$) if the new marginal cost c_1 satisfies:

$$c_0 E_H\left[\frac{\theta_i(A)}{1-\delta+\delta\theta_i(A)}\right] > c_1 \tag{D.34}$$

For an advertised ($A = 1$) upsize, this condition is guaranteed after a negative cost shock under Assumption 1.1 or under Assumption 1.3. This concludes the proof of the first part of the proposition.

If we impose Assumption 1.1, the advertised case admits a global characterization. Under full attention, the perceived size is equal to true size in every period, so demand no longer varies with t:

$$D_t(p, s, 1) = \int_{p-v(s)}^{\infty} f(\varepsilon)d\varepsilon \equiv D(p, s, 1) \tag{D.35}$$

Therefore, conditional on advertising, the adjusted size solves:

$$\max_{p,s}\left(\sum_{t=1}^{\infty}\delta^{t-1}\right)(p - c_1 s)D(p, s, 1) \iff \max_{p,s}(p - c_1 s)\int_{p-v(s)}^{\infty} f(\varepsilon)d\varepsilon \tag{D.36}$$

The first-order conditions are:

$$\int_{p-v(s)}^{\infty} f(\varepsilon)d\varepsilon - (p - c_1 s)f(p - v(s)) = 0 \tag{D.37}$$

$$-c_1\int_{p-v(s)}^{\infty} f(\varepsilon)d\varepsilon + (p - c_1 s)f(p - v(s))v'(s) = 0 \tag{D.38}$$

Combining these two first-order conditions gives: $v'(s) = c_1$. With a strictly concave v (Assumption 2) the advertised-size optimum is unique, and since $v'(s_0) = c_0$ and $c_1 < c_0$, strict concavity implies $s > s_0$.

Thus, for a firm that decides to implement an advertised size change after a cost decrease, we know that the globally optimal size is larger than the original one under Assumption 1.1.

Proof of Proposition 3. We consider a cost increase: $c_1 > c_0$, and adopt Assumption 2. From our derivations in the previous proof, we know that the firm has a strict local incentive to decrease size under advertising choice A if:

$$c_0 E_H\left[\frac{\theta_i(A)}{1-\delta+\delta\theta_i(A)}\right] < c_1. \tag{D.39}$$

This is just the opposite of Equation D.34. With $c_1 > c_0$, $E_H\left[\frac{\theta_i(A)}{1-\delta+\delta\theta_i(A)}\right] \leq 1$ is sufficient to guarantee the condition. This is what is in Assumption 3.